



深度学习平台与应用

第五讲：卷积神经网络

范琦

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2024年10月9日

大 纲

- 卷积神经网络-引言
- 卷积神经网络

■ 图像分类



假设有一个标签集合
{人, 狗, 猫, 汽车, ...}

模型的分类预测

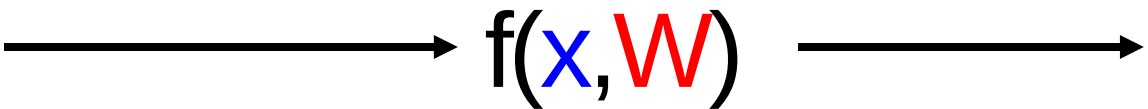
猫

课程回顾：线性分类器



输入图像

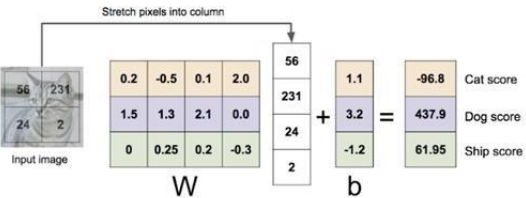
$$f(x, W) = Wx + b$$



571.3	猫
1.25	狗
-132.2	船

Algebraic Viewpoint

$$f(x, W) = Wx$$



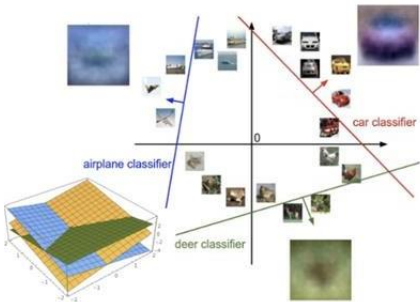
Visual Viewpoint

One template per class



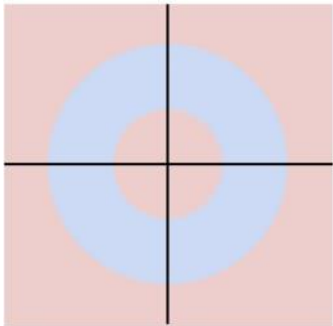
Geometric Viewpoint

Hyperplanes cutting up space



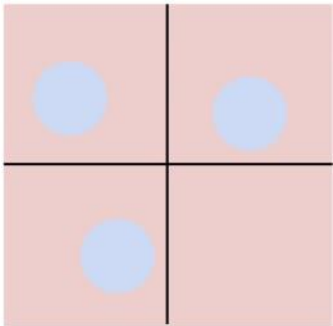
Class 1:
 $1 \leq L2 \text{ norm} \leq 2$

Class 2:
Everything else



Class 1:
Three modes

Class 2:
Everything else



课程回顾：损失函数与优化

- 我们有训练数据：

$$\{(x_i, y_i)\}_{i=1}^N$$

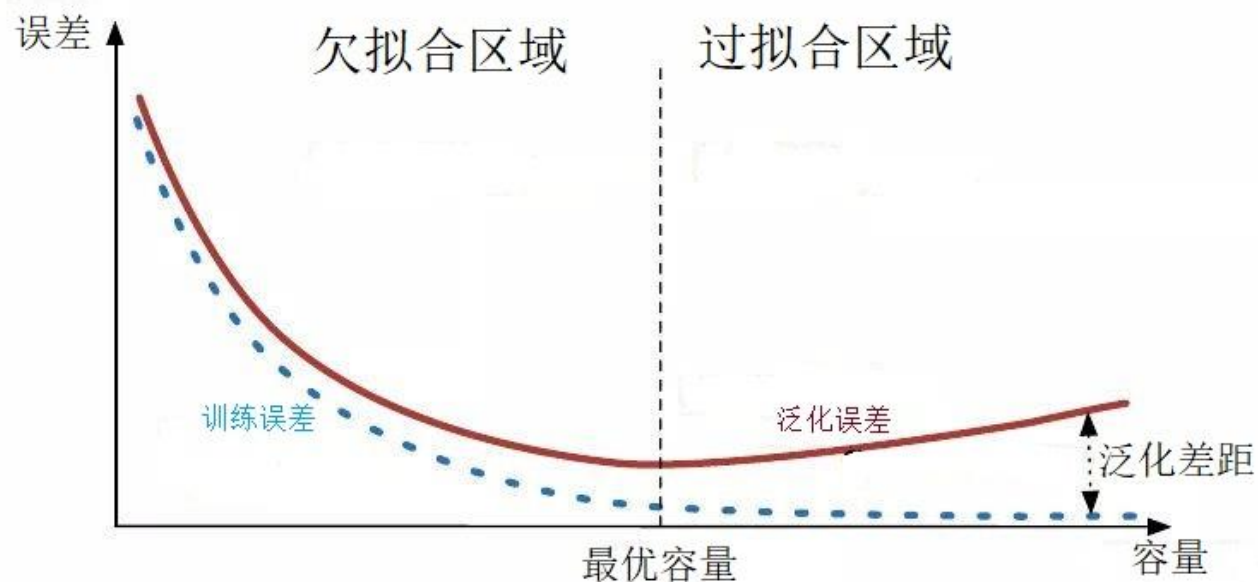
- 我们有线性分类预测：

$$s = f(x; W) = Wx$$

- 我们有损失函数：

$$L = \frac{1}{N} \sum_{i=1}^N L_i + R(W)$$

- 优化模型来降低模型在数据上的损失：SGD, SGD+Momentum, RMSProp, Adam

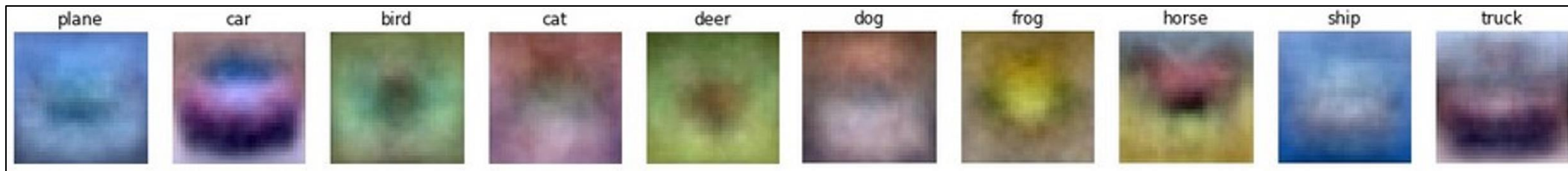
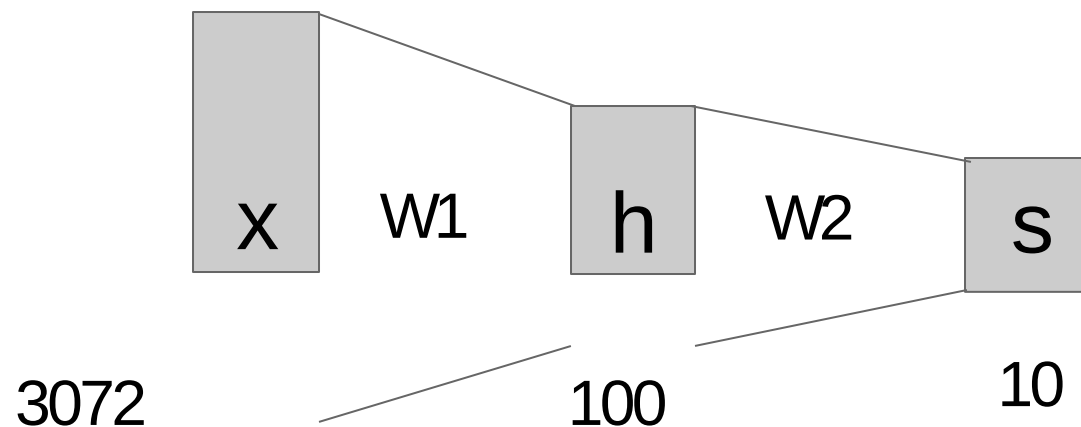


- 线性分类器

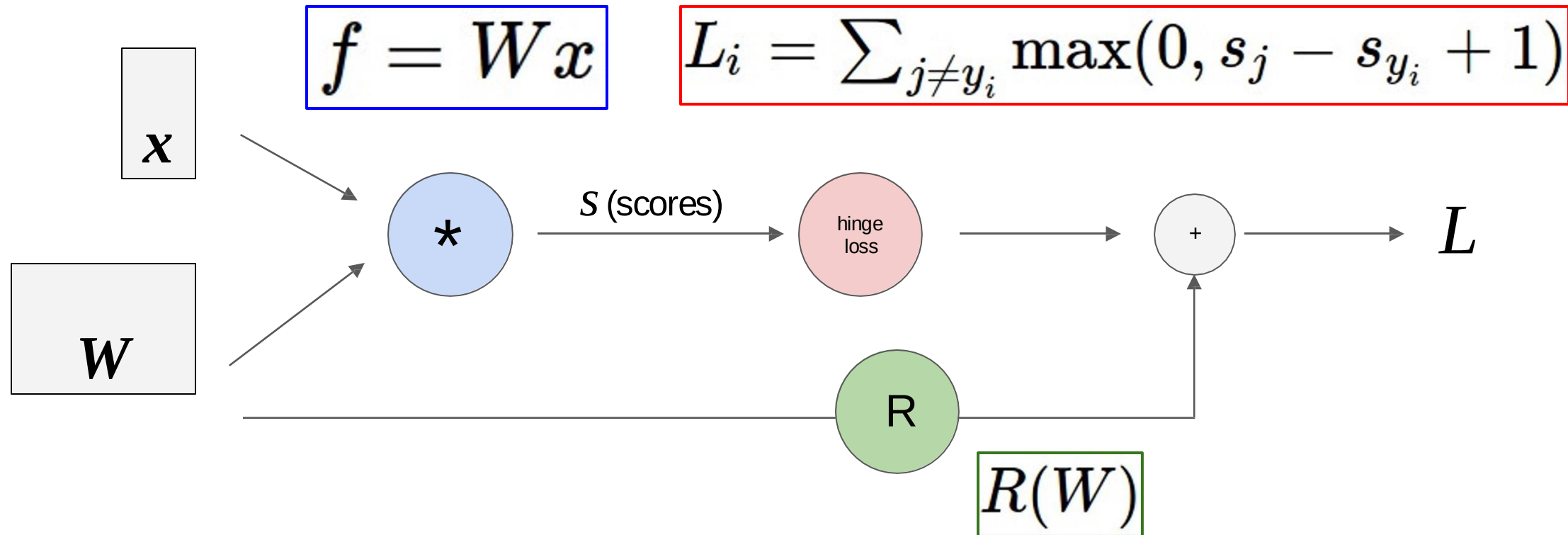
$$f = Wx$$

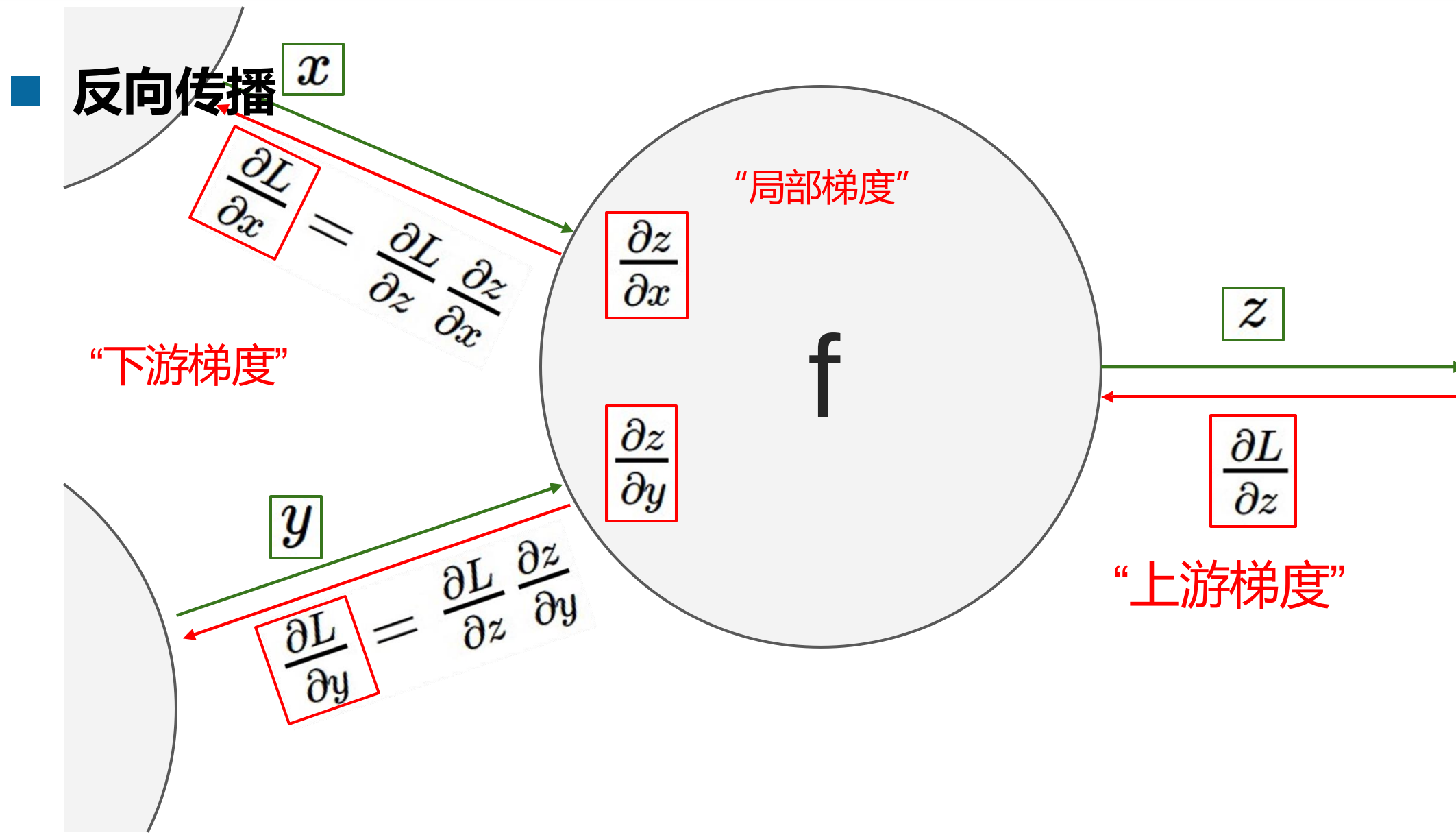
- 2 层神经网络

$$f = W_2 \max(0, W_1 x)$$

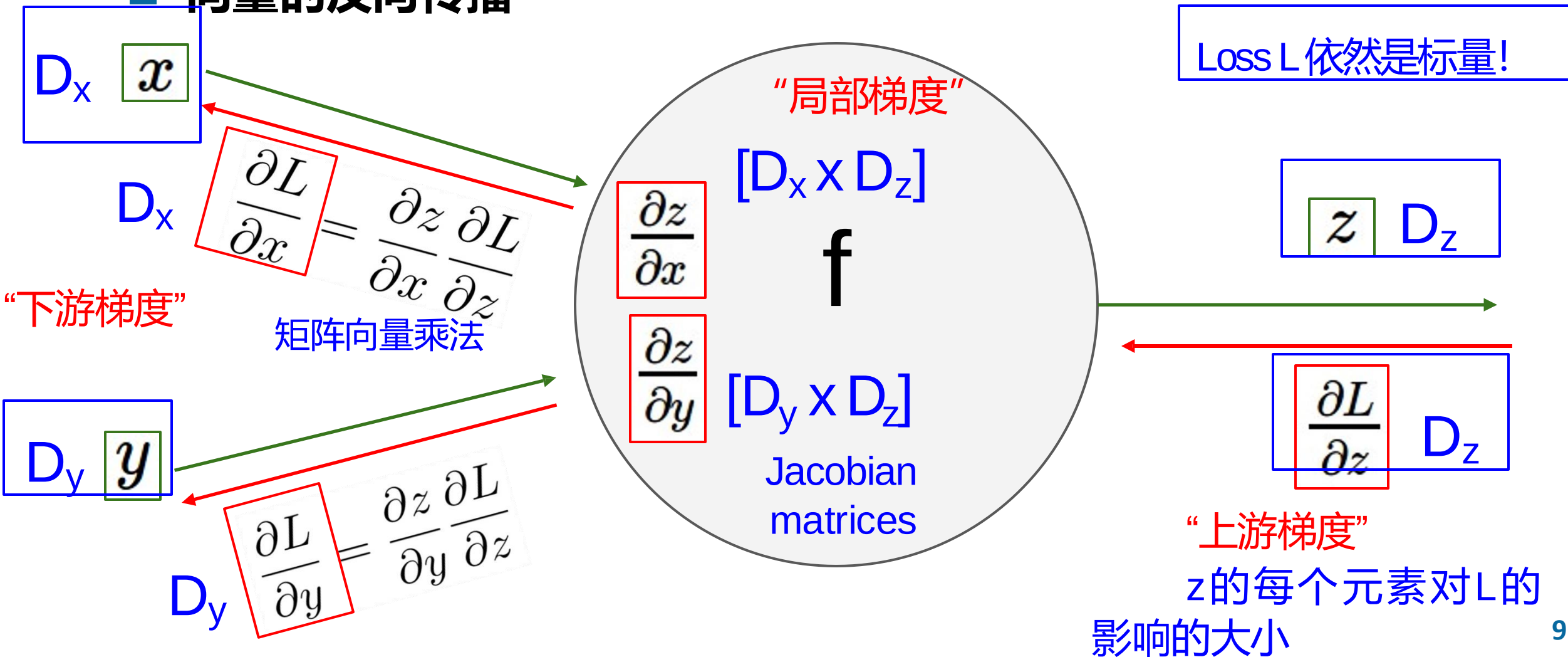


■ 计算图+反向传播





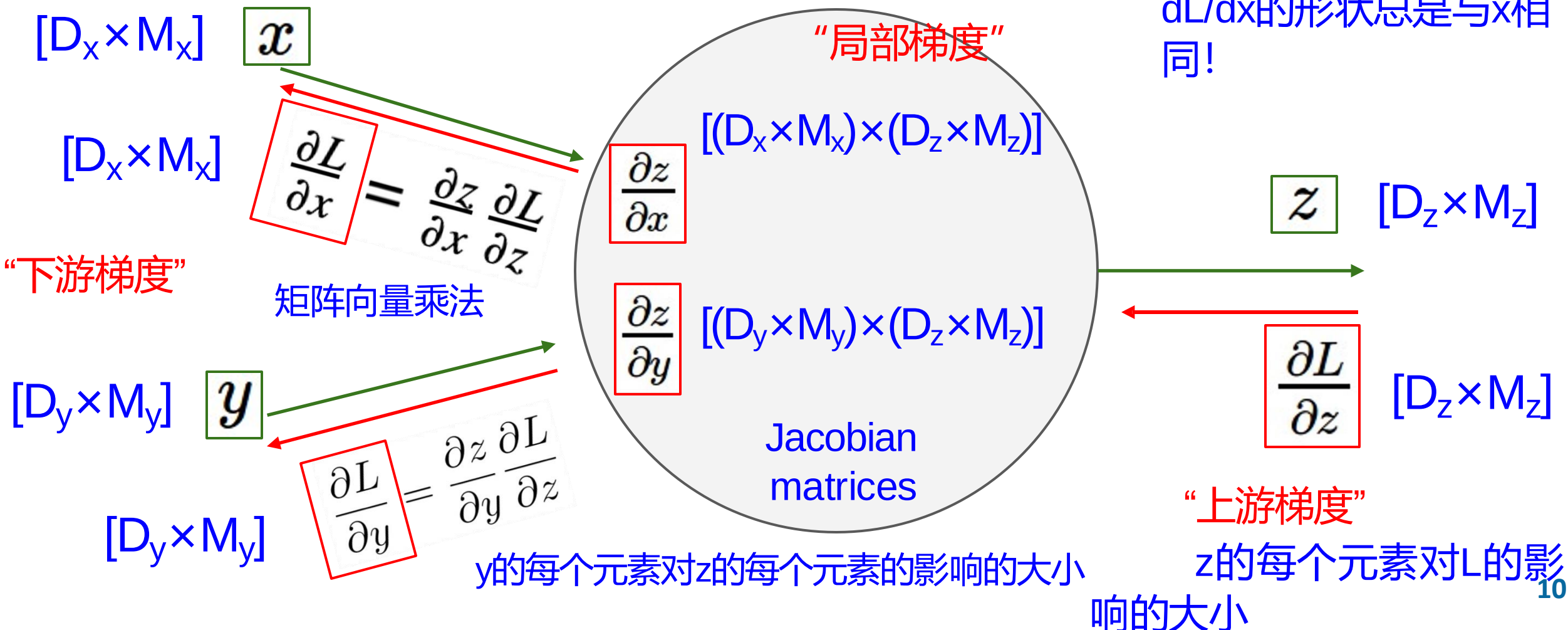
■ 向量的反向传播



■ 矩阵的反向传播

Loss L 依然是标量!

dL/dx 的形状总是与 x 相同!



- 图像分类是最核心的计算机视觉任务

假设有一个标签集合
{人, 狗, 猫, 汽车, ...}



模型的分类预测

Class
Scores

■ 基于图像像素的图像分类

假设有一个标签集合
{飞机, 狗, 猫, 汽车, ...}



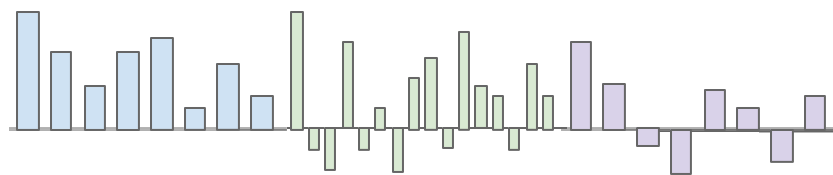
$$f(x) = Wx$$

Class
Scores



■ 基于图像特征图像分类

假设有一个标签集合
{飞机, 狗, 猫, 汽车, ...}



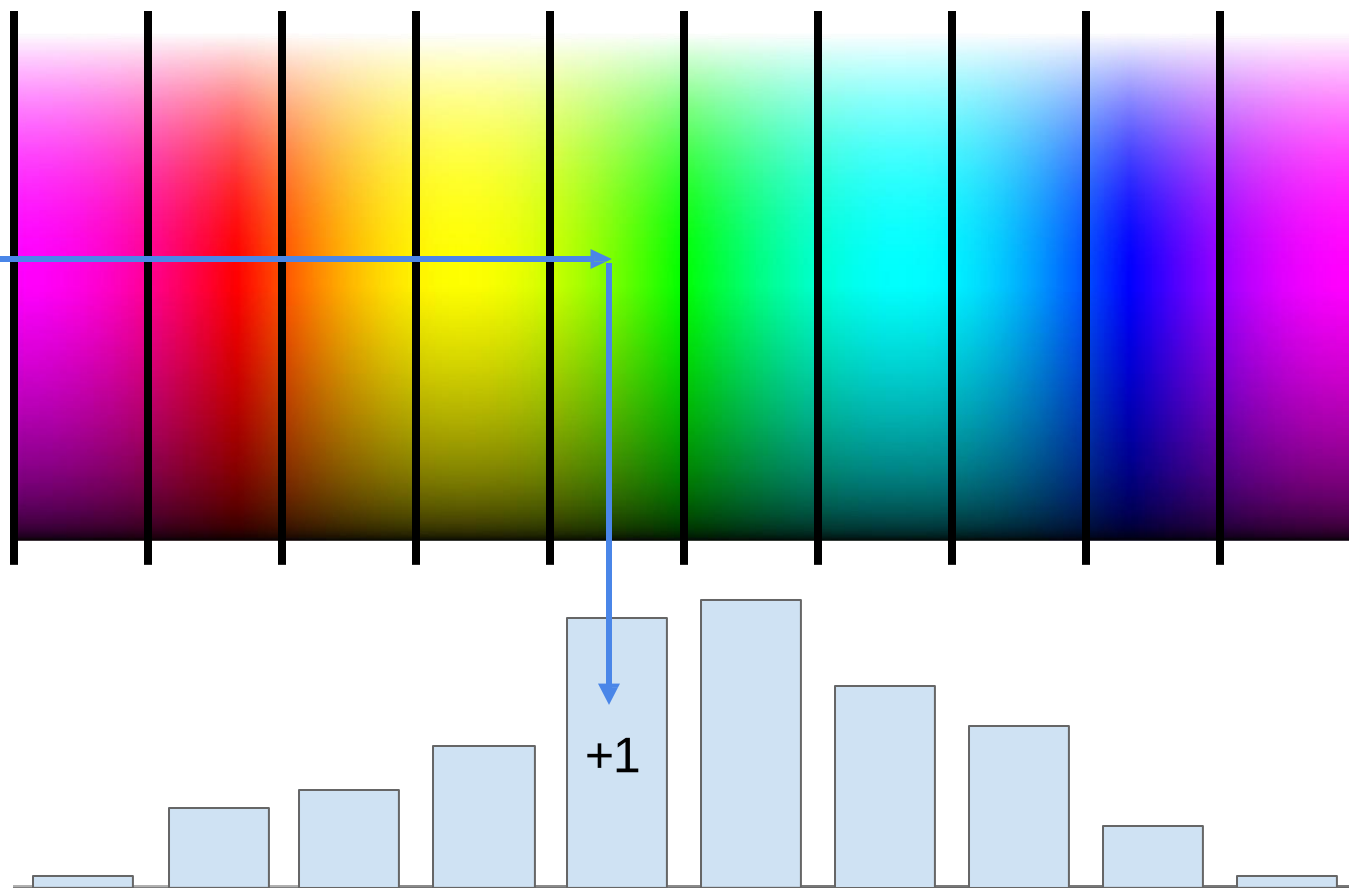
特征表示

$$f(x) = Wx$$



Class
Scores

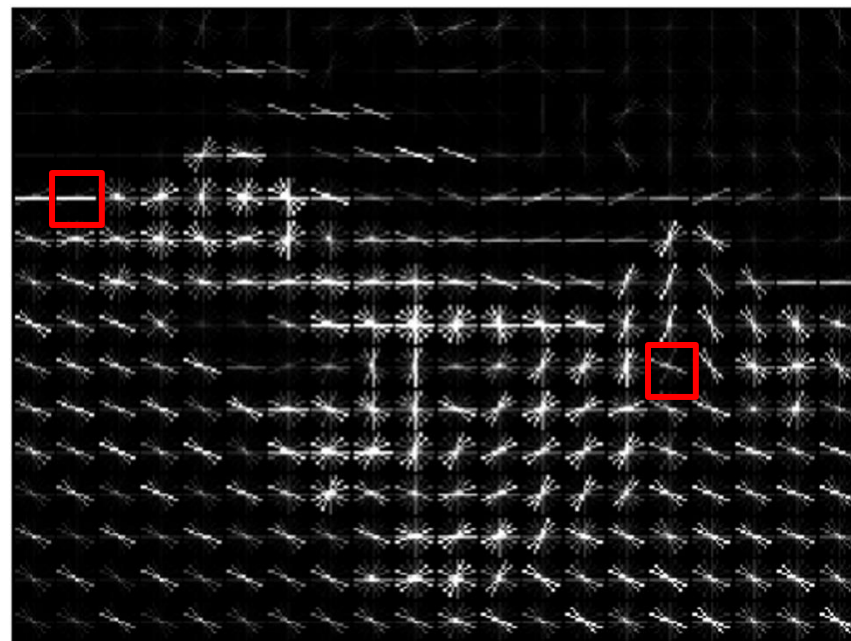
■ 特征提取：颜色直方图 (Color Histogram)



■ 方向梯度直方图 (Histogram of Oriented Gradients, HoG)



HoG: 将图像以8x8为基础区域进行划分。在每个基础区域内，将边缘方向量化为9个数字



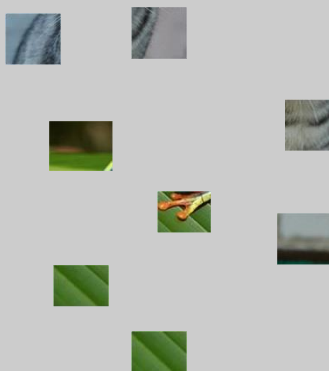
例子：320x240的图像被划分为40x30的区域；每个区域中有9个数字，因此特征向量有 $30 \times 40 \times 9 = 10800$ 个数字

■ 词袋 (Bag of Words)

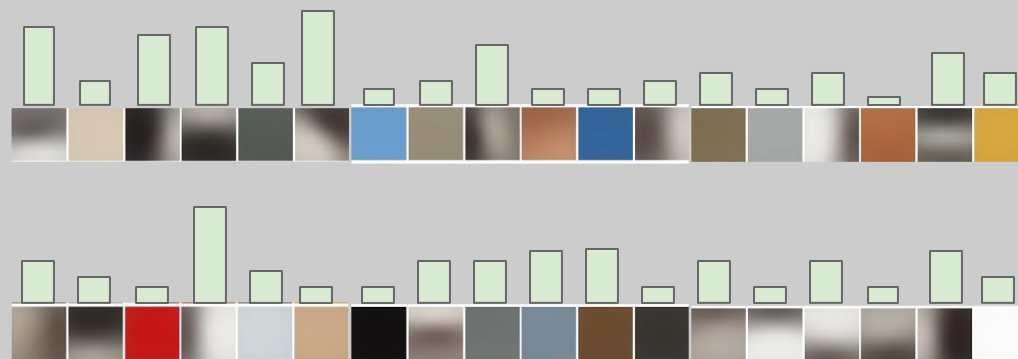
Step 1: 构建 codebook



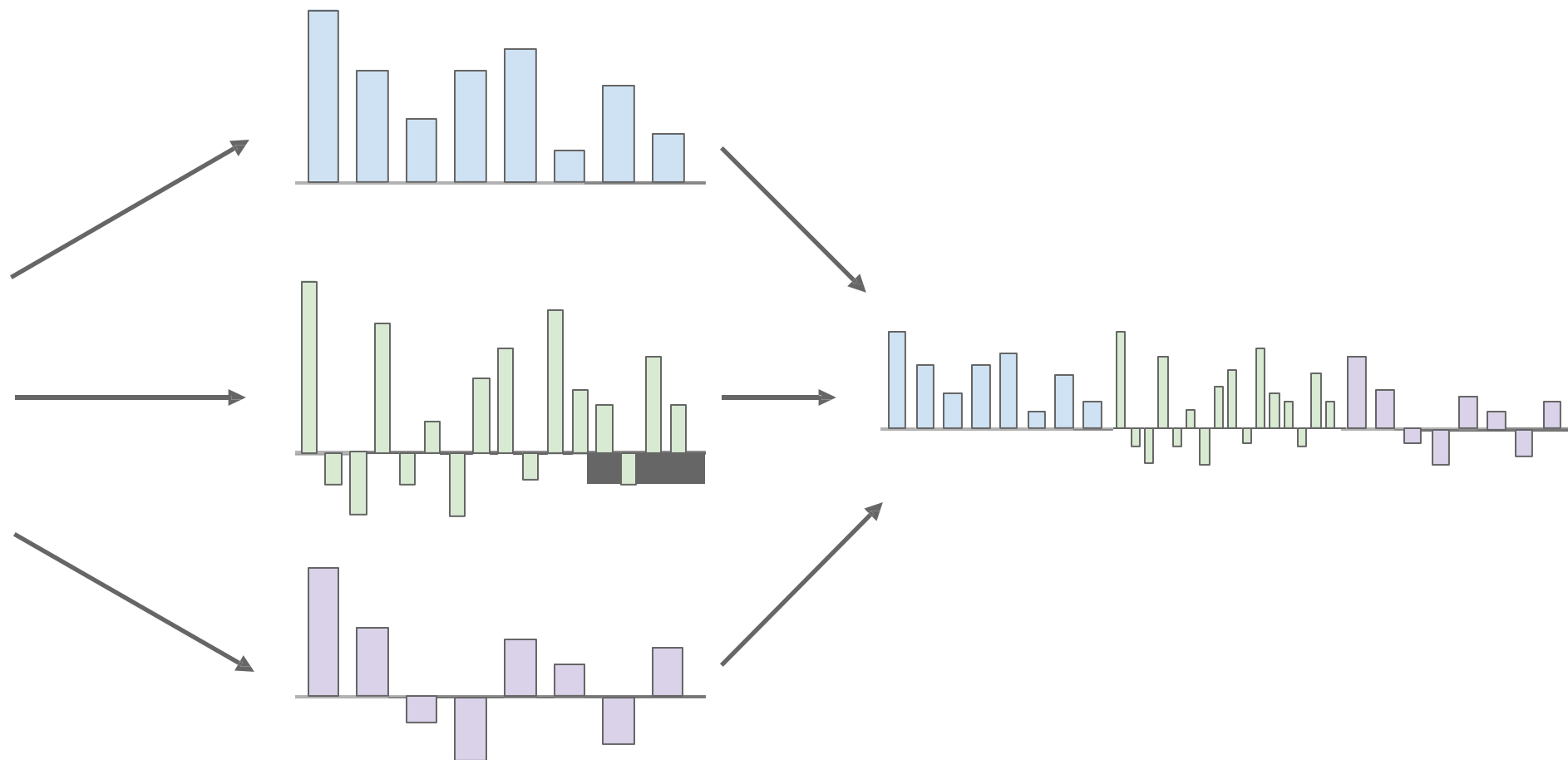
提取随机图像块



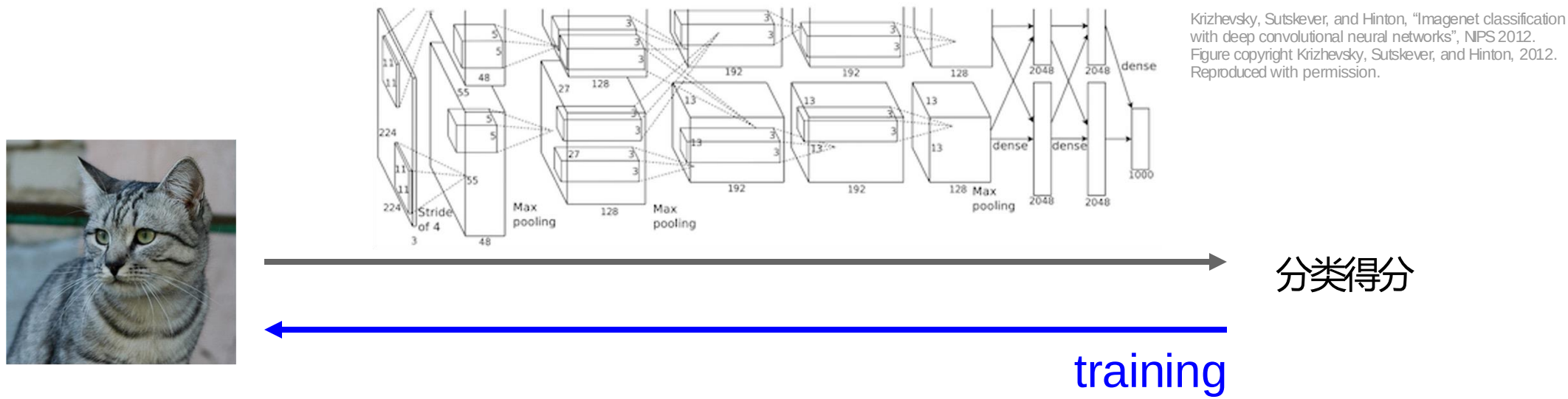
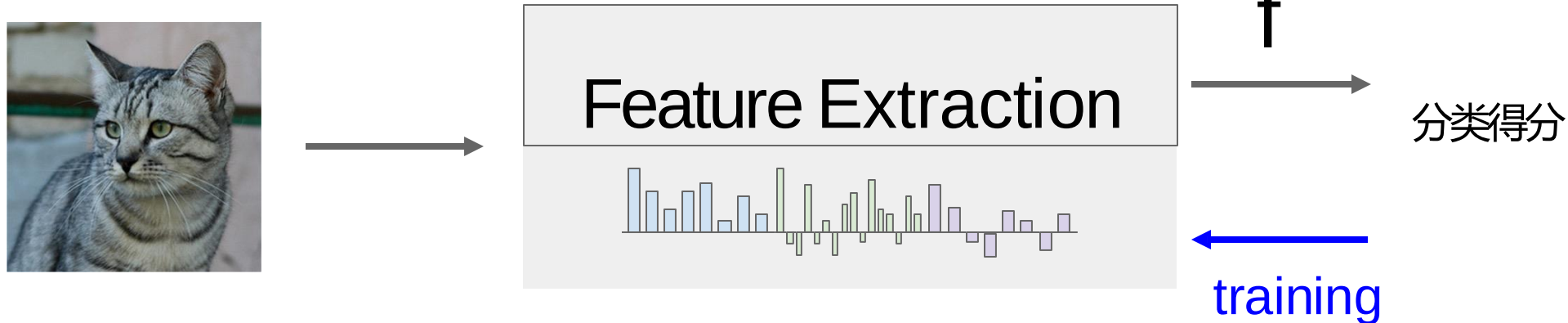
聚类图像块，形成
“visual words”
的codebook



■ 图像特征



■ 手工图像特征与卷积神经网络

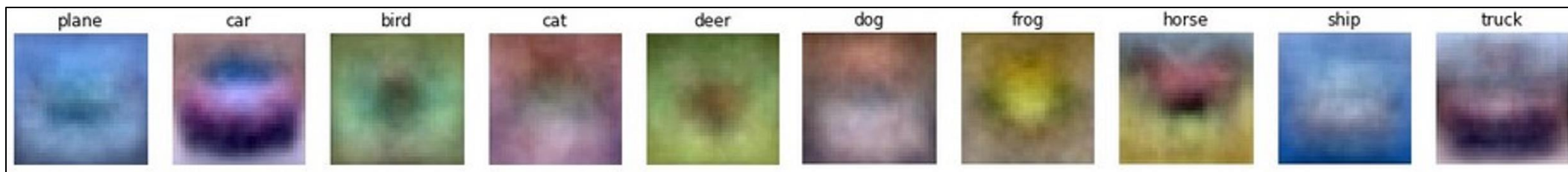
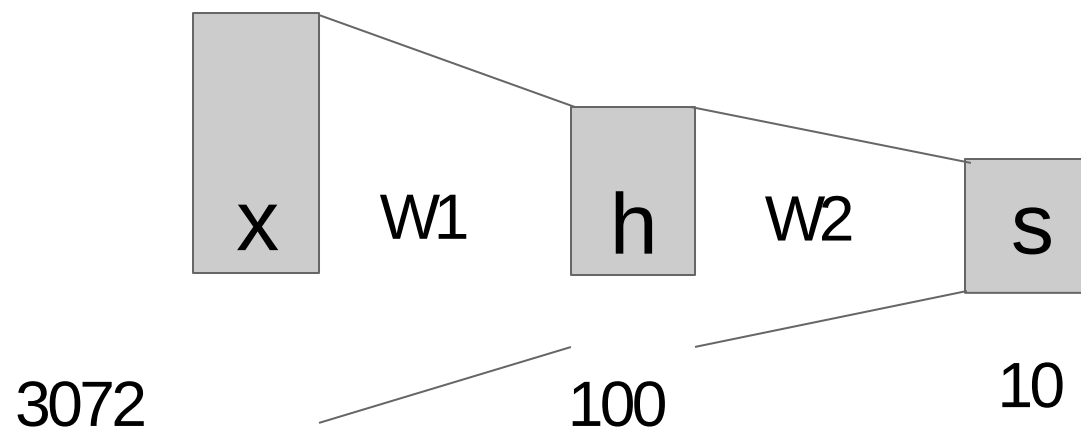


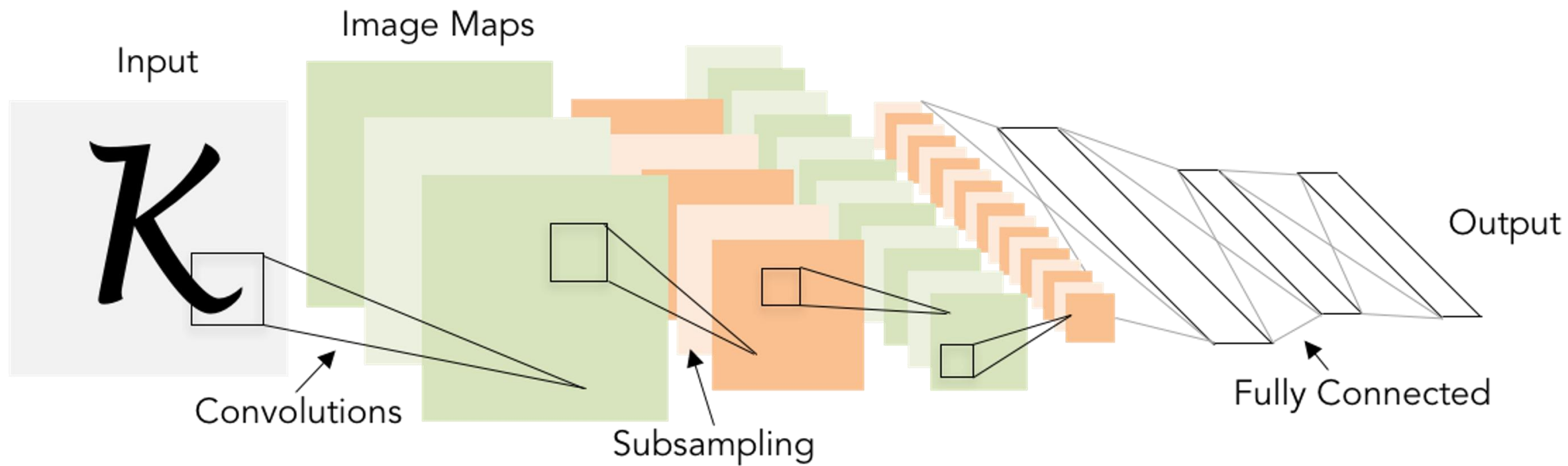
- 线性分类器

$$f = Wx$$

- 2 层神经网络

$$f = W_2 \max(0, W_1 x)$$

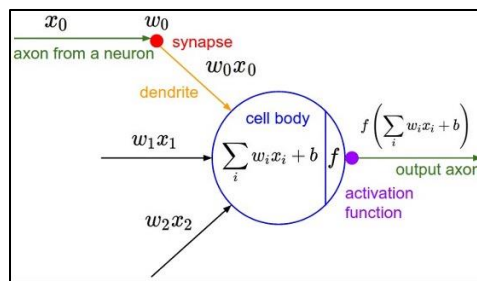




- 神经网络的发展历史
- 第一个感知机：Mark I 感知机

- 识别字母：

$$f(x) = \begin{cases} 1 & \text{if } w \cdot x + b > 0 \\ 0 & \text{otherwise} \end{cases}$$



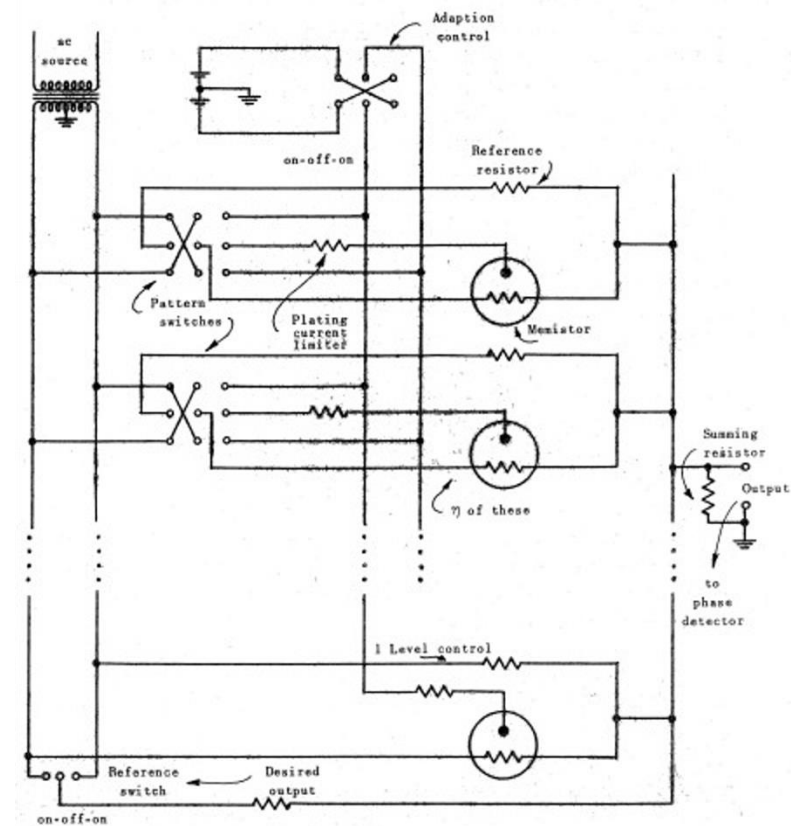
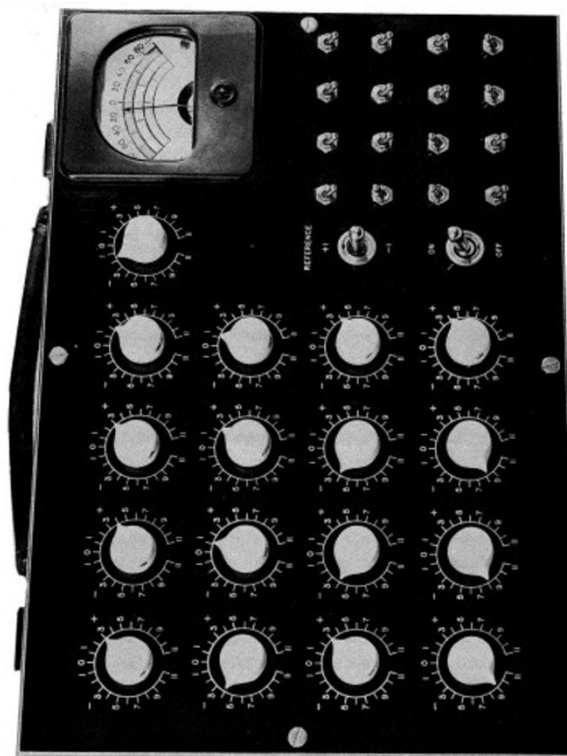
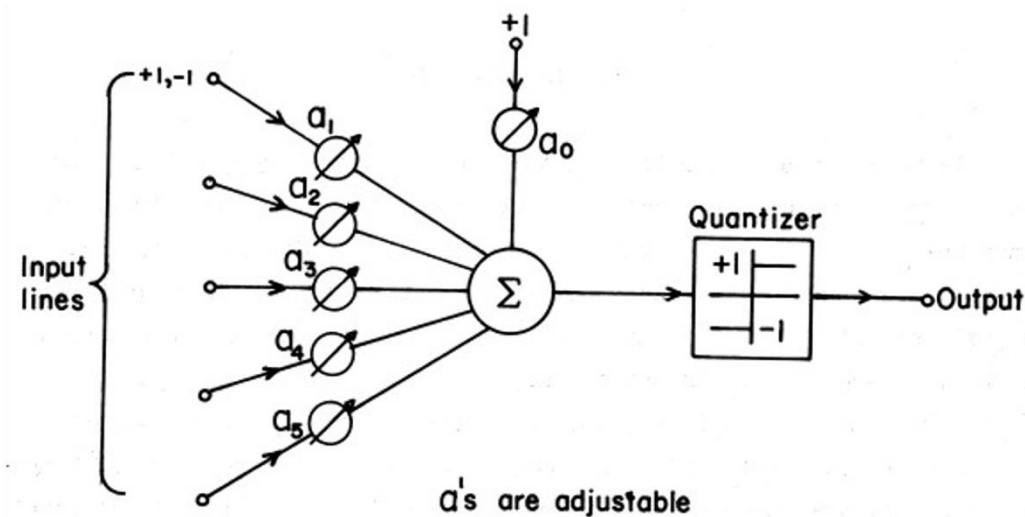
- 网络权重更新规则：

$$w_i(t+1) = w_i(t) + \alpha(d_j - y_j(t))x_{j,i}$$

- Frank Rosenblatt, ~1957: Perceptron

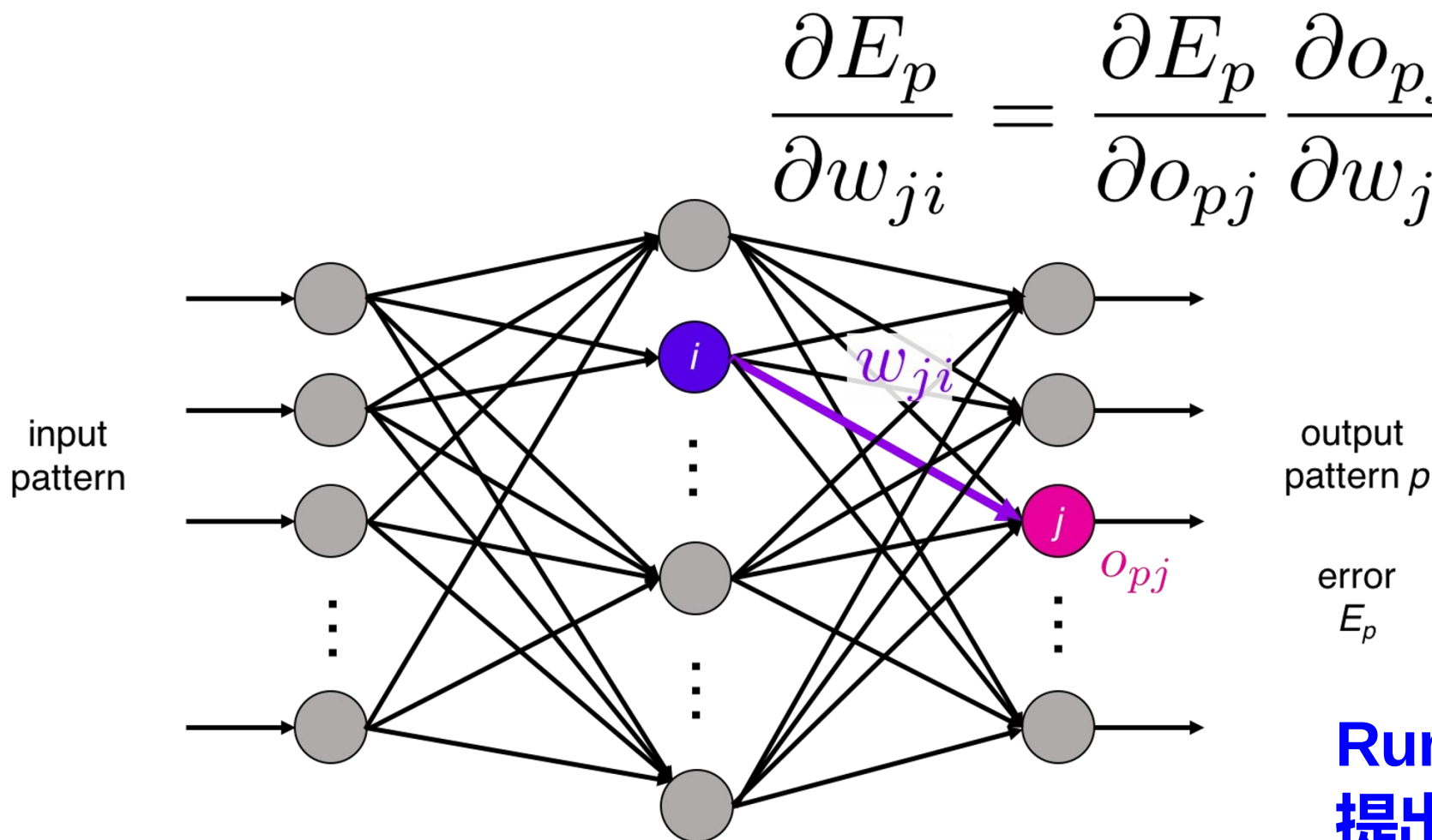


■ 神经网络的发展历史



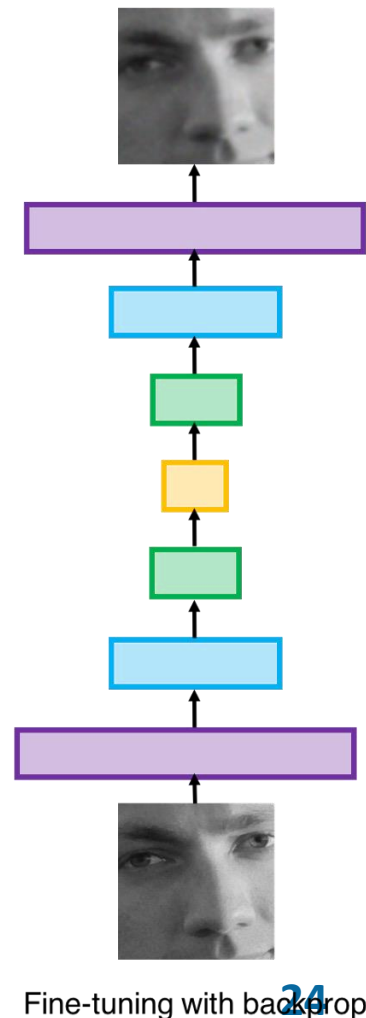
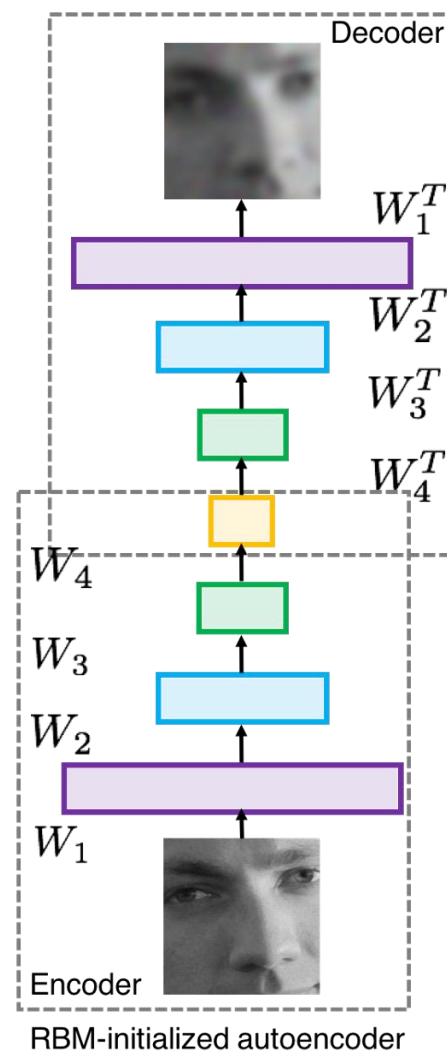
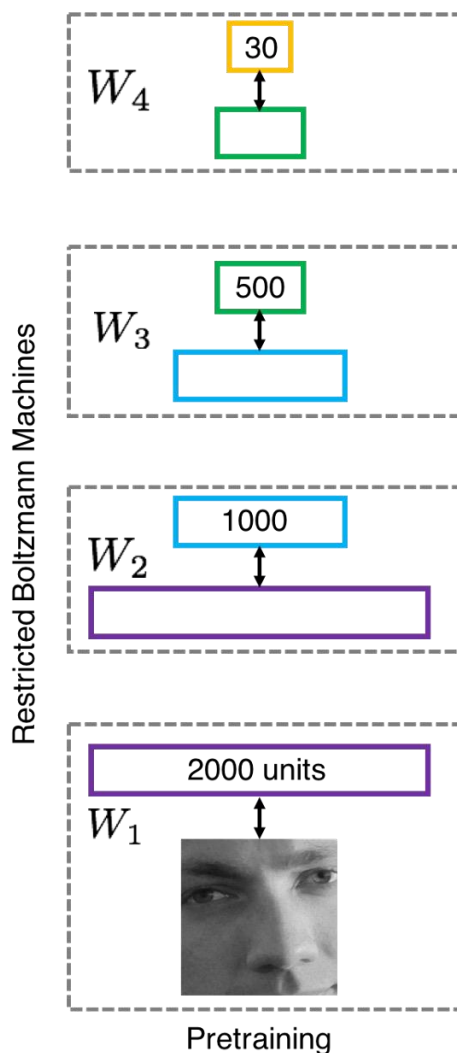
Widrow and Hoff, ~1960 : Adaline/Madaline

■ 神经网络的发展历史



Rumelhart et al., 1986:
提出反向传播

- 神经网络的发展历史
- [Hinton and Salakhutdinov 2006]
- 受限玻尔兹曼机
- 重振深度学习研究



神经网络的发展历史

Acoustic Modeling using Deep Belief Networks

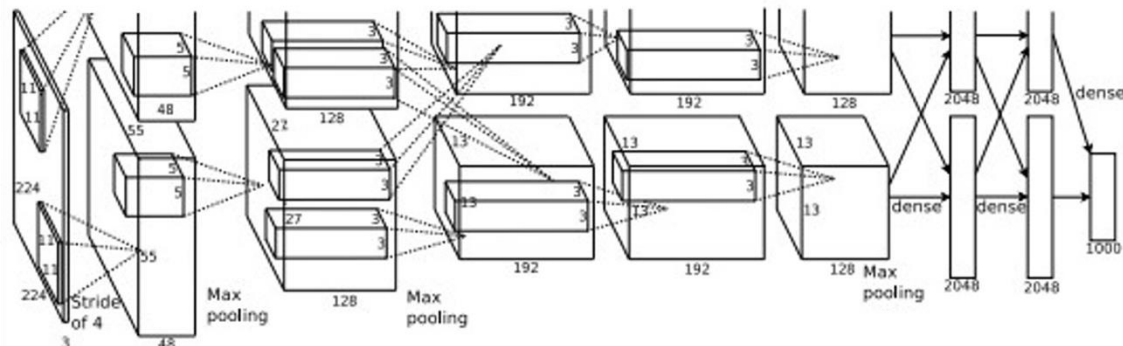
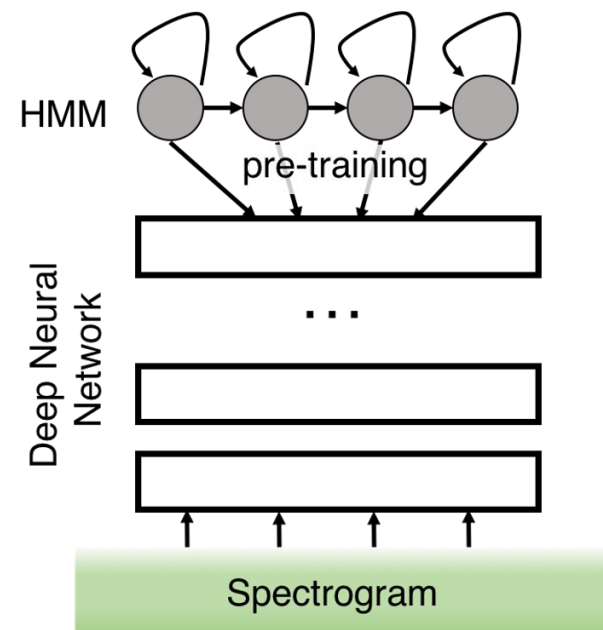
Abdel-rahman Mohamed, George Dahl, Geoffrey Hinton, 2010

Context-Dependent Pre-trained Deep Neural Networks
for Large Vocabulary Speech Recognition

George Dahl, Dong Yu, Li Deng, Alex Acero, 2012

Imagenet classification with deep convolutional
neural networks

Alex Krizhevsky, Ilya Sutskever, Geoffrey E Hinton, 2012



■ 卷积神经网络的发展历史

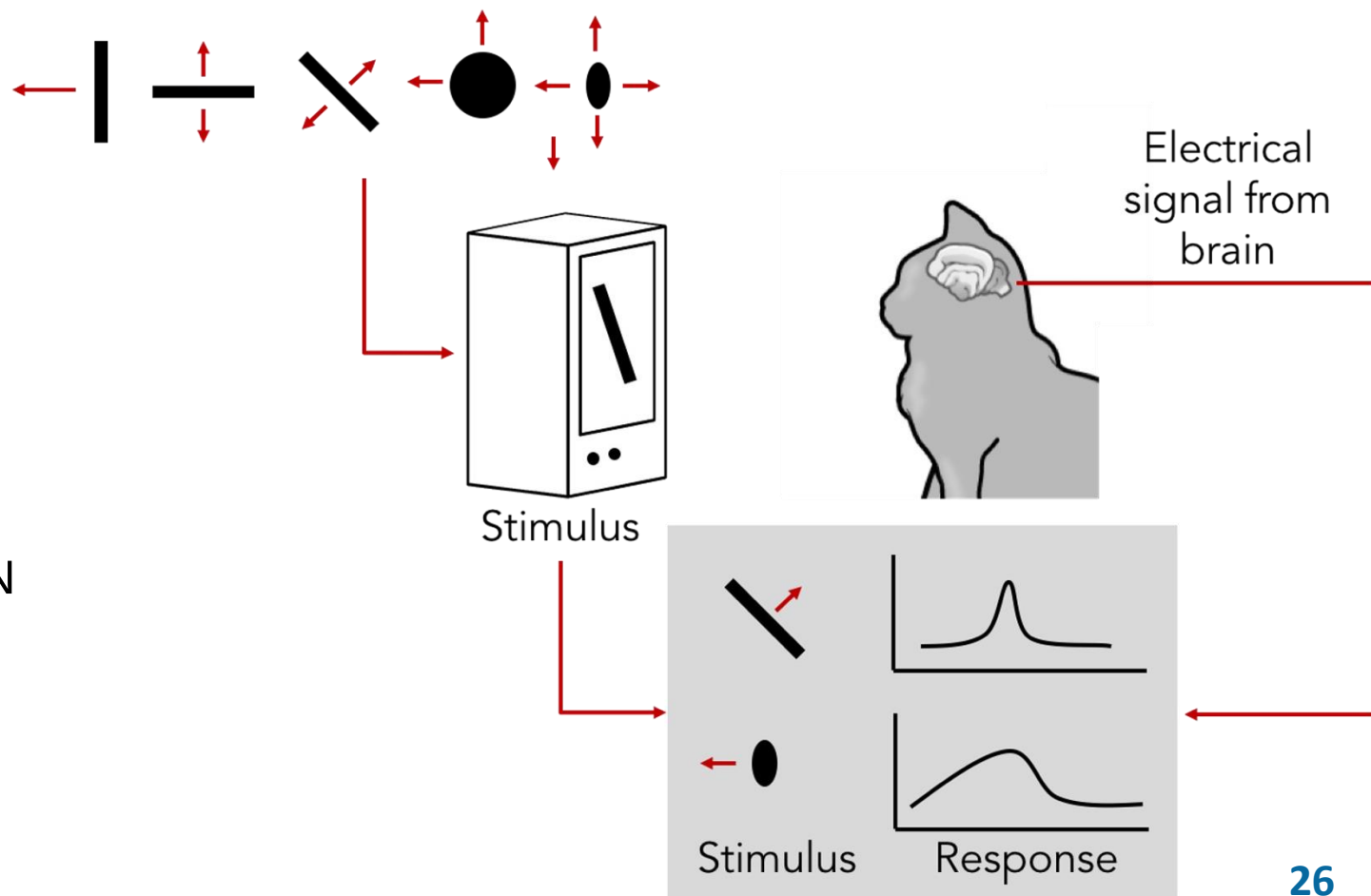
Hubel & Wiesel,
1959

RECEPTIVE FIELDS OF SINGLE NEURONES IN
THE CAT'S STRIATE CORTEX

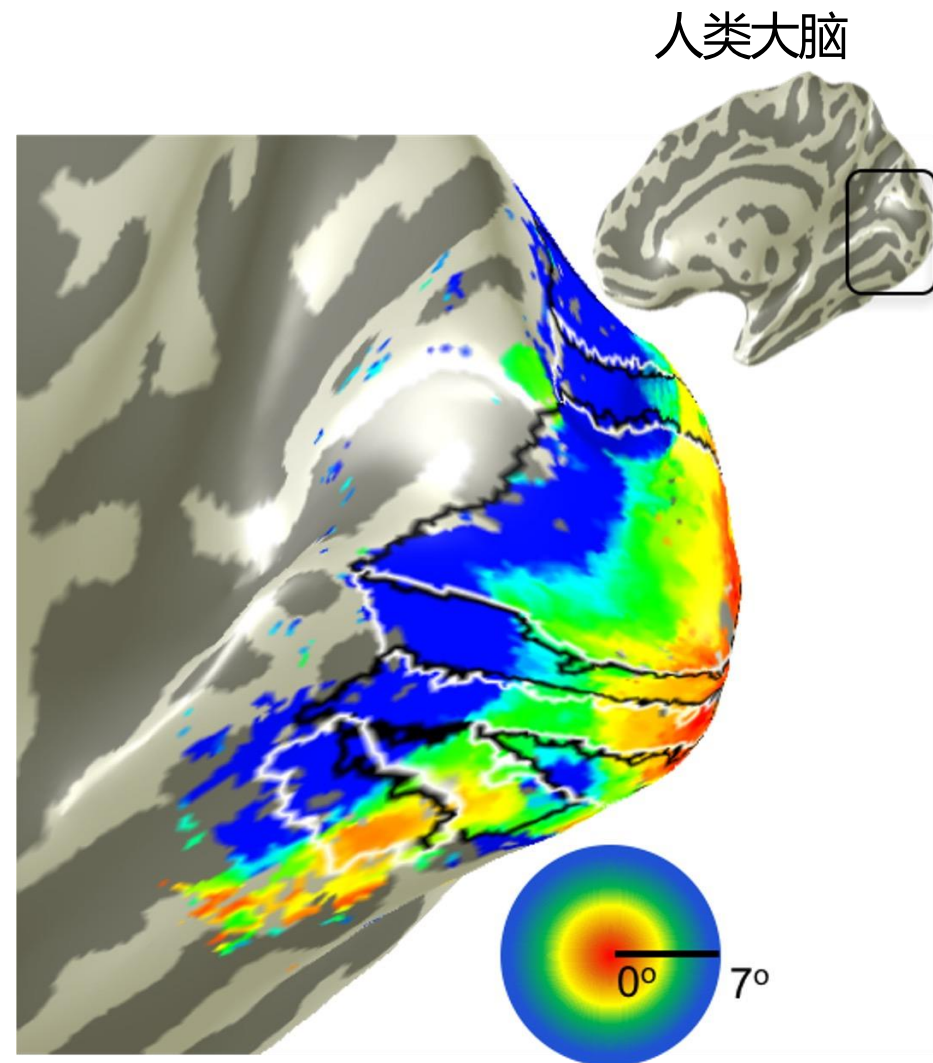
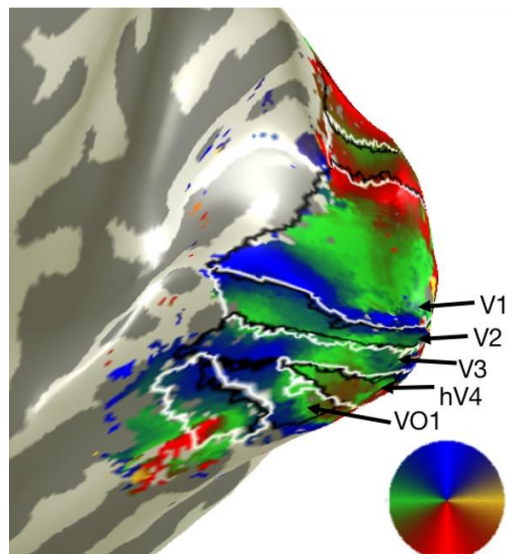
1962

RECEPTIVE FIELDS, BINOCULAR INTERACTION
AND FUNCTIONAL ARCHITECTURE IN
THE CAT'S VISUAL CORTEX

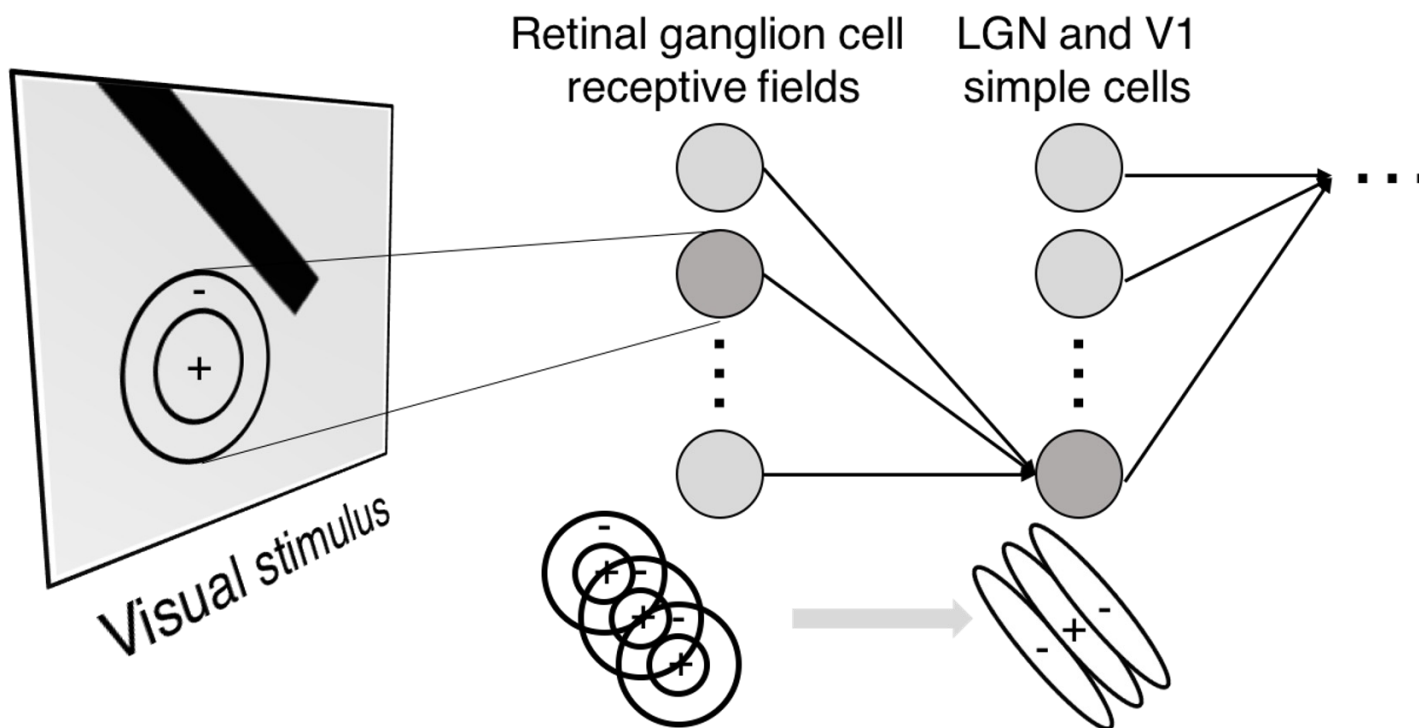
1968...



- 卷积神经网络的发展历史
- 皮层地形图：大脑皮层中的相邻细胞被视野中的相邻区域激活



- 卷积神经网络的发展历史
- 视觉分级感知



Simple cells:
Response to light
orientation

Complex cells:
Response to light
orientation and movement

Hypercomplex cells:
response to movement
with an end point



No response



Response
(end point)

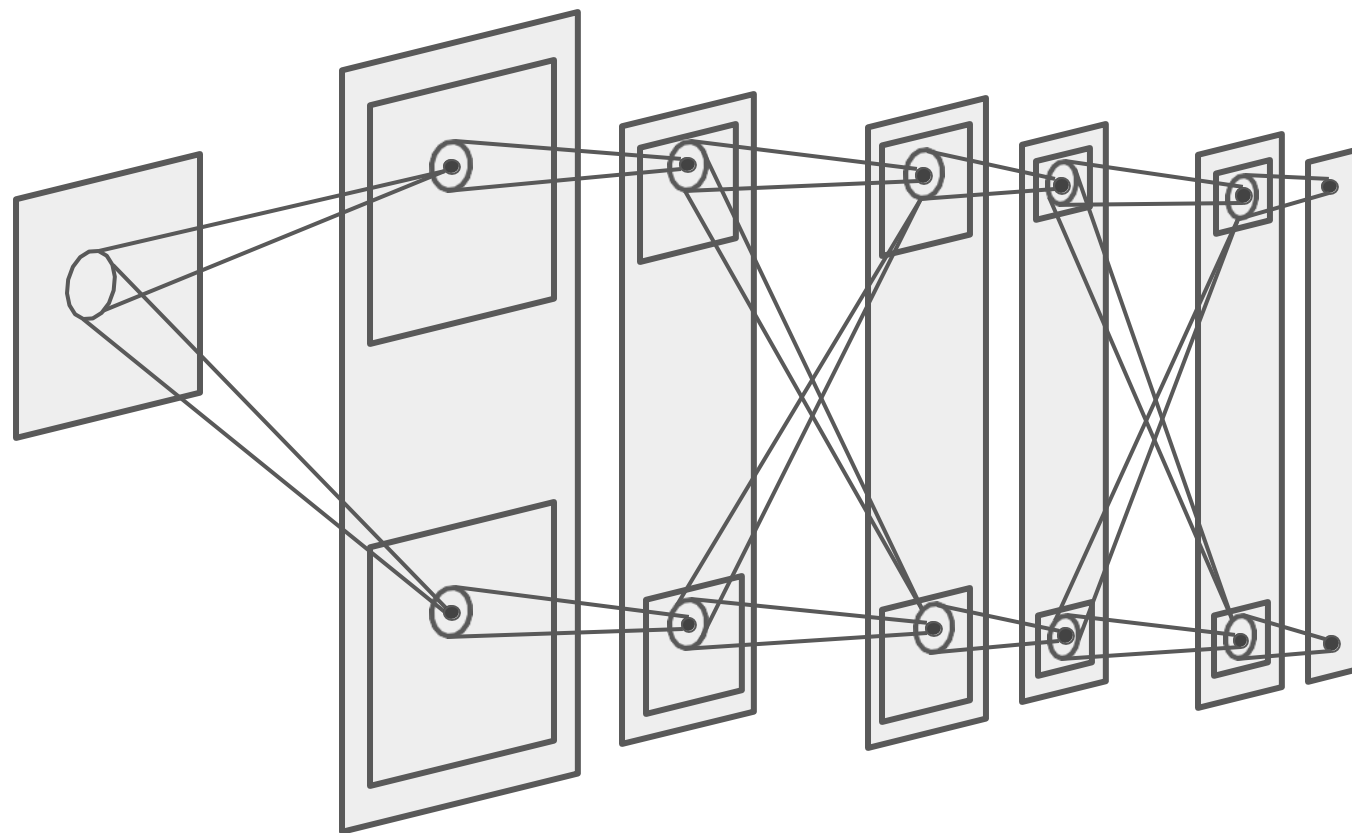
■ 卷积神经网络的发展历史

Neocognitron
[Fukushima 1980]

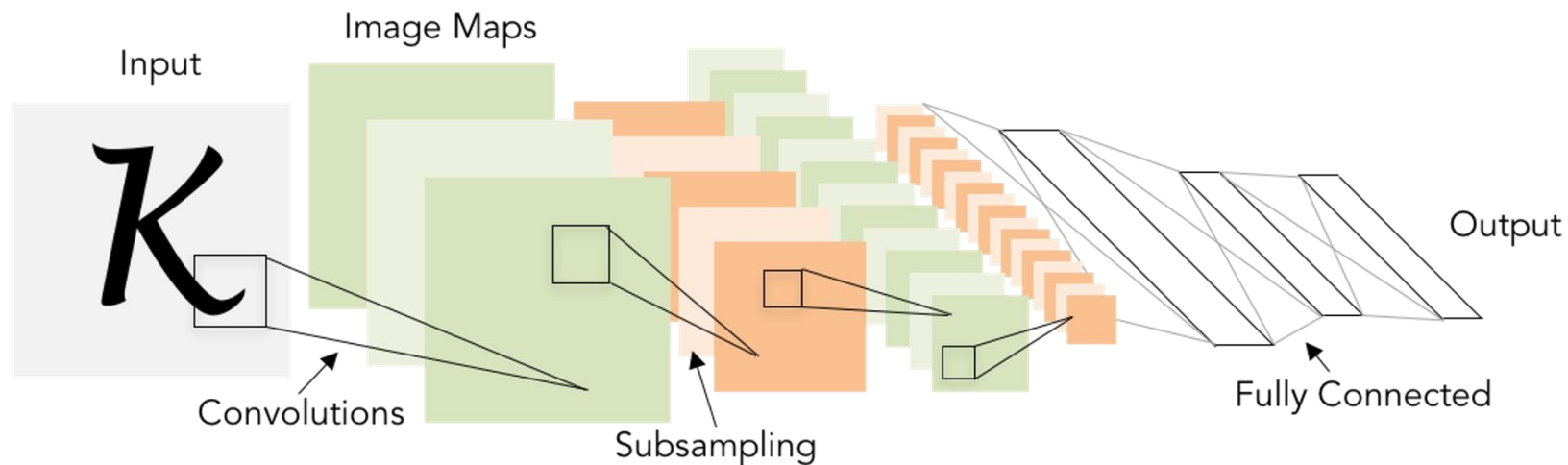
三明治结构 (SCSCSC...)

simple cells: modifiable parameters

complex cells: perform pooling



- 卷积神经网络的发展历史
- 将基于梯度优化的卷积网络应用到文档识别
- [LeCun, Bottou, Bengio, Haffner 1998]



■ 卷积神经网络的发展历史

ImageNet Classification with Deep Convolutional Neural Networks
[Krizhevsky, Sutskever, Hinton, 2012]

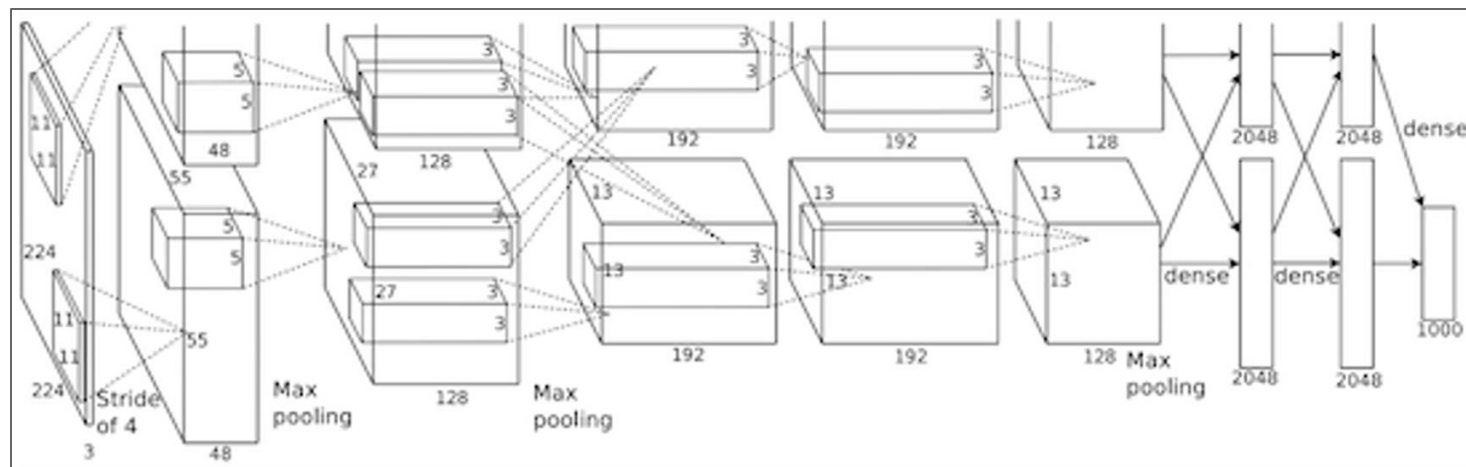
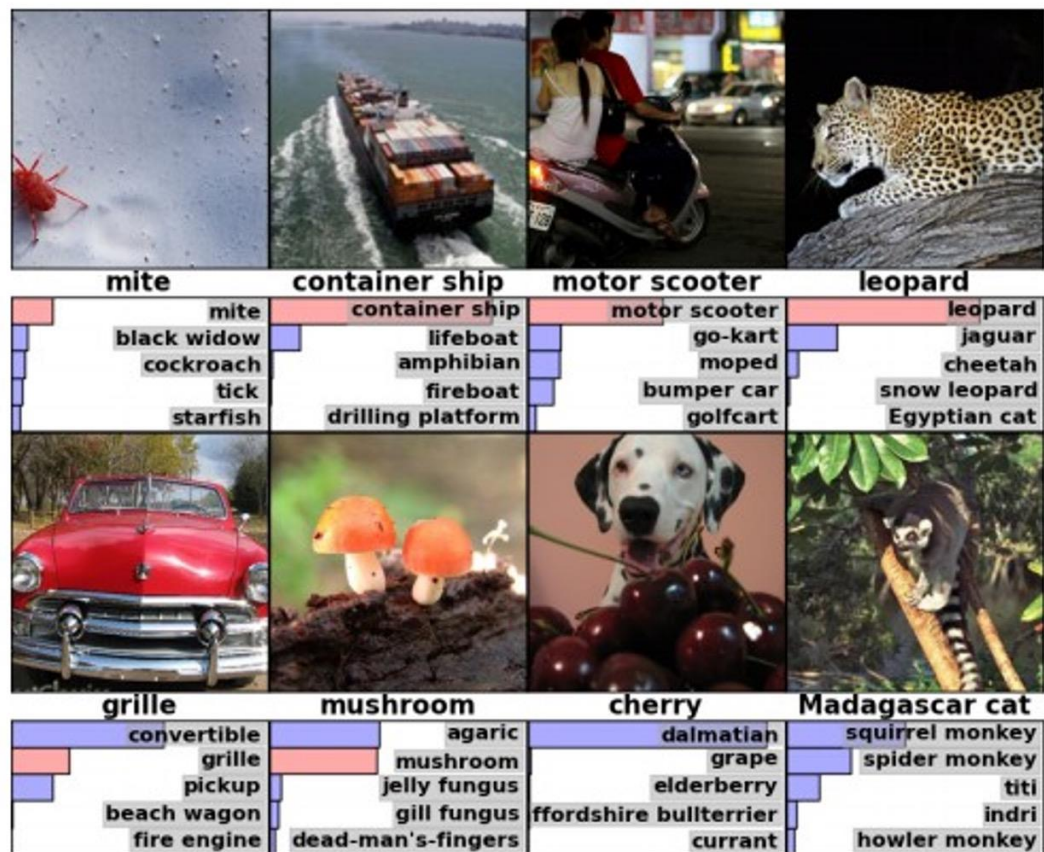


Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

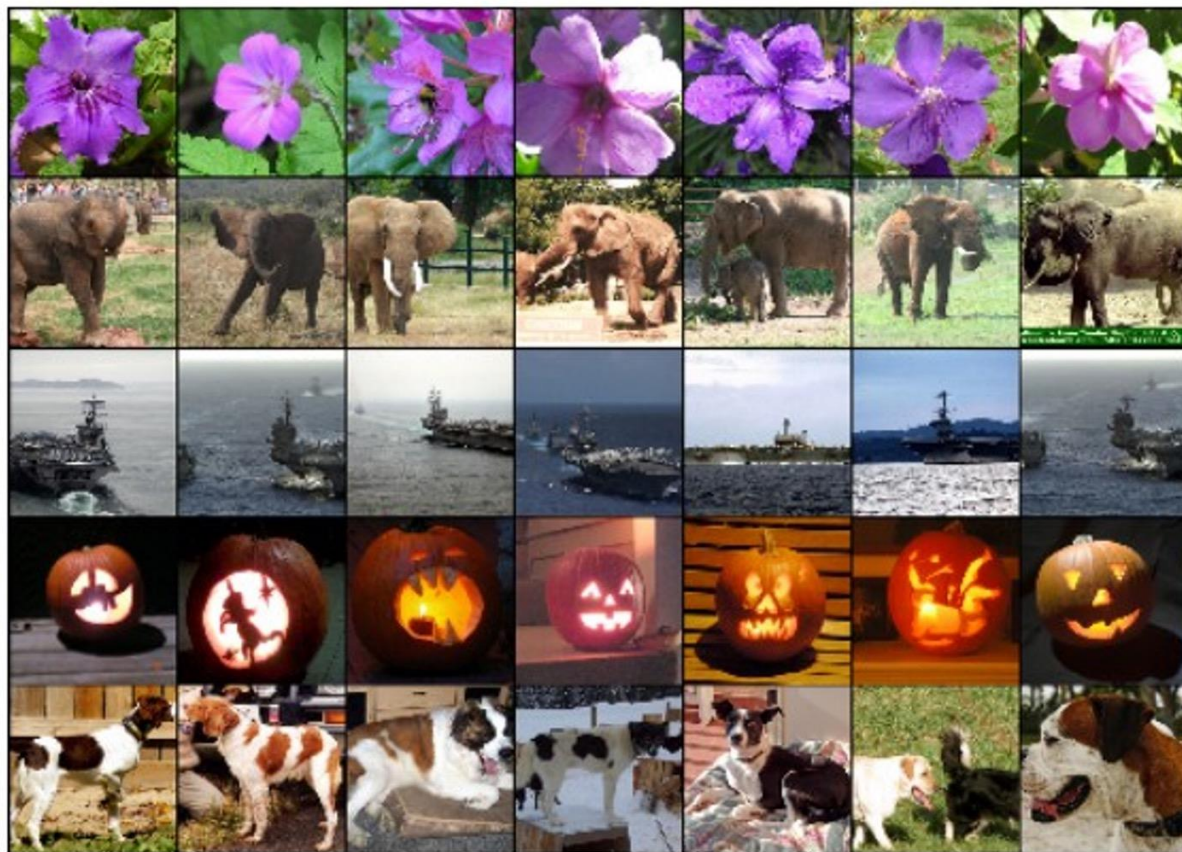
“AlexNet”

■ 卷积神经网络的广泛应用

Classification

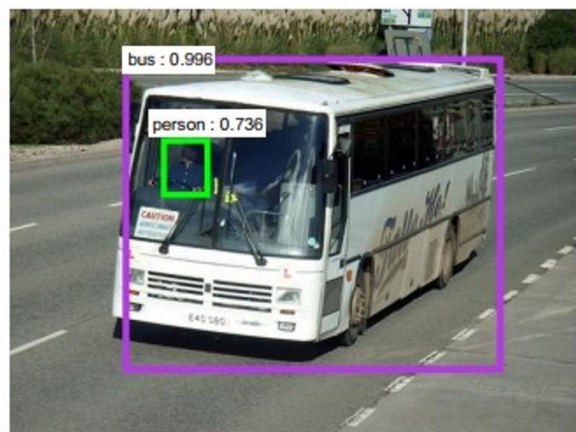
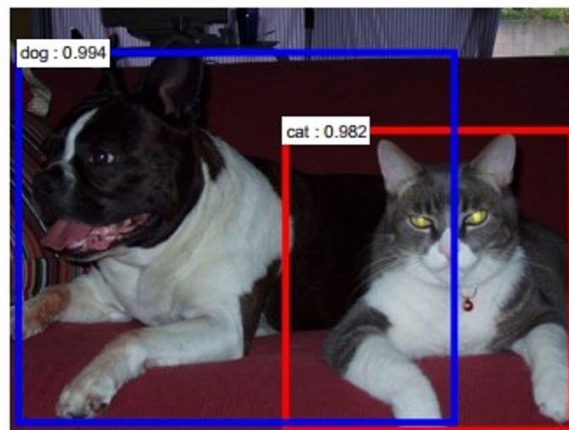
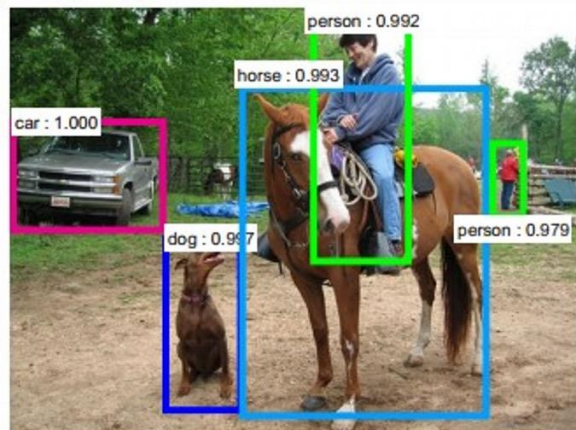


Retrieval

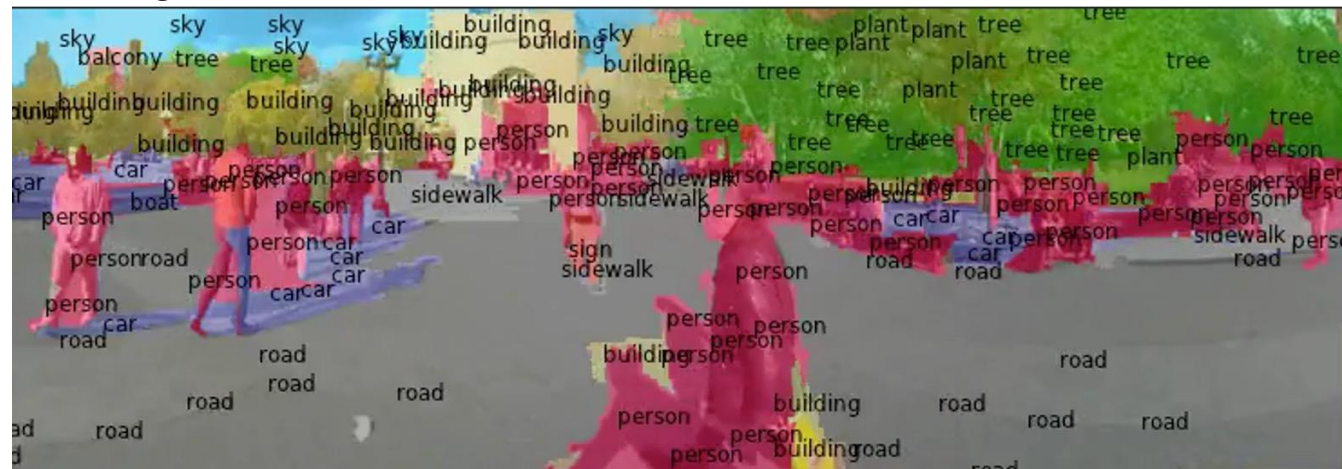


卷积神经网络的广泛应用

Detection



Segmentation



■ 卷积神经网络的广泛应用

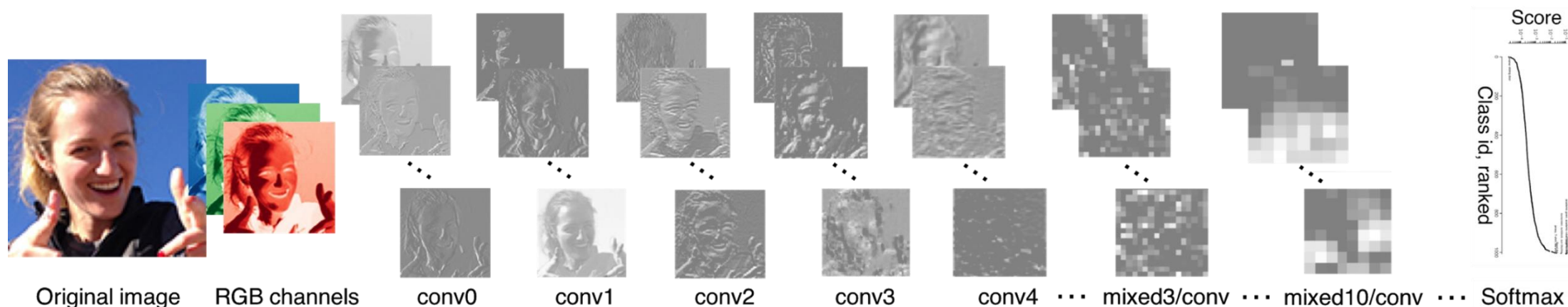


自动驾驶汽车

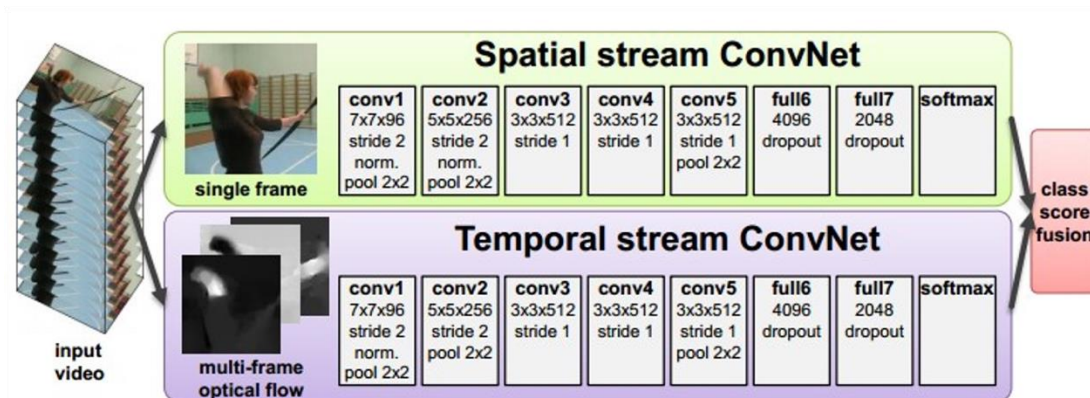


NVIDIA Tesla 系列显卡

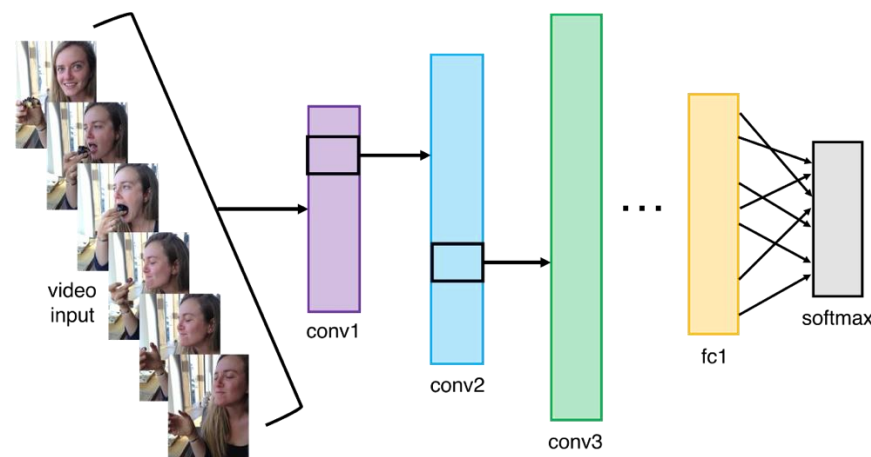
卷积神经网络的广泛应用



[Taigman et al. 2014]



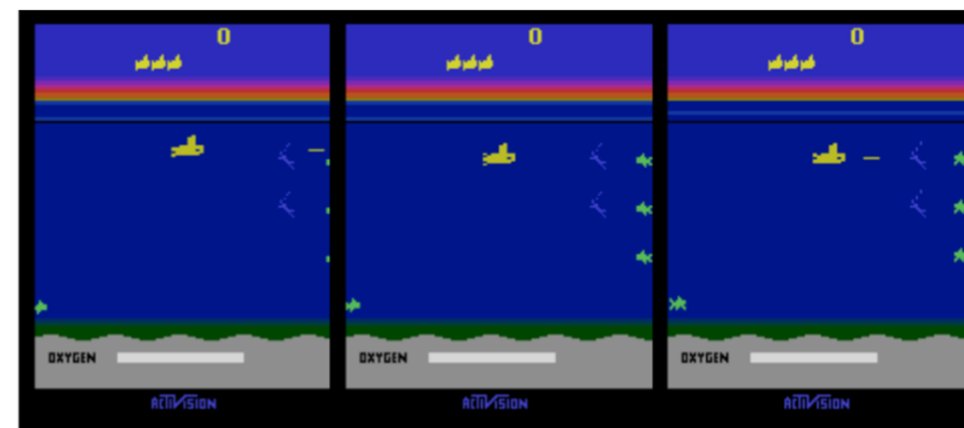
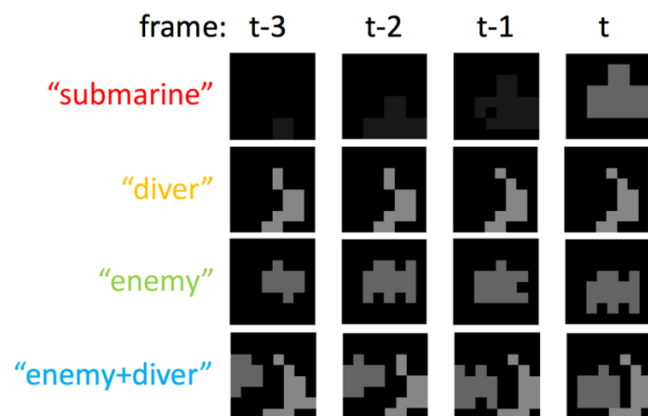
[Simonyan et al. 2014]



■ 卷积神经网络的广泛应用

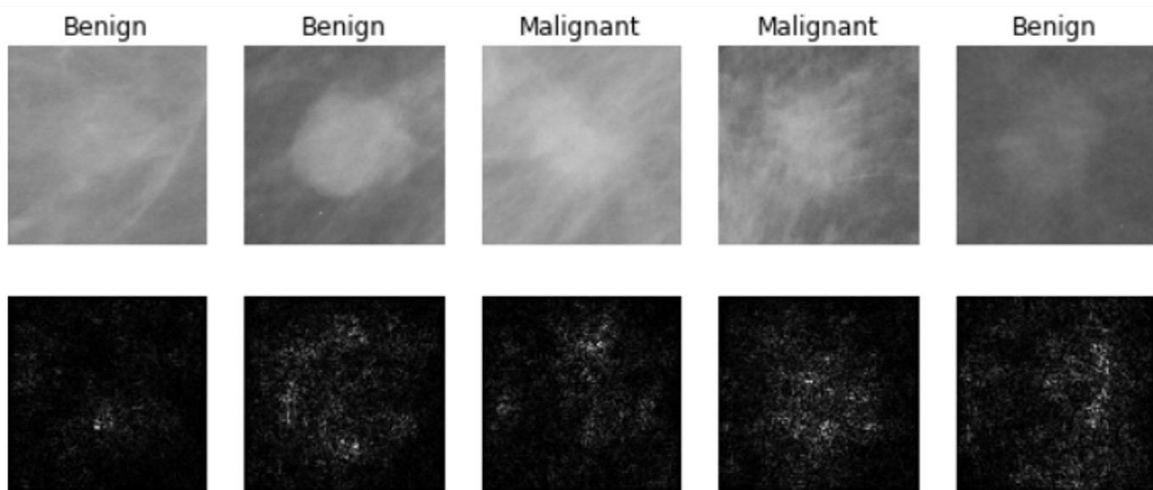


[Toshev, Szegedy 2014]



[Guo et al. 2014]

■ 卷积神经网络的广泛应用



[Levy et al. 2016]



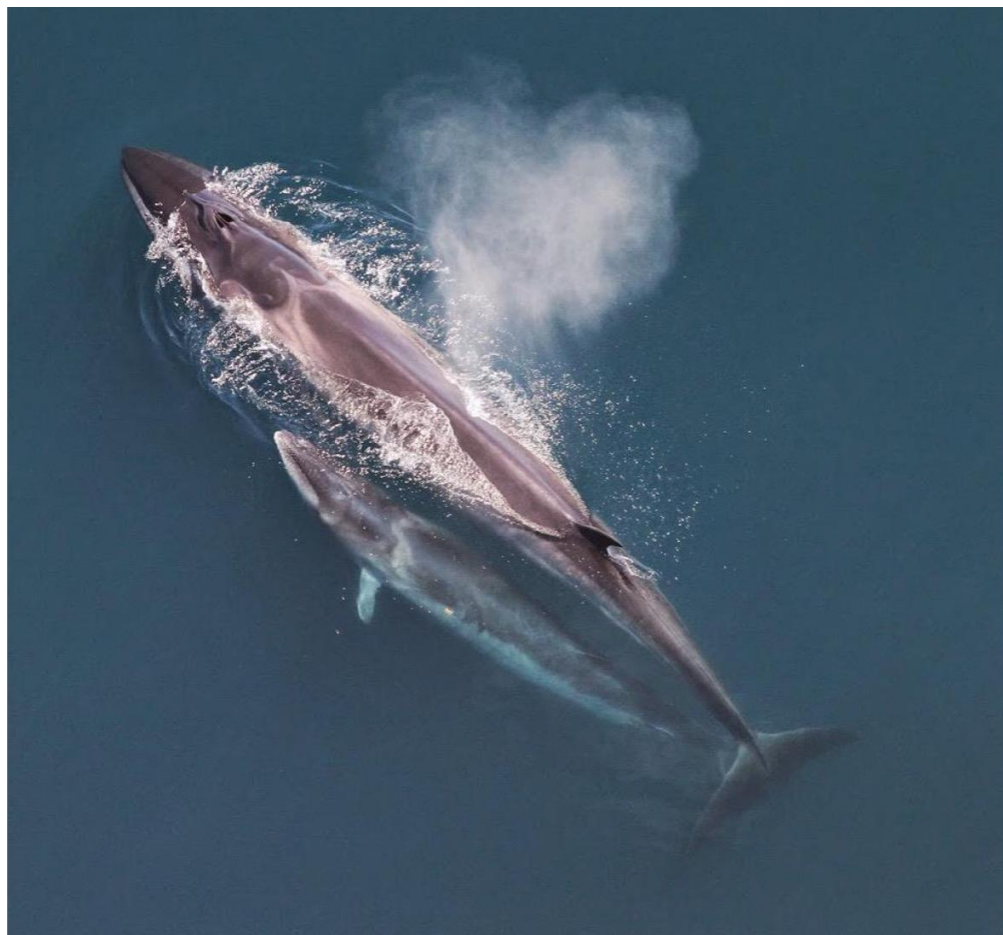
[Dieleman et al. 2014]



[Sermanet et al. 2011]

[Ciresan et al.]

■ 卷积神经网络的广泛应用



Kaggle Challenge



Mnih and Hinton, 2010

■ 卷积神经网络的广泛应用

No errors



A white teddy bear sitting in the grass



A man riding a wave on top of a surfboard

Minor errors



A man in a baseball uniform throwing a ball



A cat sitting on a suitcase on the floor

Somewhat related



A woman is holding a cat in her hand

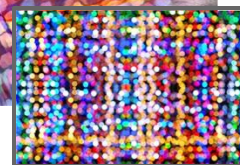
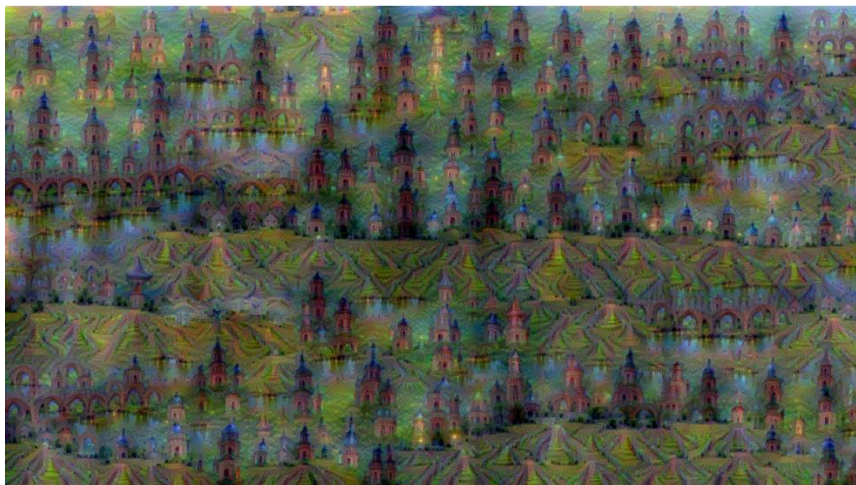
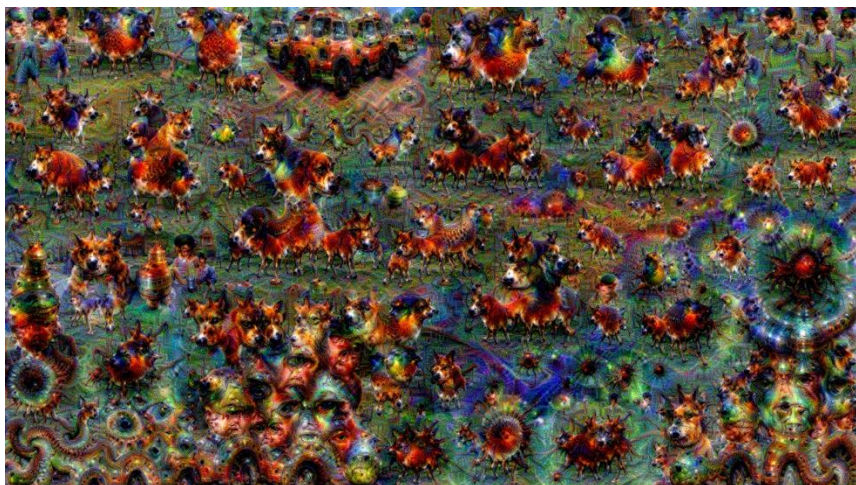


A woman standing on a beach holding a surfboard

图像描述

[Vinyals et al., 2015]
[Karpathy and Fei-Fei, 2015]

■ 卷积神经网络的广泛应用

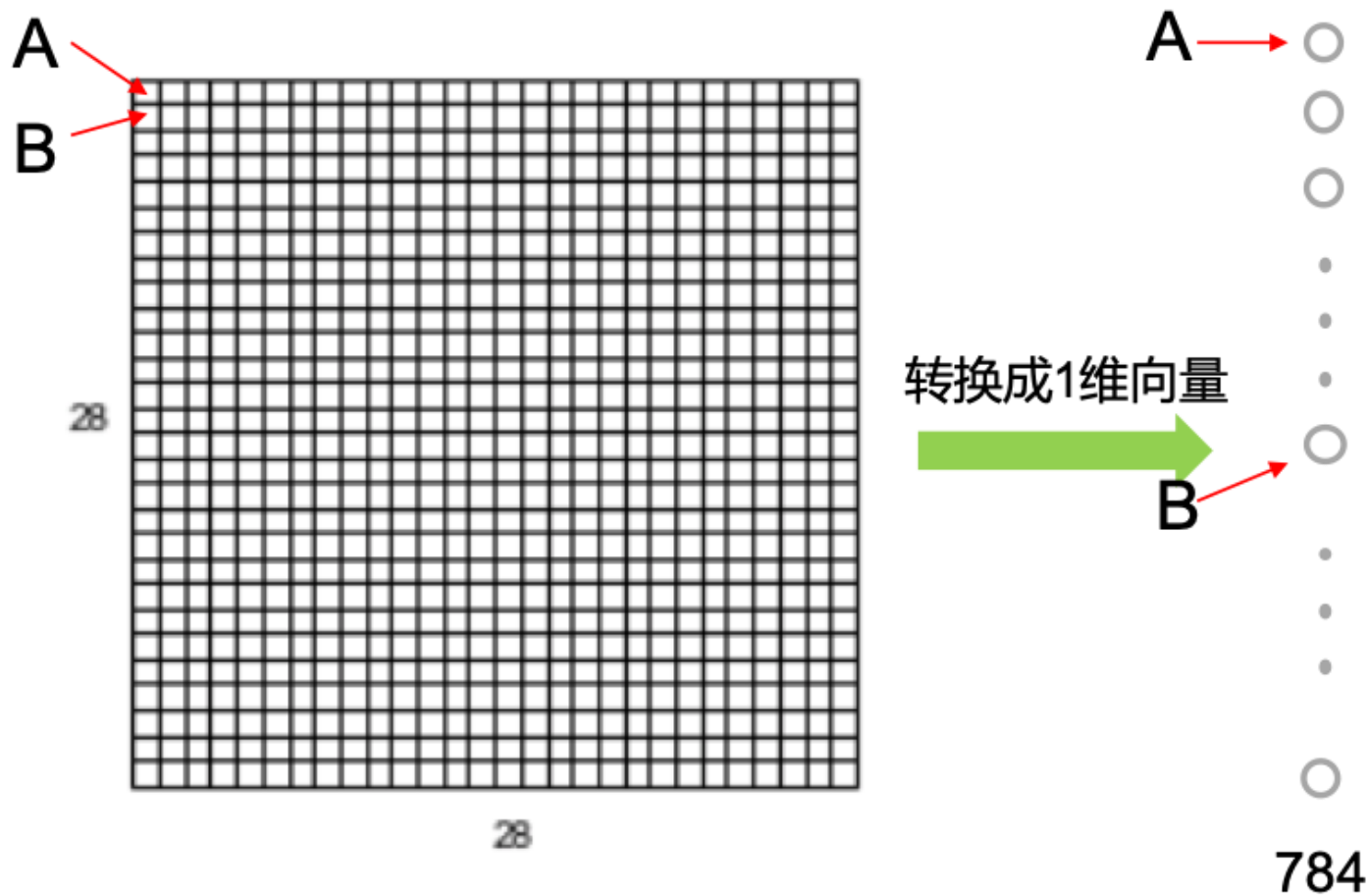


大 纲

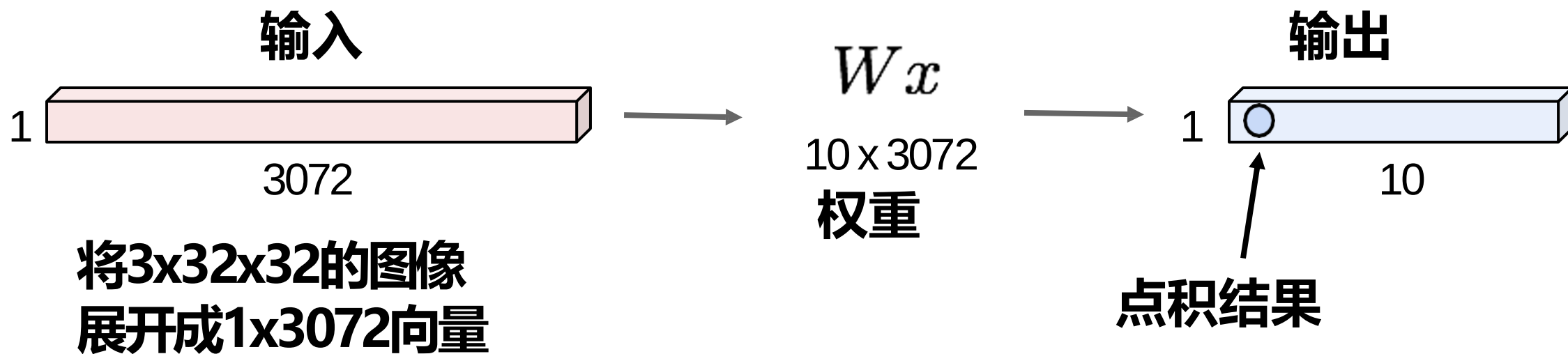
- 卷积神经网络-引言

- 卷积神经网络

■ 全连接层（参数量，优化，空间信息）

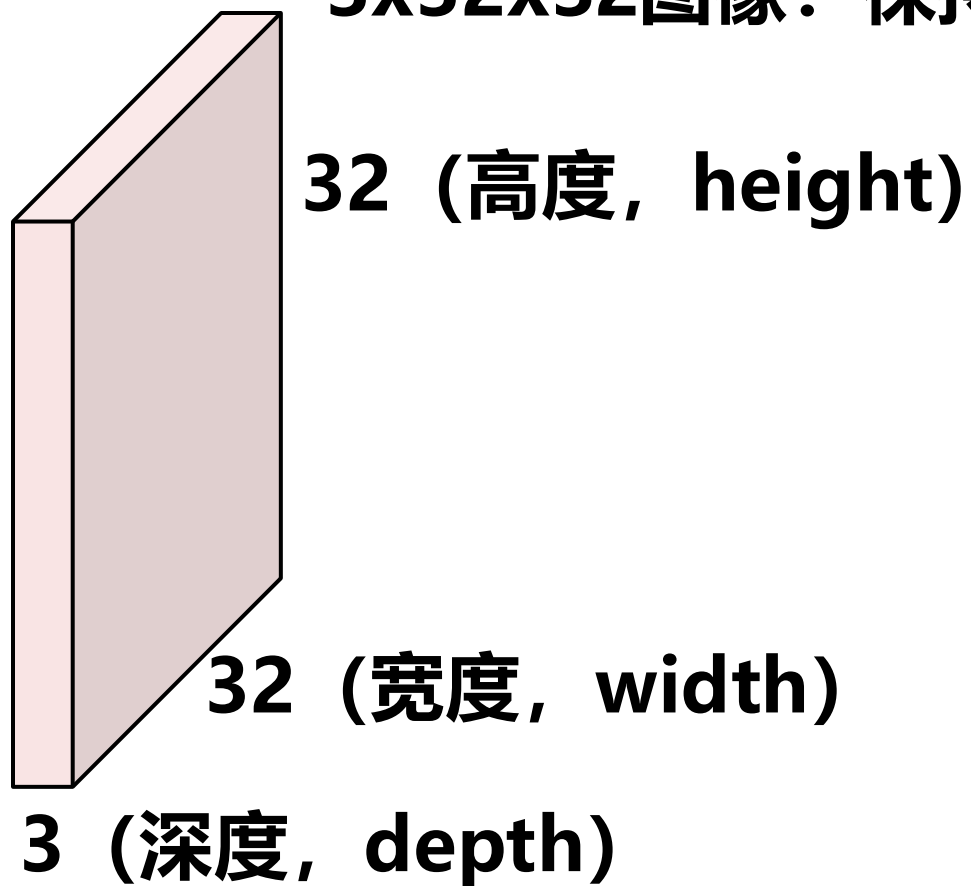


■ 全连接层（参数量，优化，空间信息）

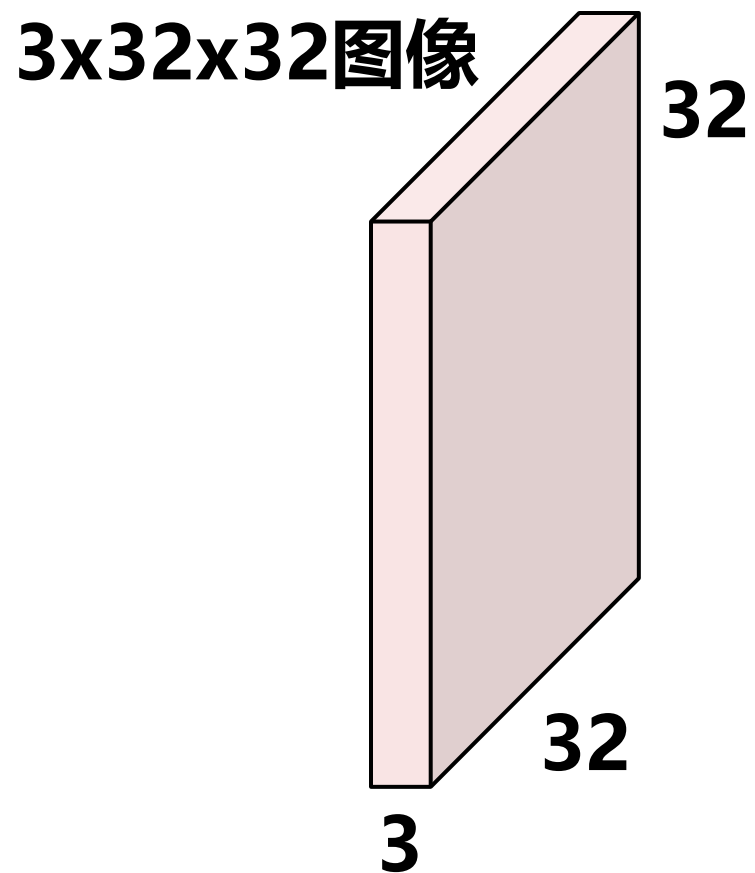


■ 卷积层

3x32x32图像：保持图像的空间结构



■ 卷积层

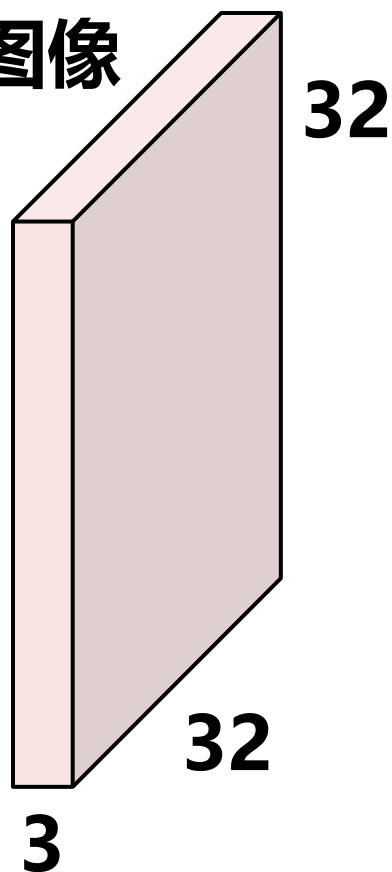


3x5x5滤波器



■ 卷积层

3x32x32图像



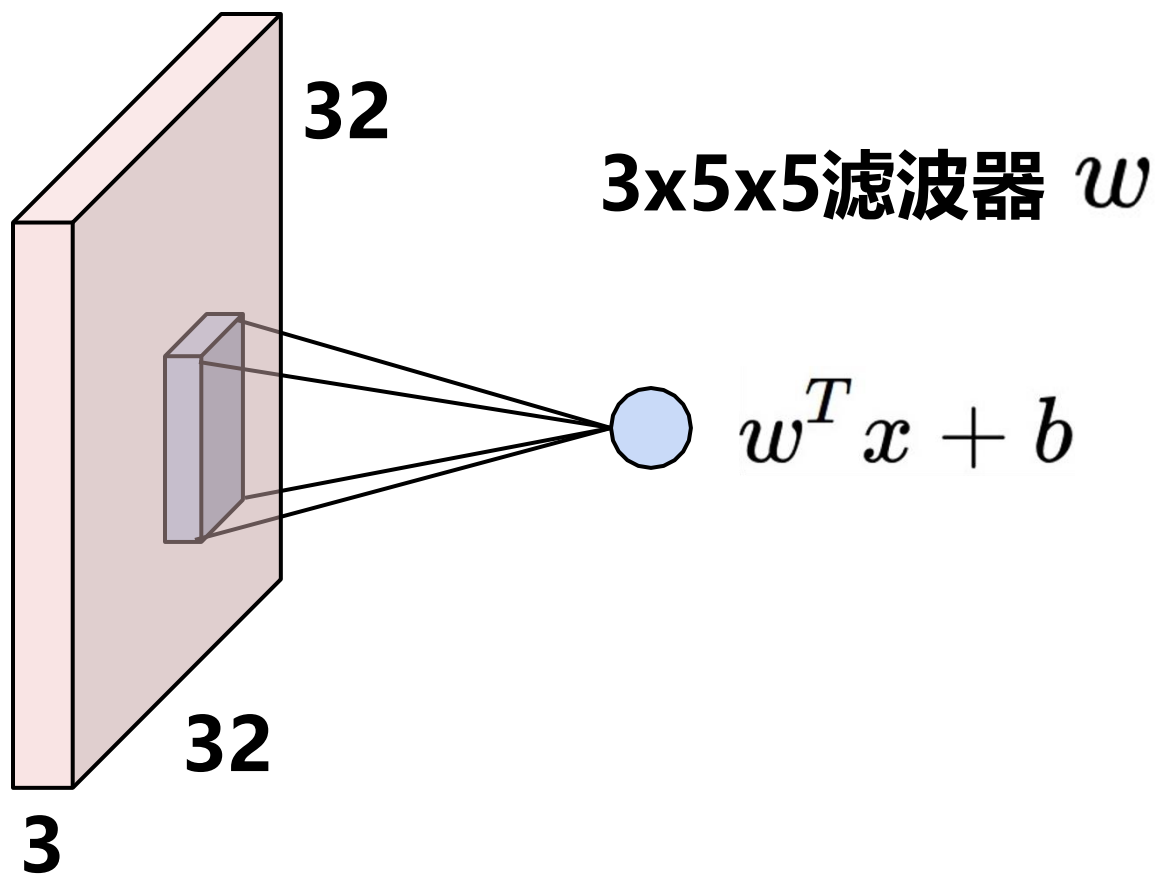
滤波器的深度始终与输入的深度相同

3x5x5滤波器

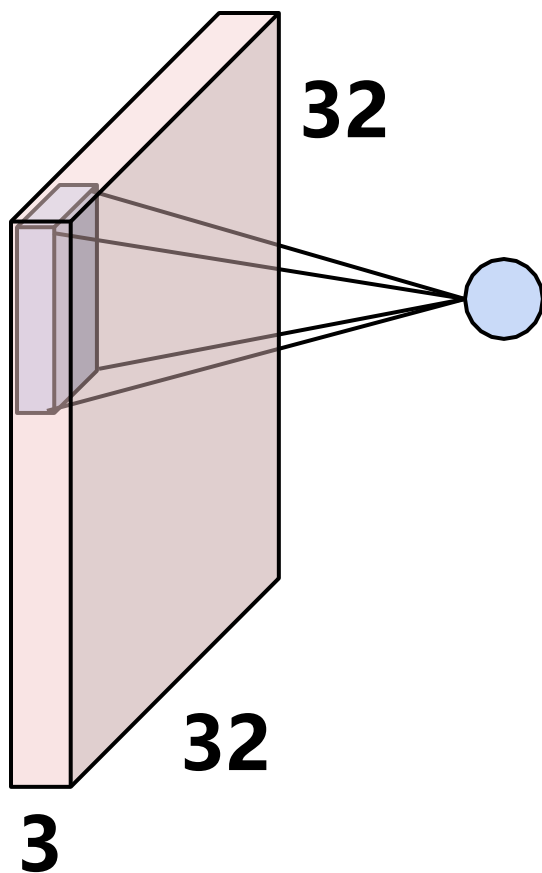


将滤波器与图像卷积，
即“在图像上空间滑动，计算点积”

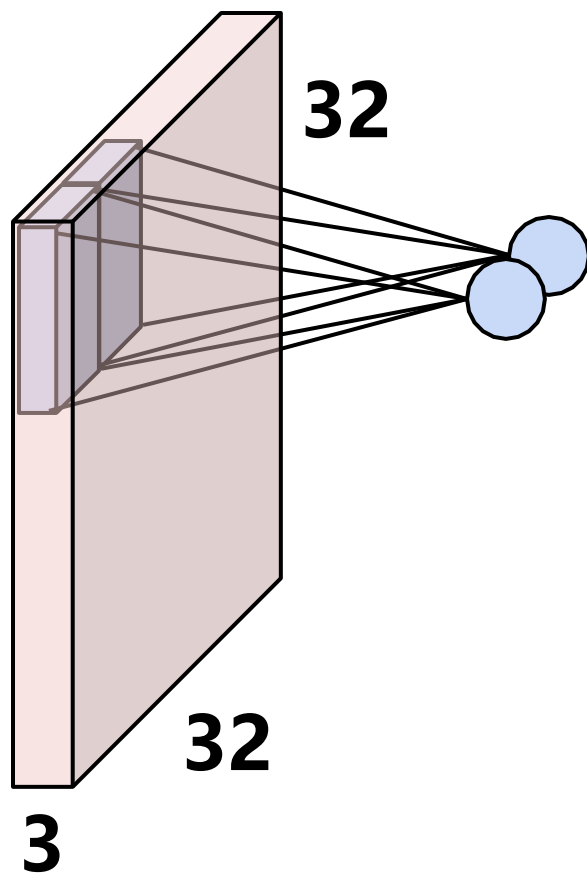
■ 卷积层



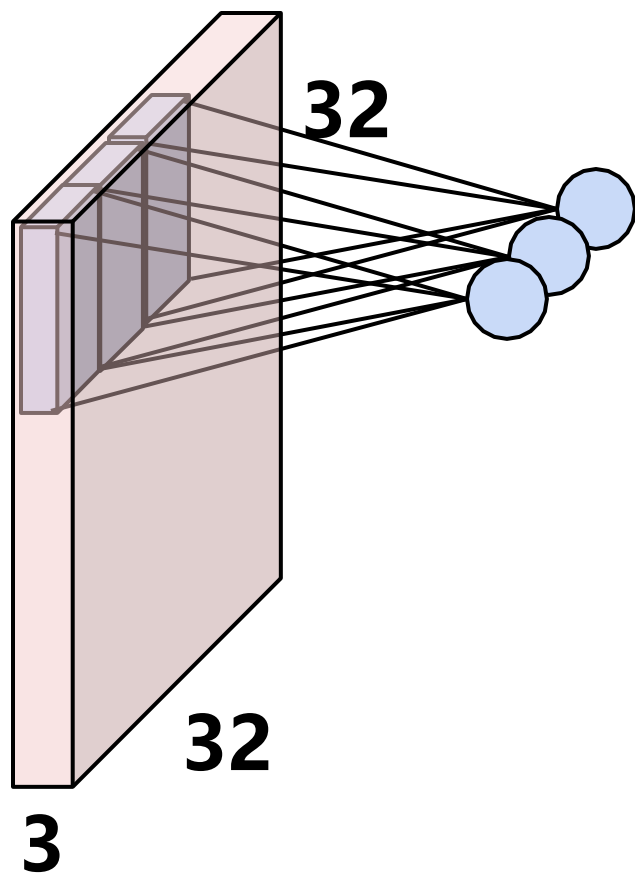
■ 卷积层



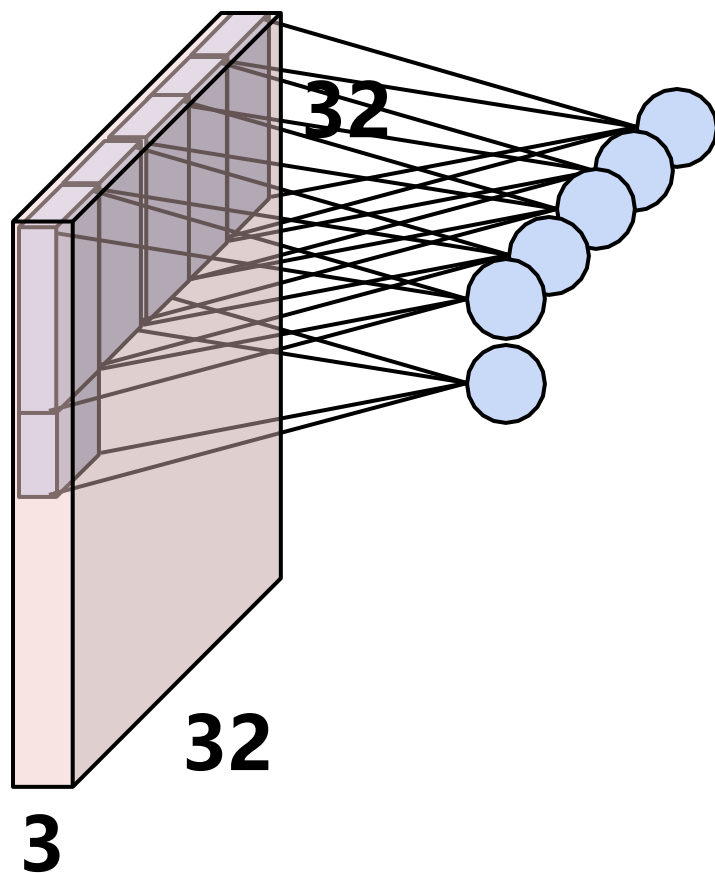
■ 卷积层



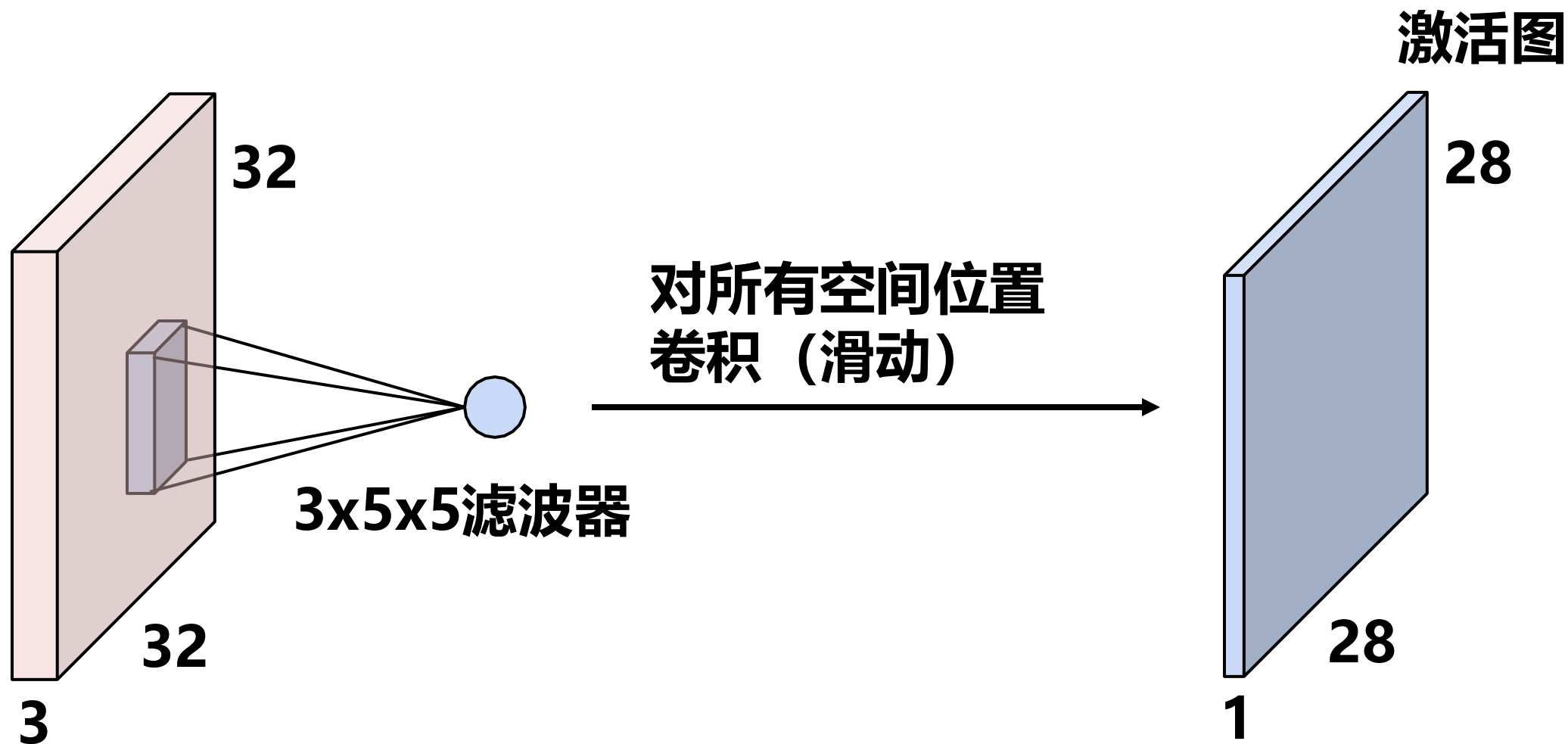
■ 卷积层



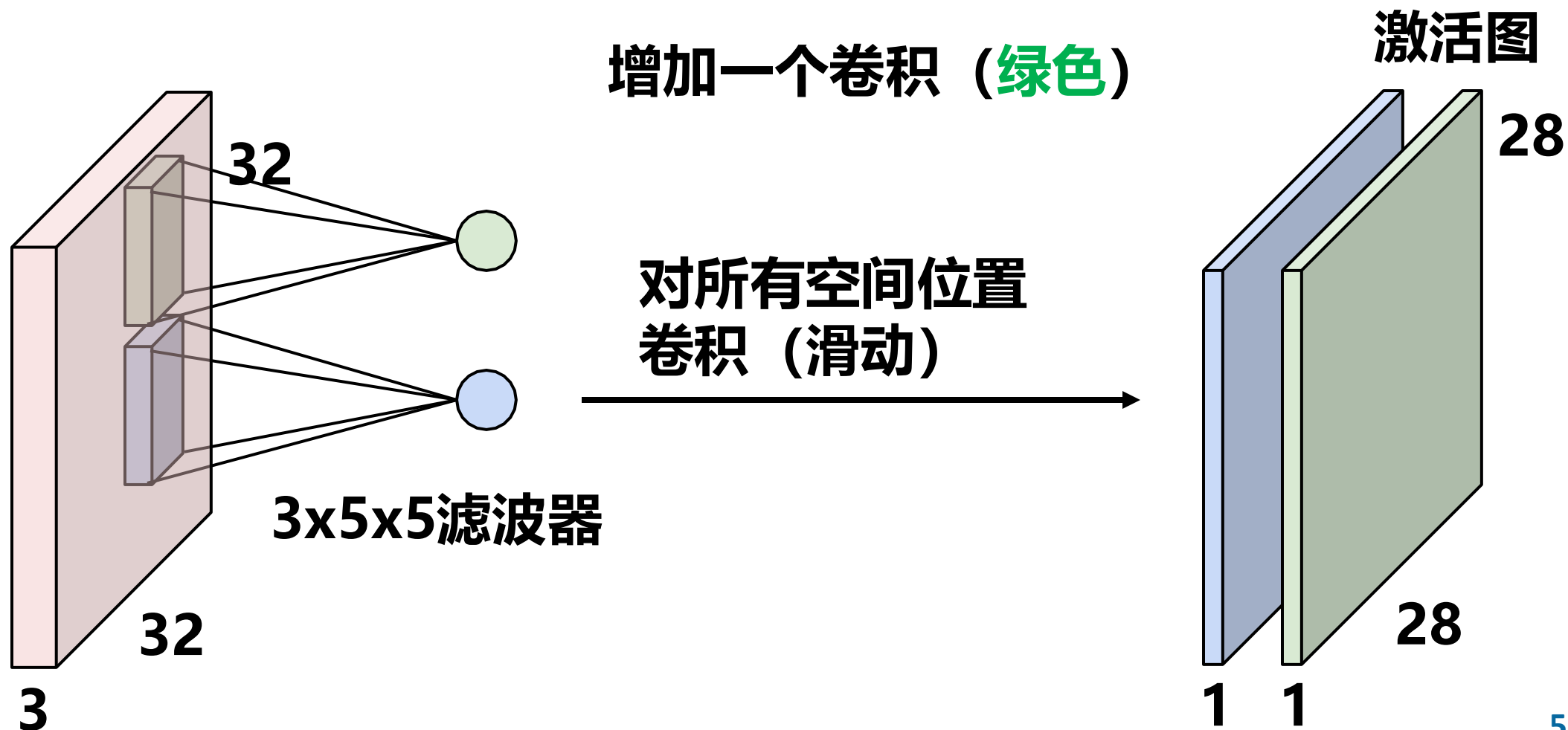
■ 卷积层



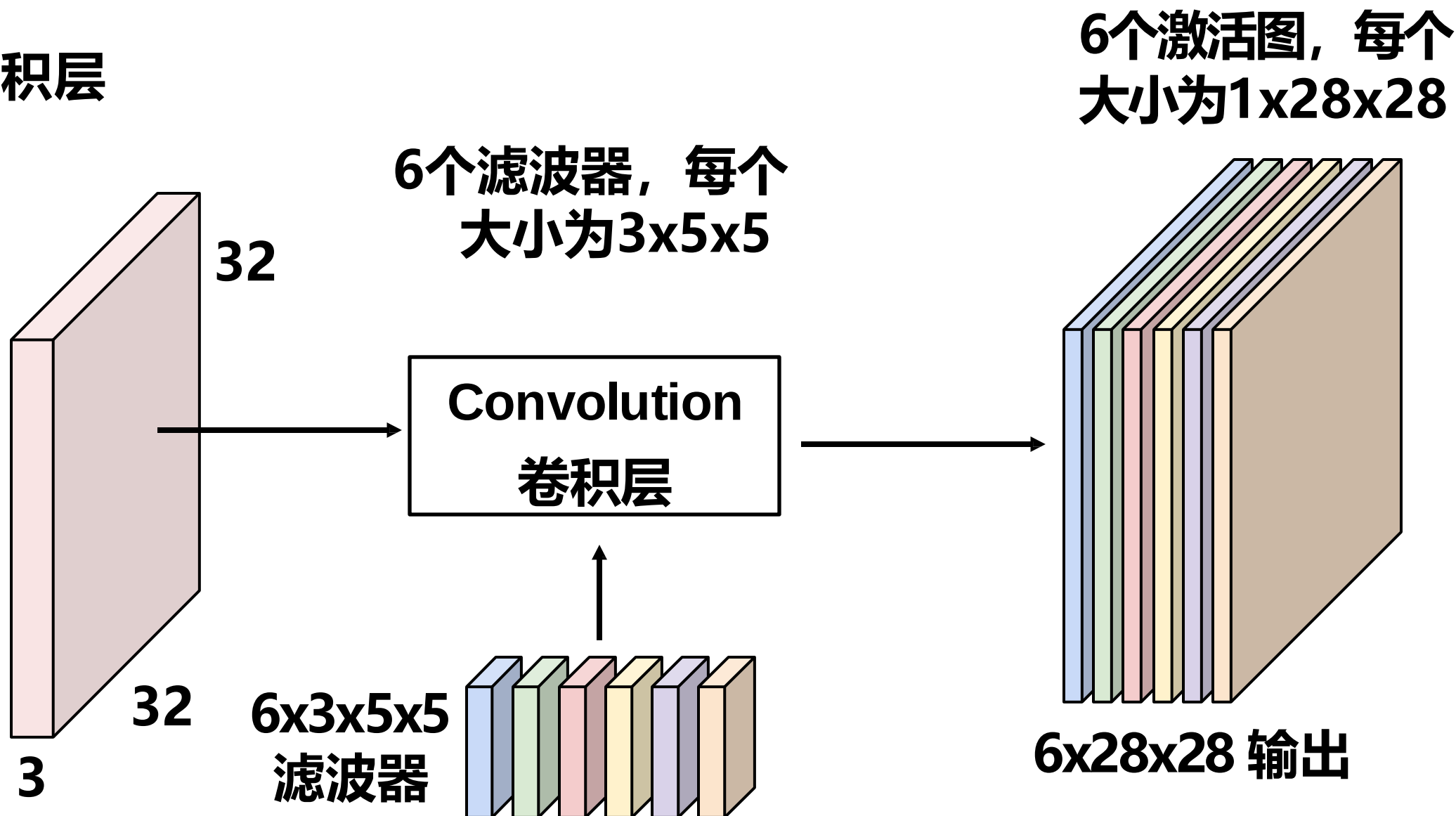
■ 卷积层



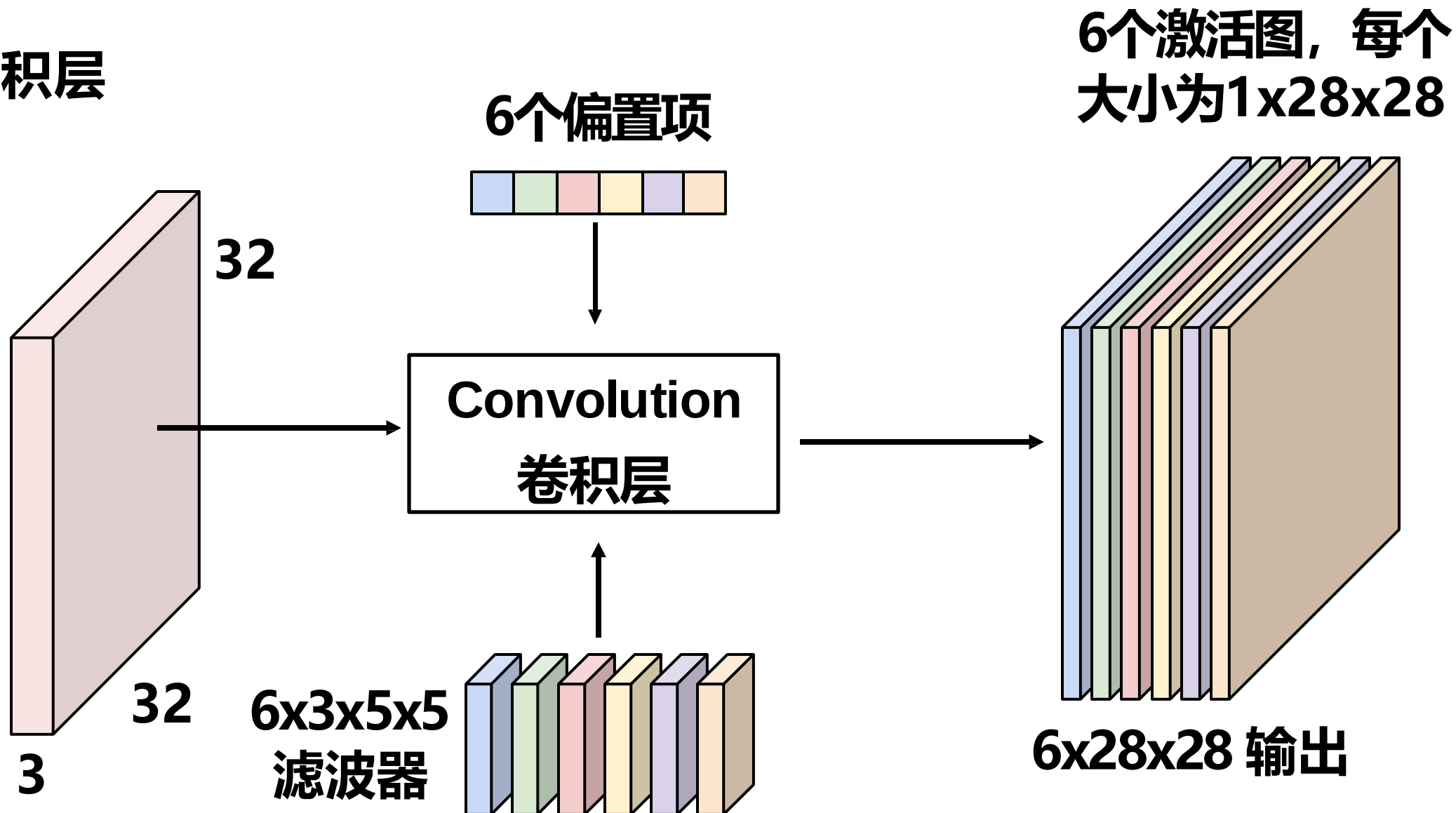
■ 卷积层



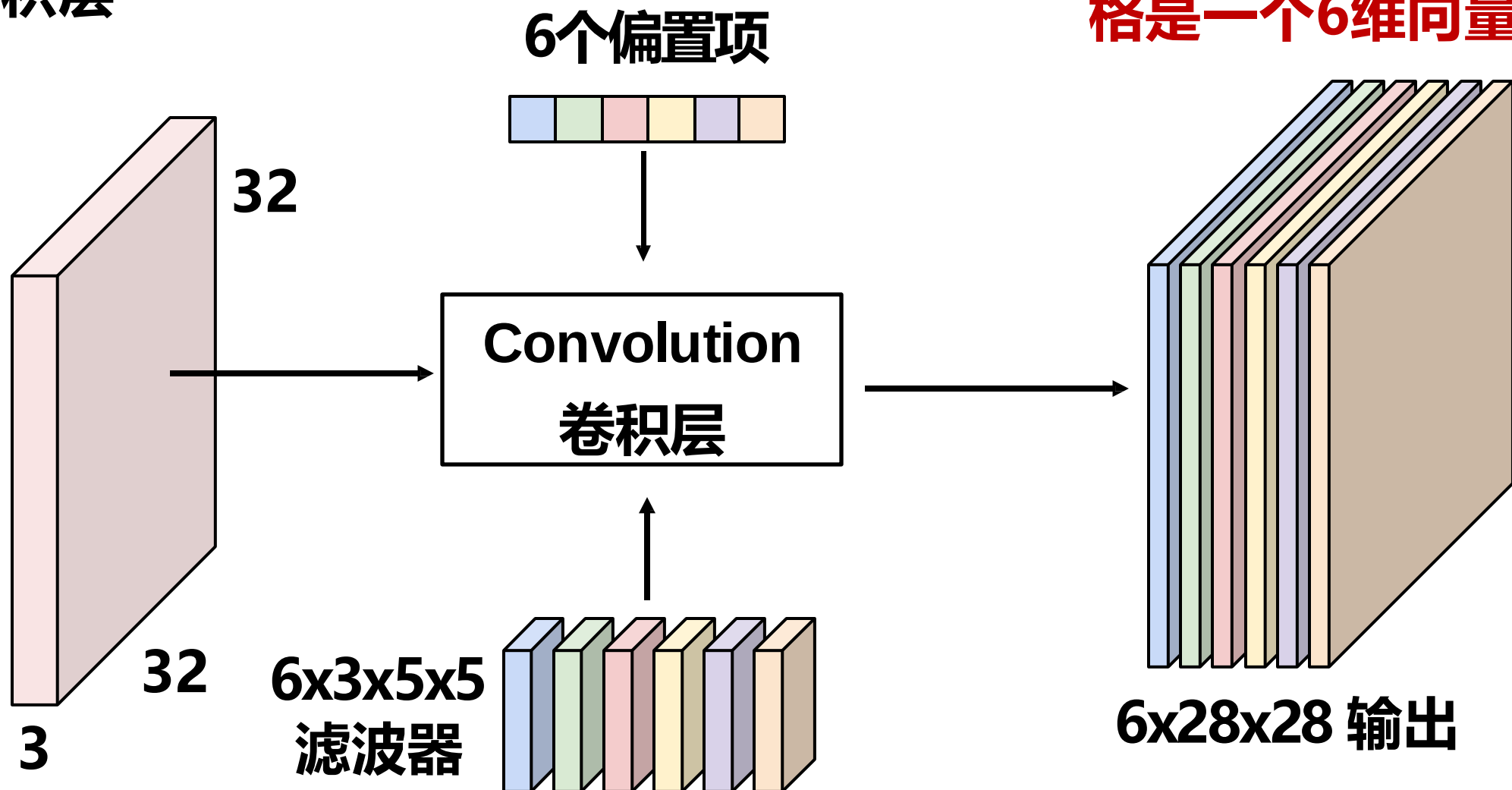
■ 卷积层



■ 卷积层

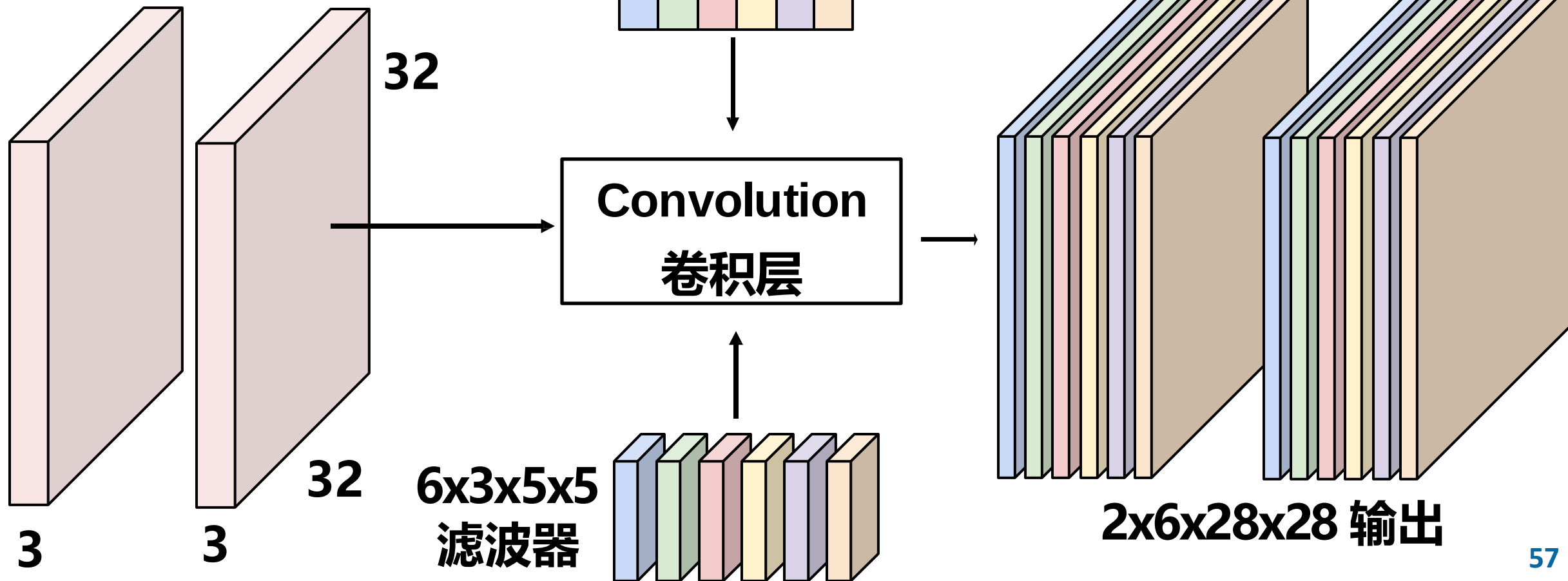


■ 卷积层

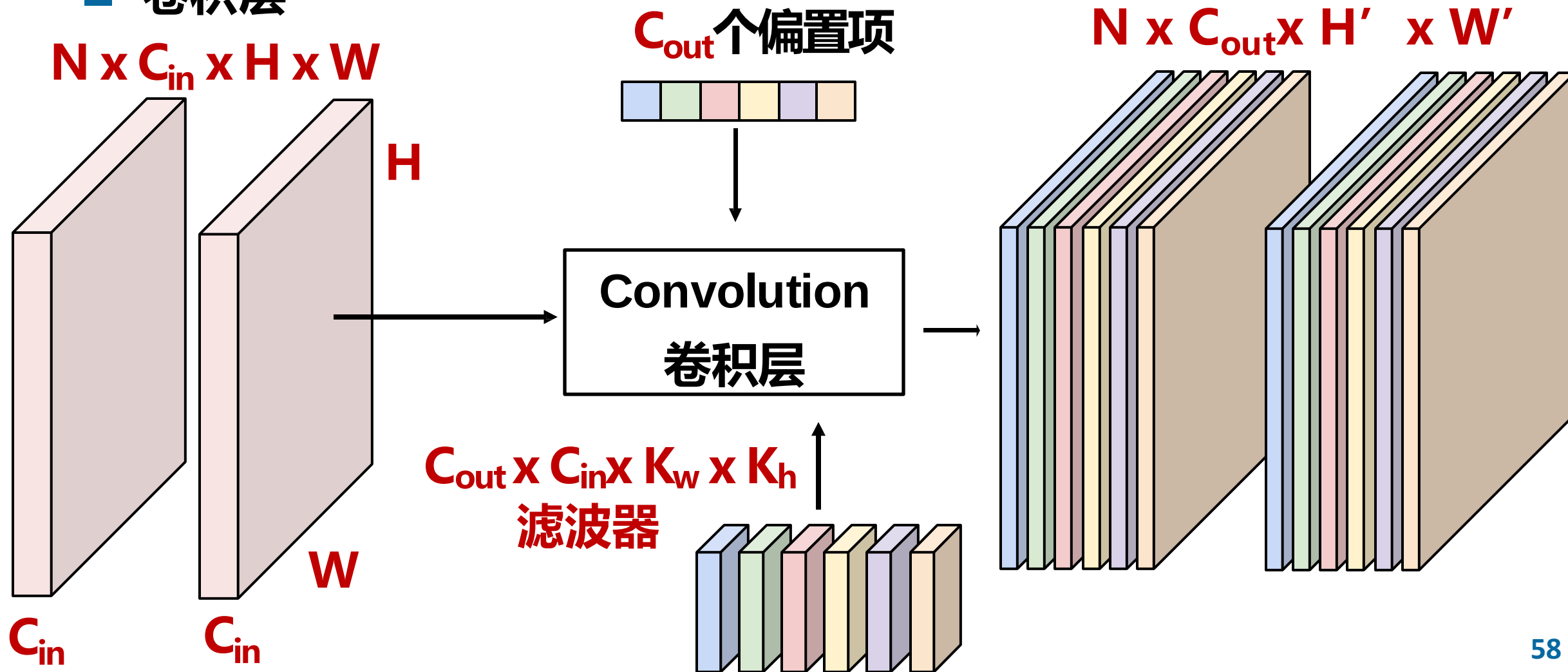


■ 卷积层

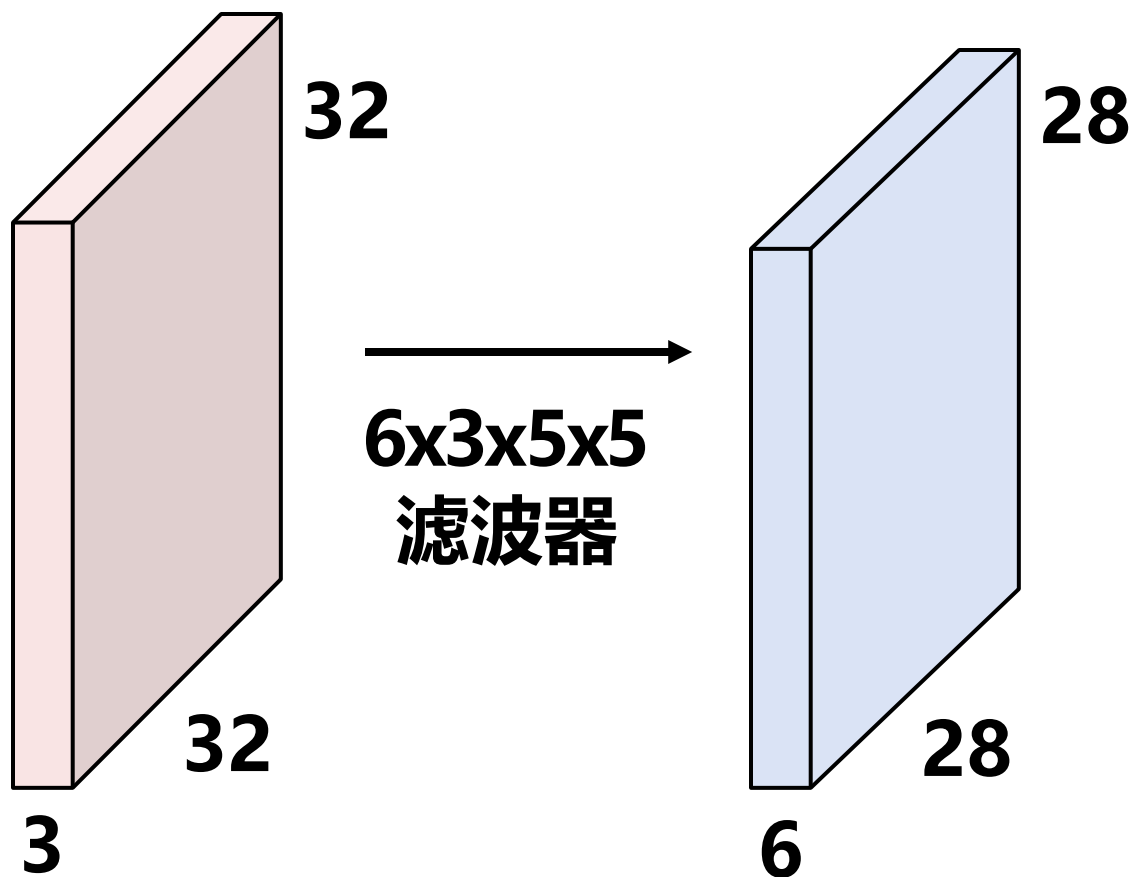
一个batch里2张图



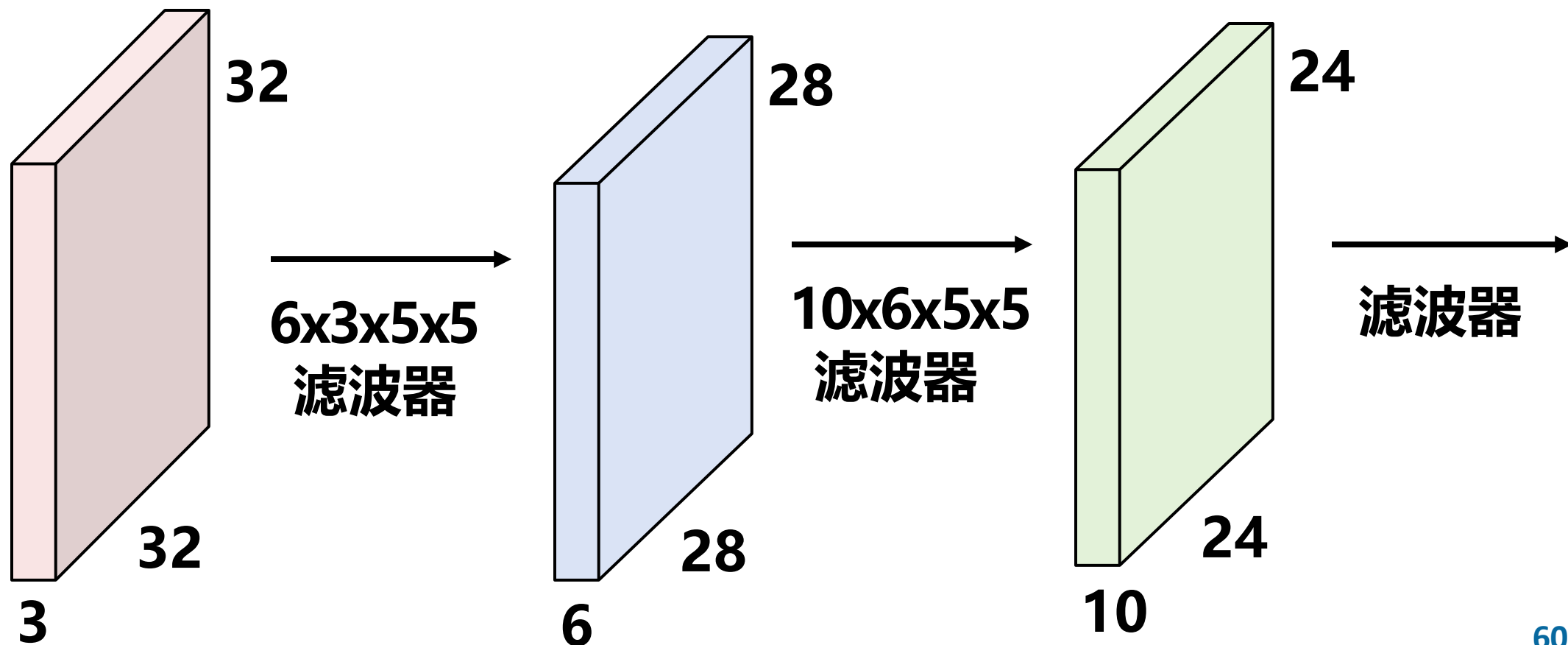
■ 卷积层



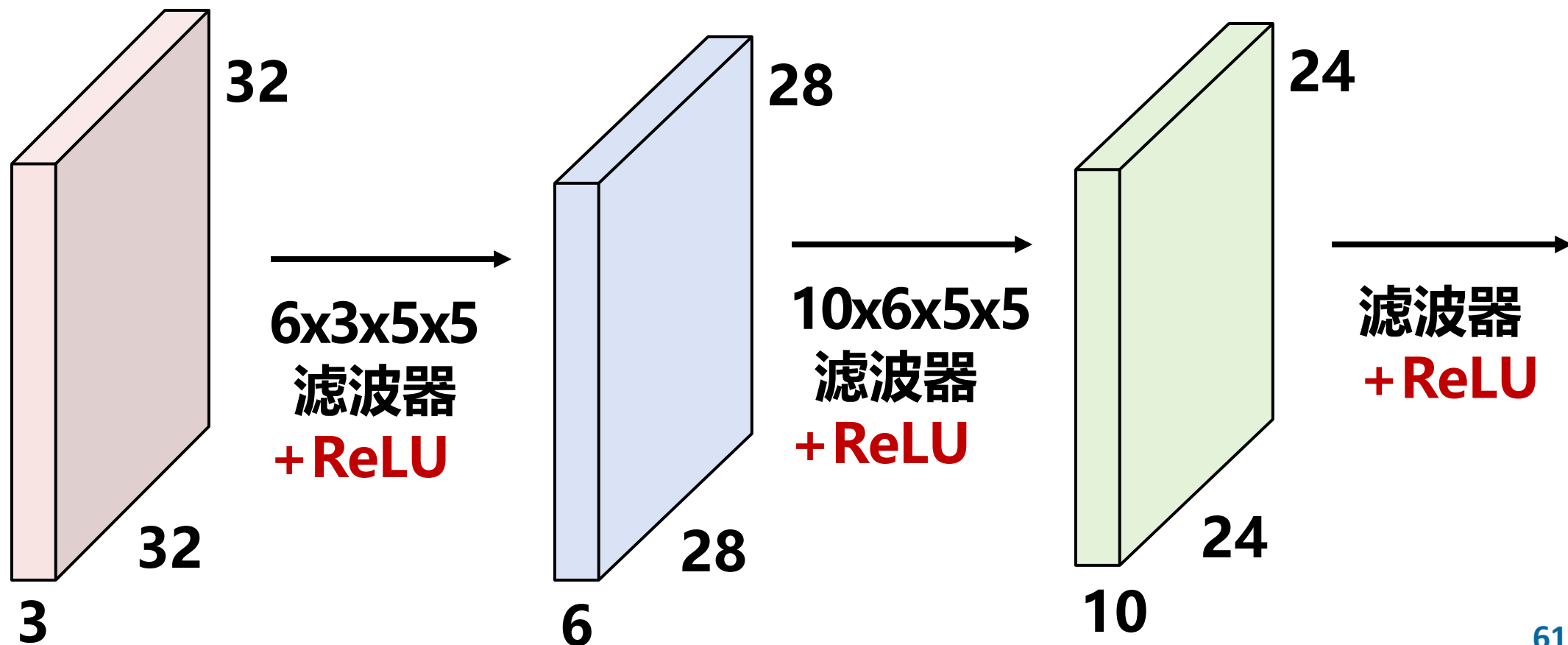
■ 卷积神经网络由一系列卷积层组成



■ 卷积神经网络由一系列卷积层组成



■ 卷积神经网络由一系列卷积层组成



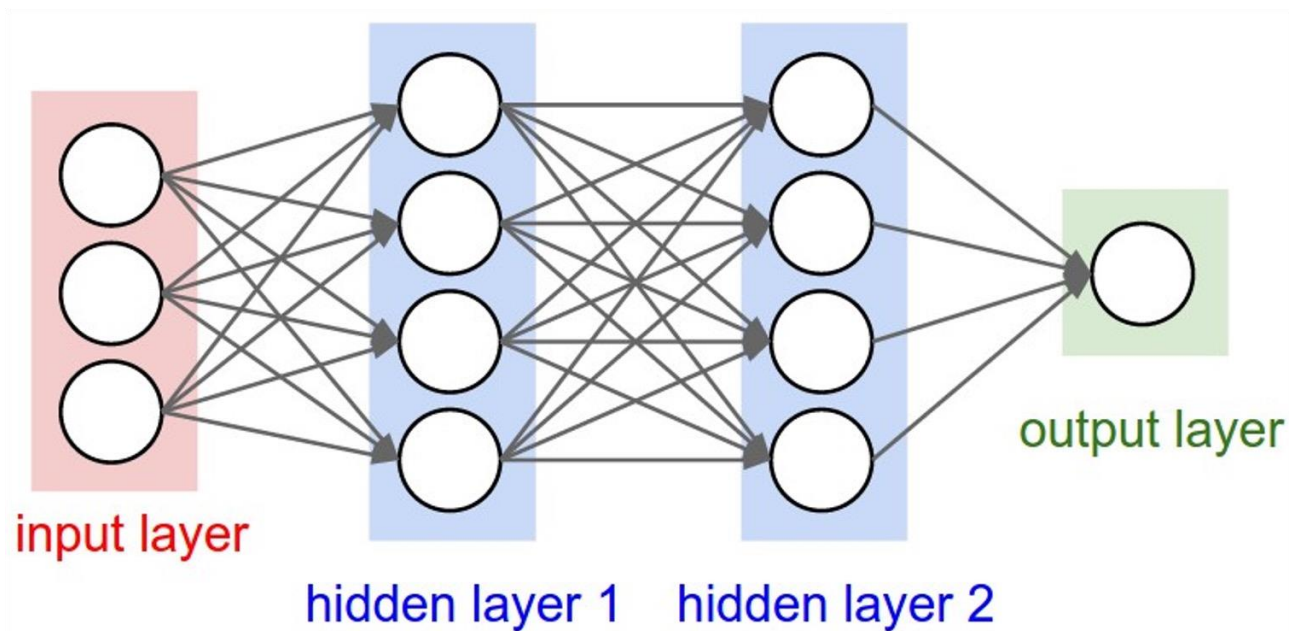
■ 线性分类器学到了什么？

$$f(x, W) = Wx + b$$

每个类别的模板



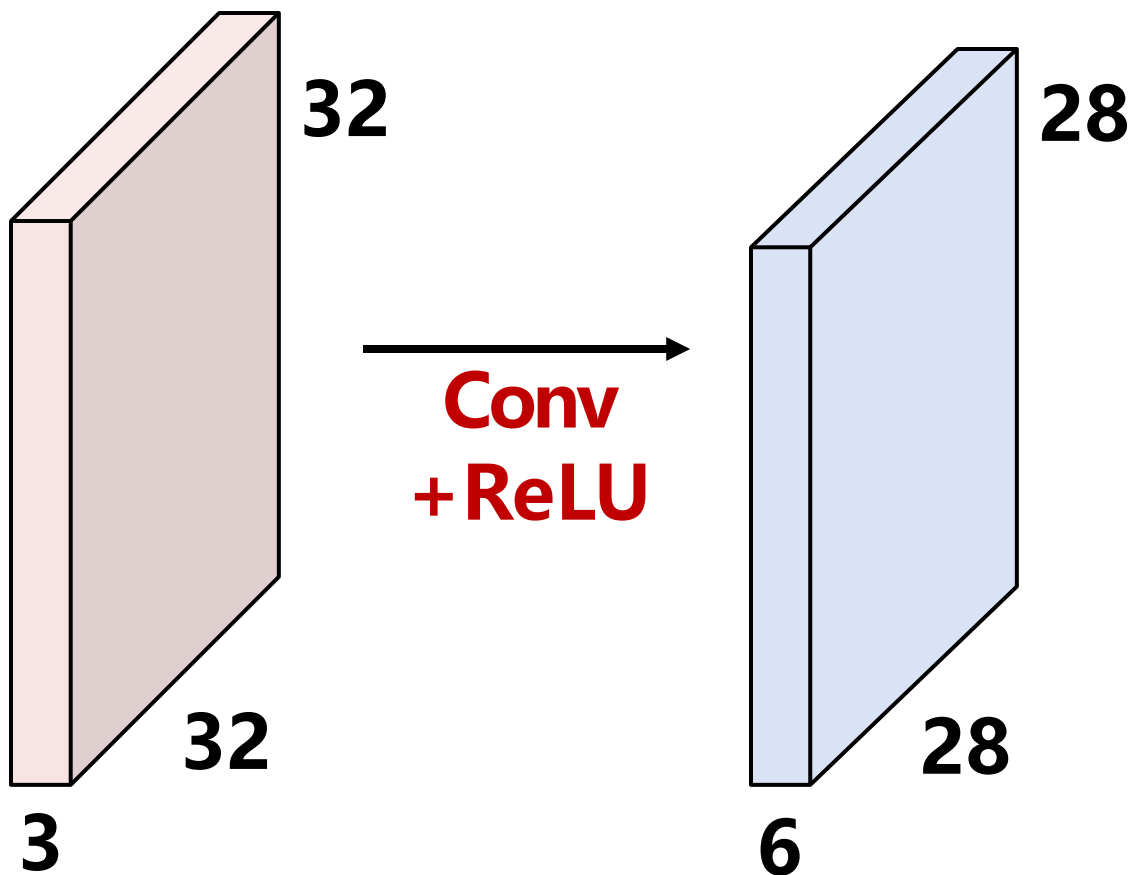
■ 全连接神经网络学到了什么？



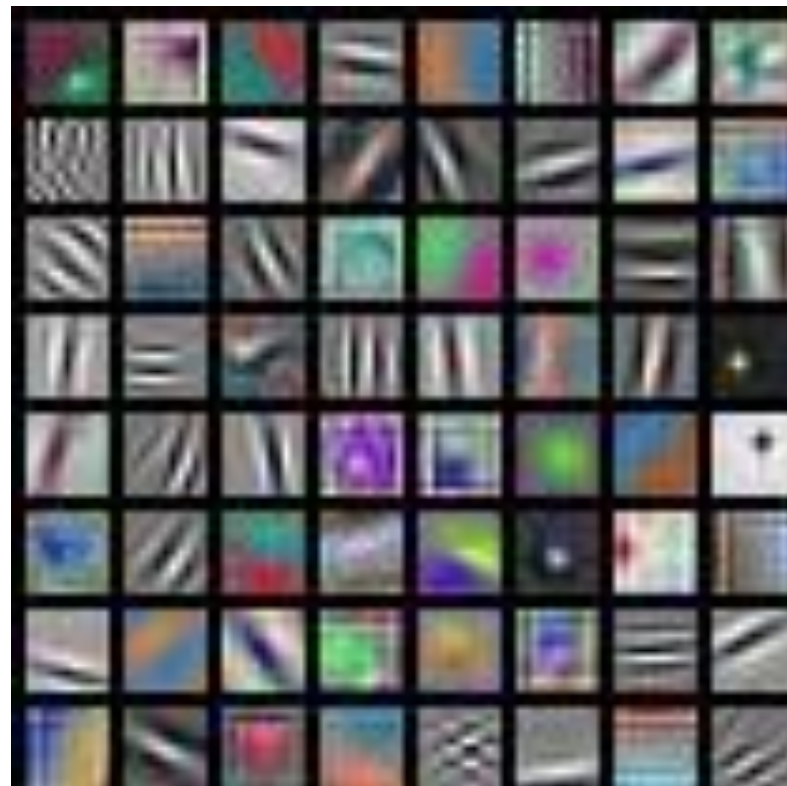
整张图像的的模板库



■ 滤波器学到了什么？

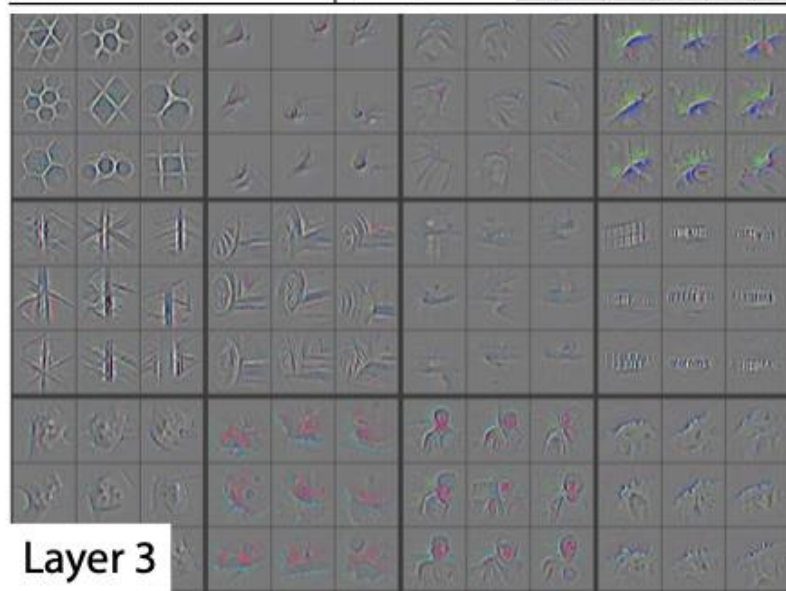
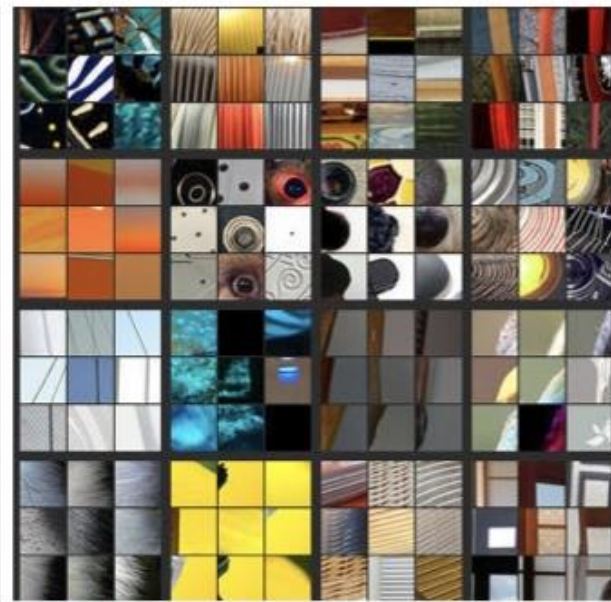
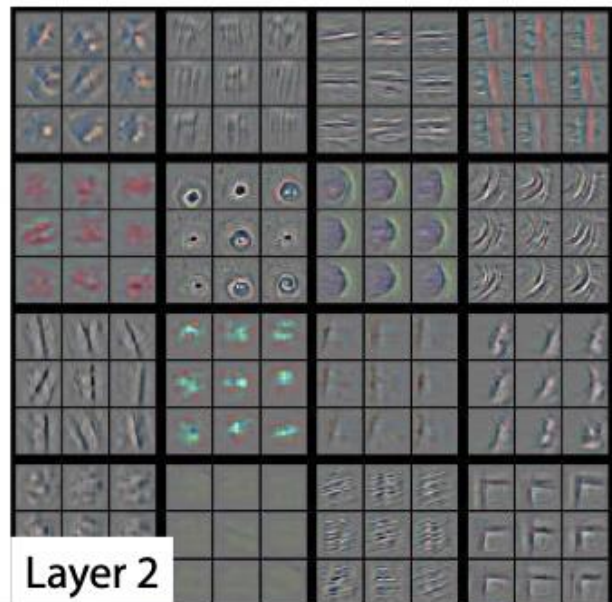


第一层卷积滤波器：局部图像模板
(通常学习边缘、颜色等纹理)



卷积神经网络

■ 滤波器

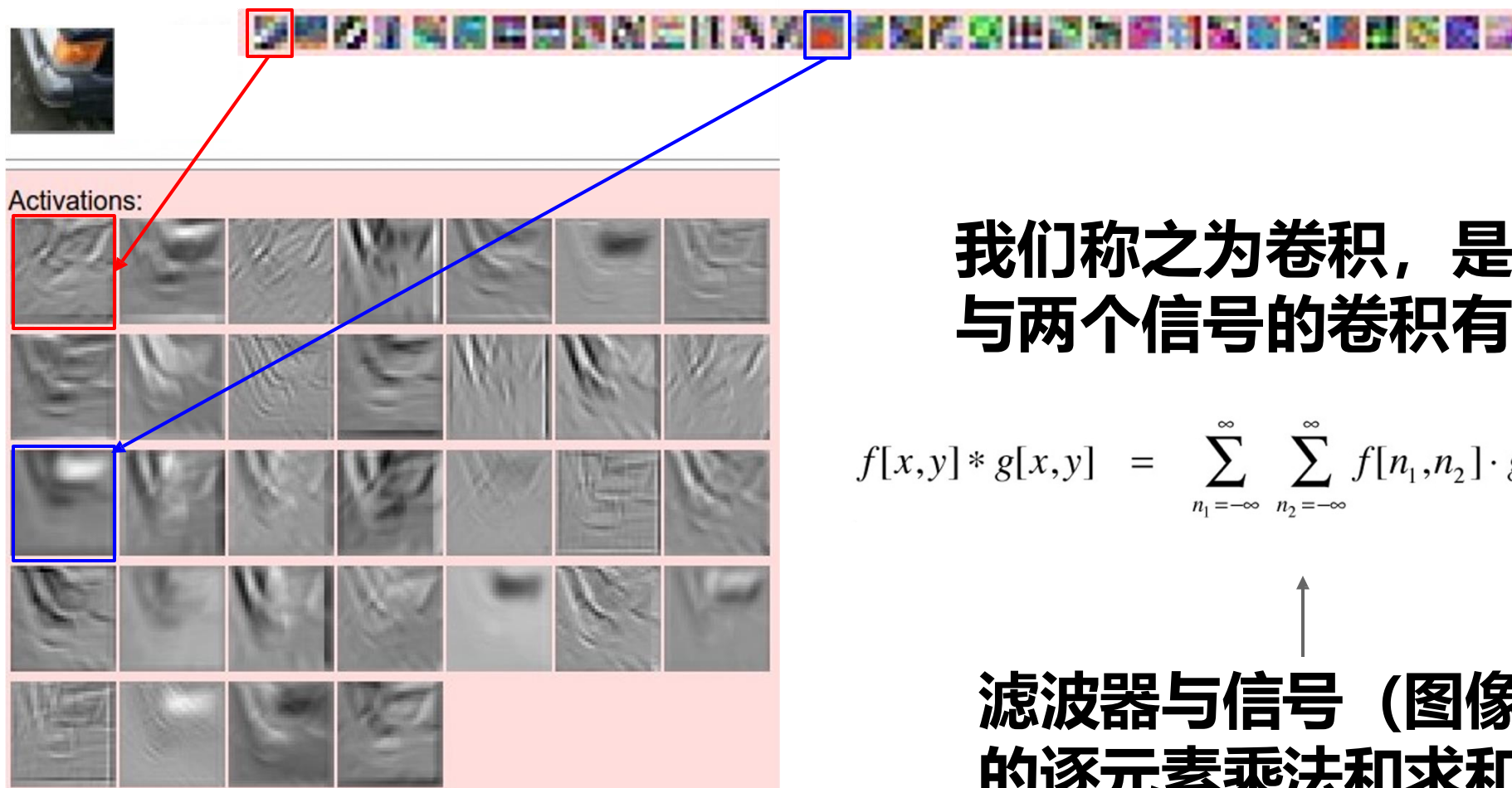


卷积神经网络



■ 卷积神经网络

一个滤波器-->一个激活图

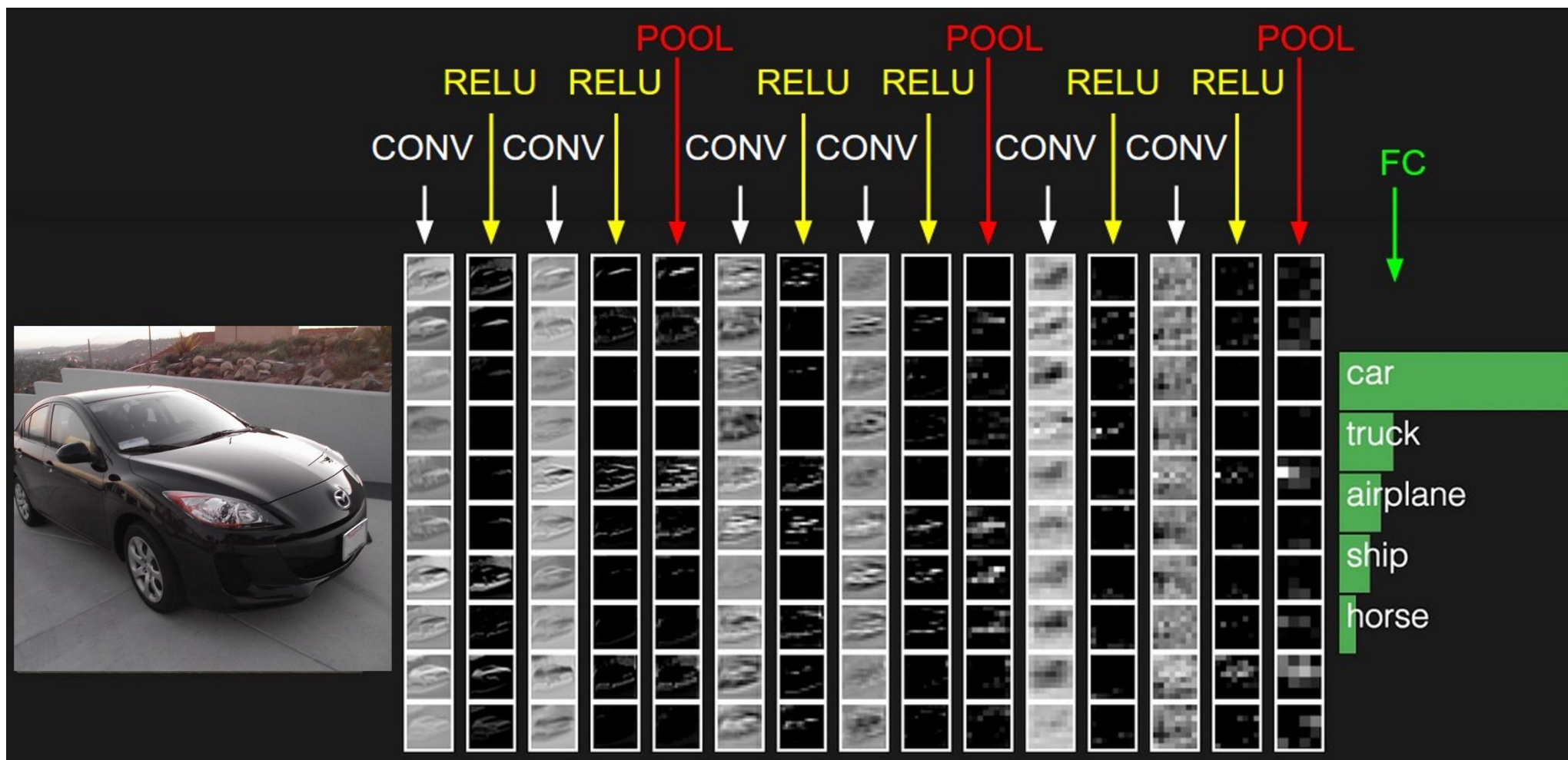


我们称之为卷积，是因为它与两个信号的卷积有关：

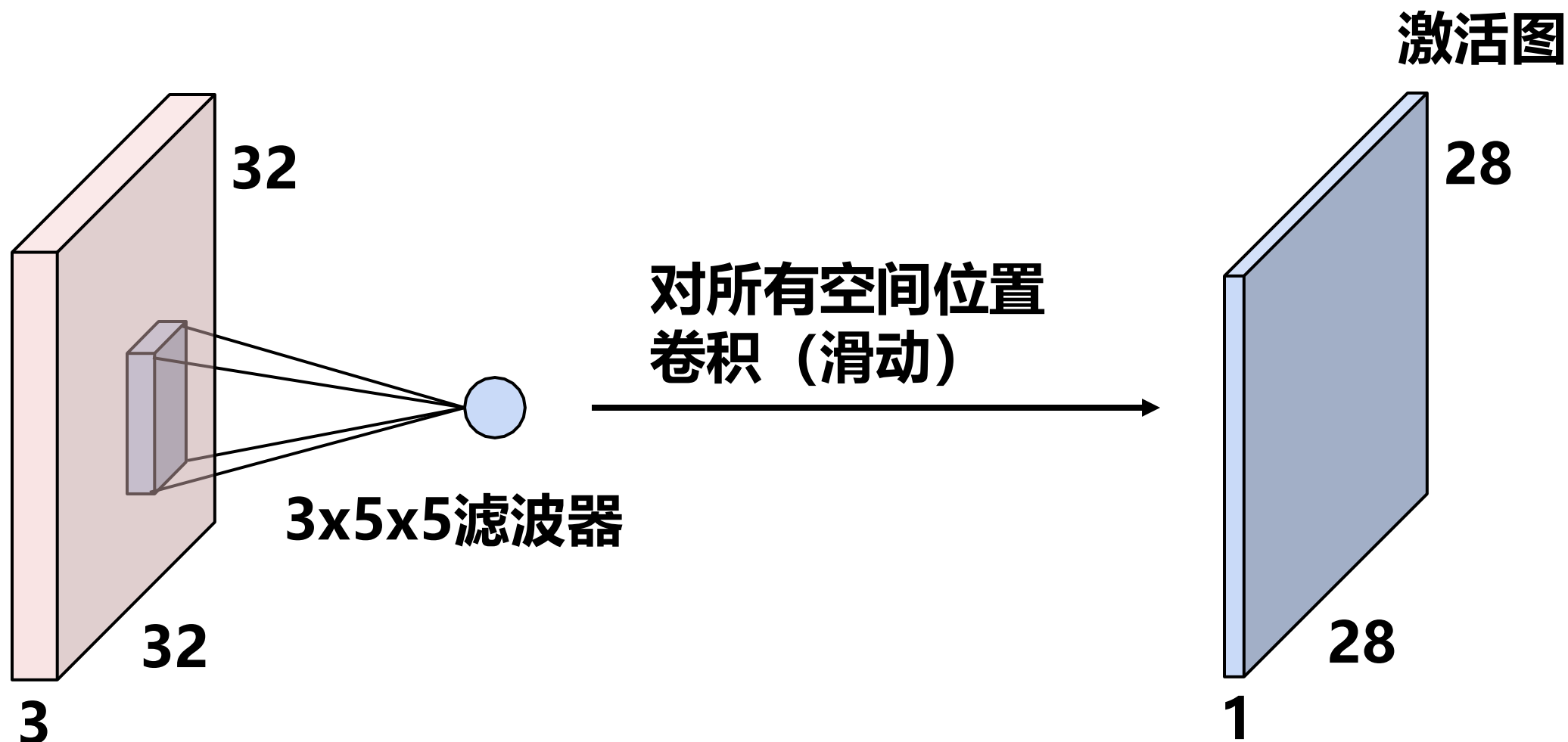
$$f[x,y] * g[x,y] = \sum_{n_1=-\infty}^{\infty} \sum_{n_2=-\infty}^{\infty} f[n_1,n_2] \cdot g[x-n_1,y-n_2]$$

↑
滤波器与信号（图像）
的逐元素乘法和求和

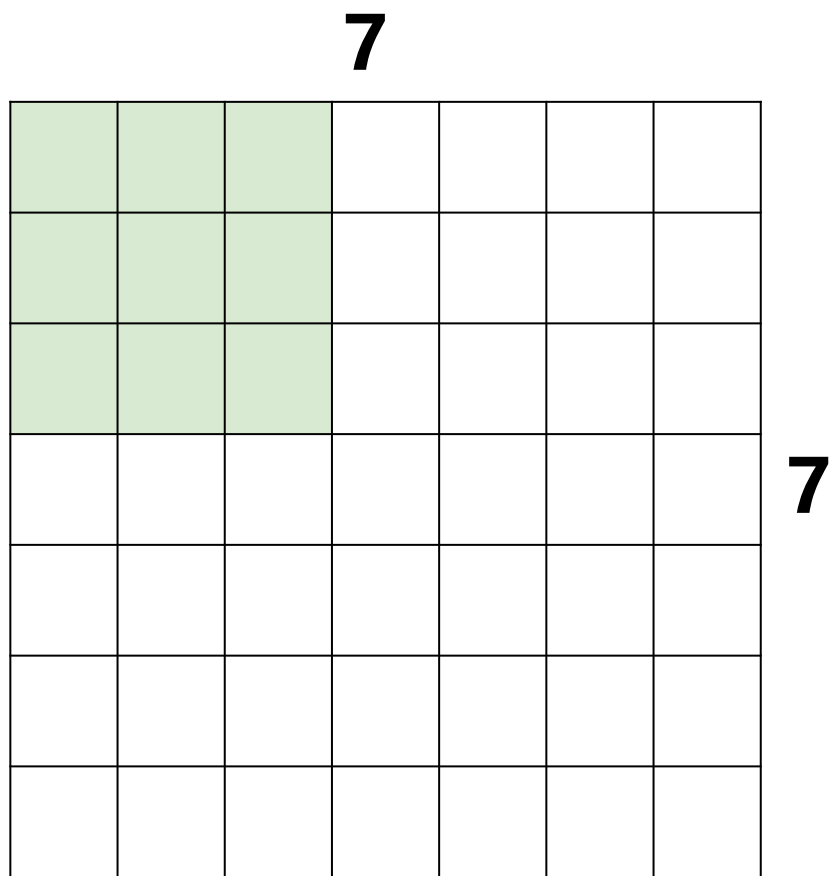
■ 卷积神经网络



■ 卷积输出的空间尺寸

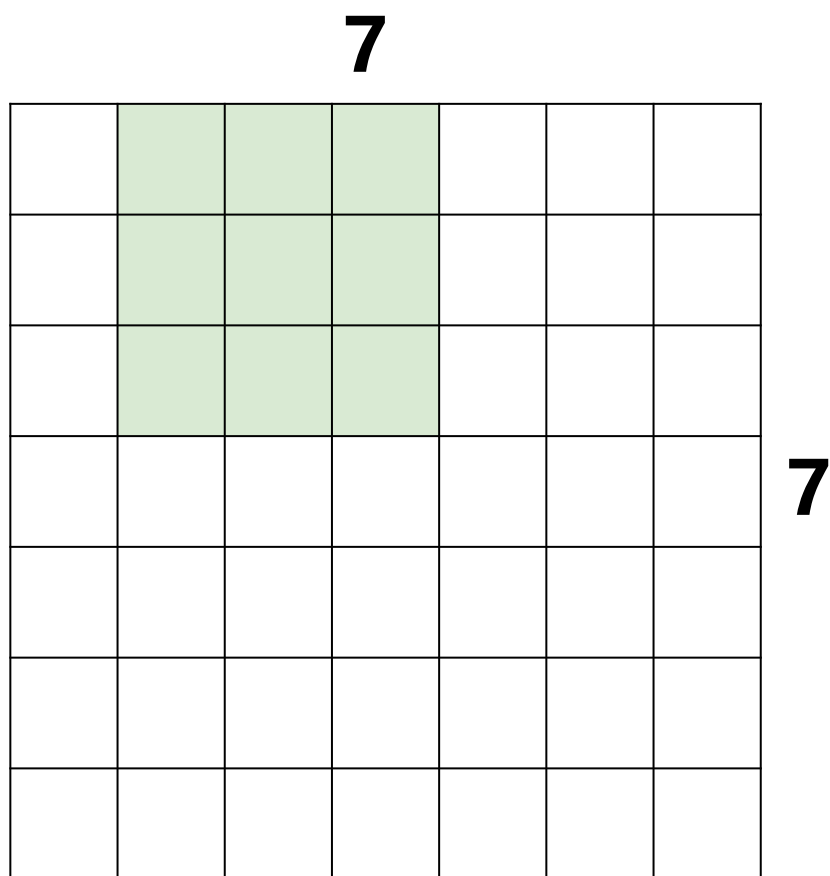


■ 卷积输出的空间尺寸



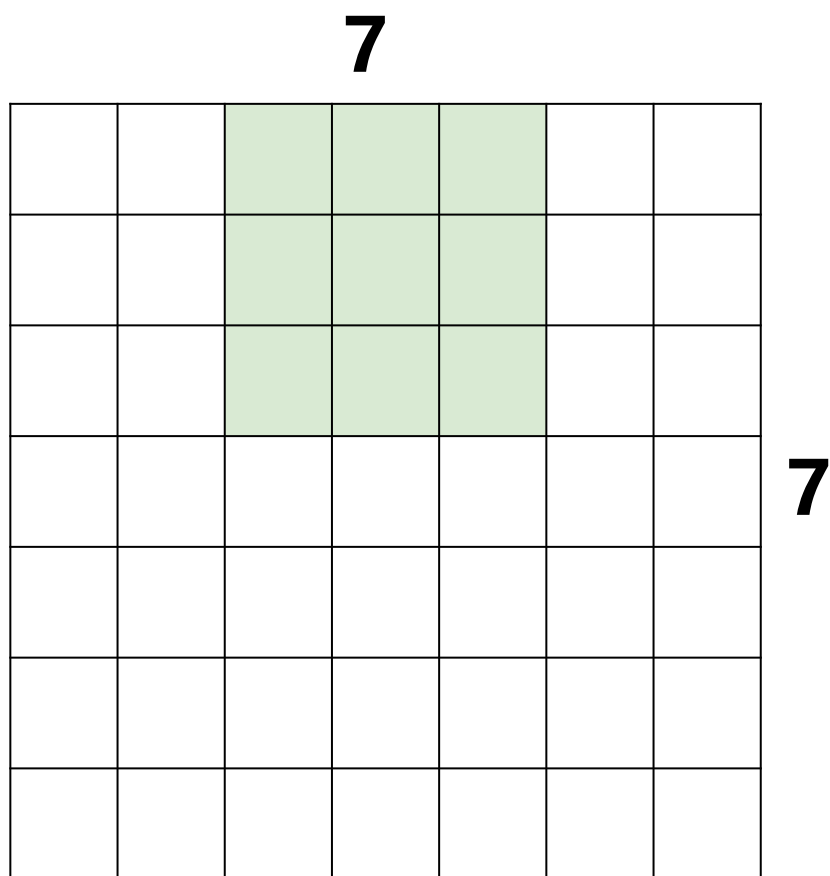
7x7输入
3x3滤波器

■ 卷积输出的空间尺寸



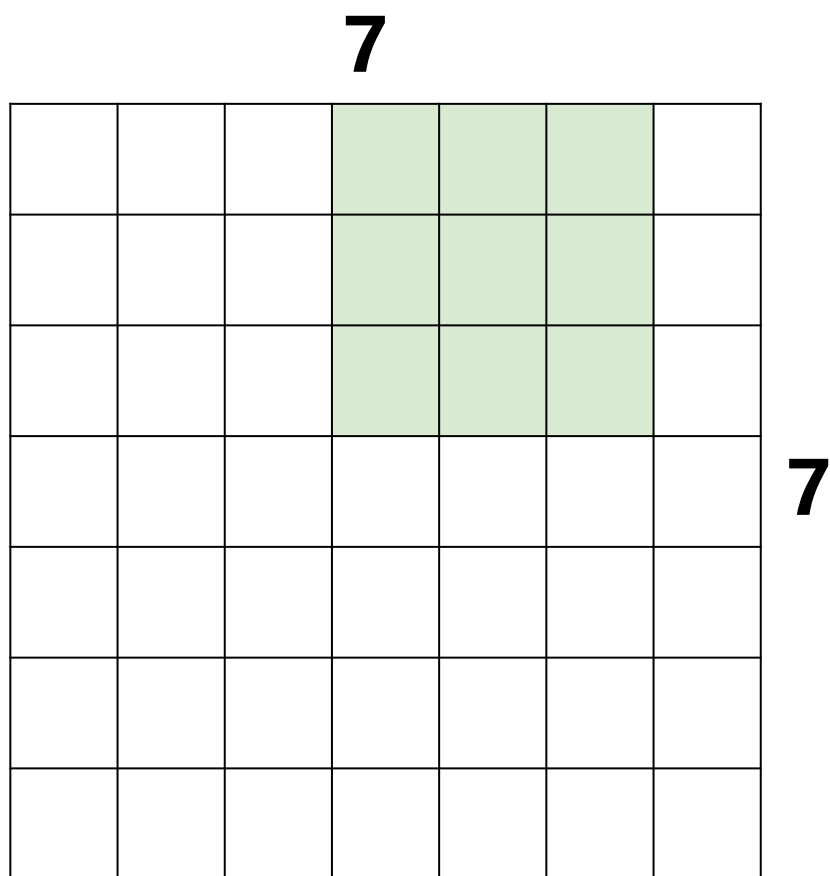
7x7输入
3x3滤波器

■ 卷积输出的空间尺寸



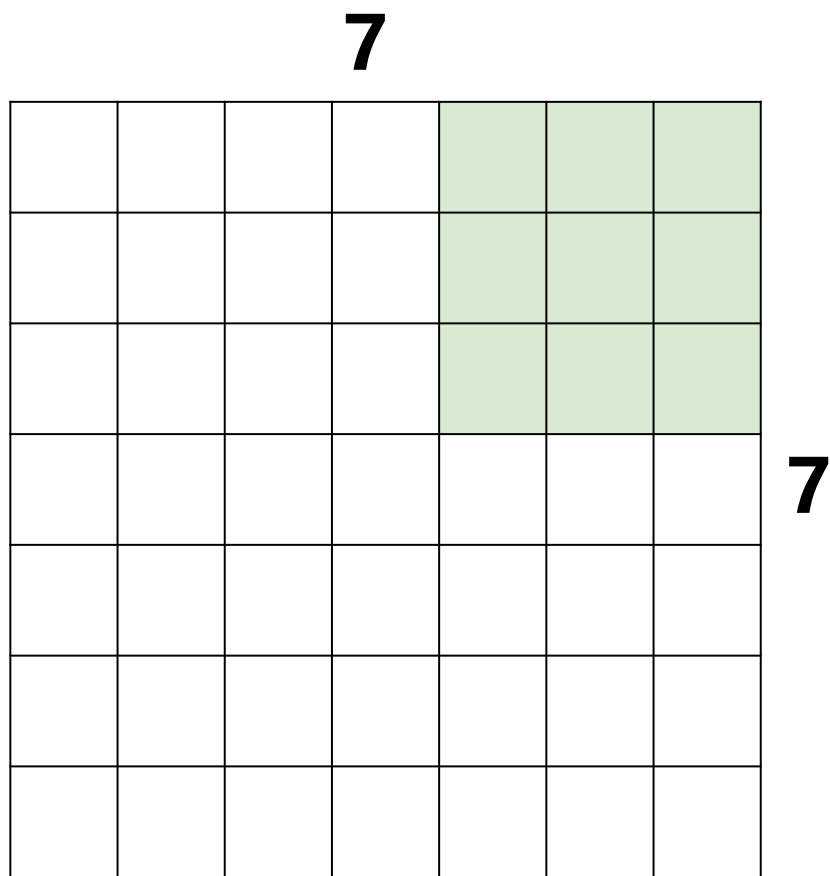
7x7输入
3x3滤波器

■ 卷积输出的空间尺寸



7x7输入
3x3滤波器

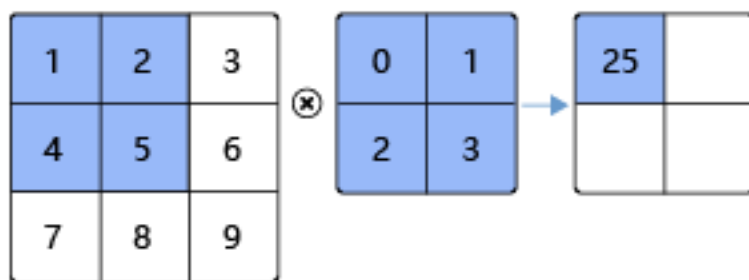
■ 卷积输出的空间尺寸



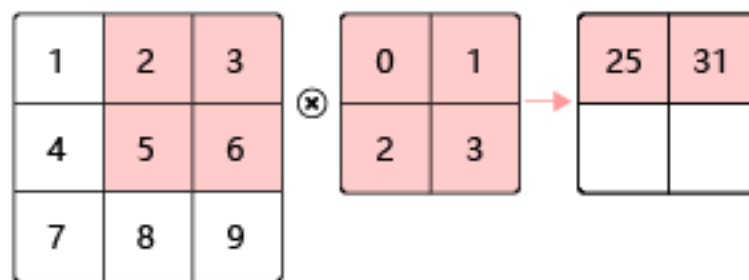
7x7输入
3x3滤波器

5x5输出

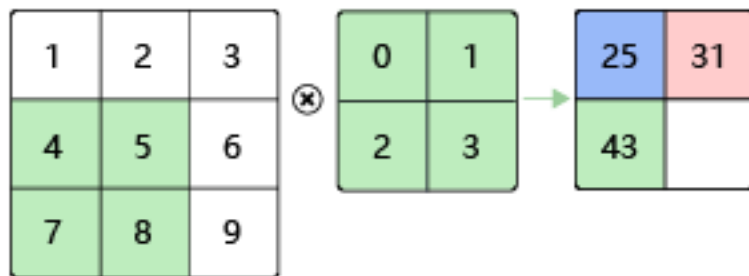
■ 卷积层



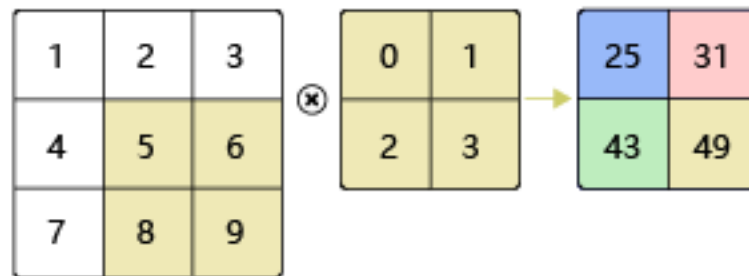
(a) $0 \times 1 + 1 \times 2 + 2 \times 4 + 3 \times 5 = 25$



(b) $0 \times 2 + 1 \times 3 + 2 \times 5 + 3 \times 6 = 31$

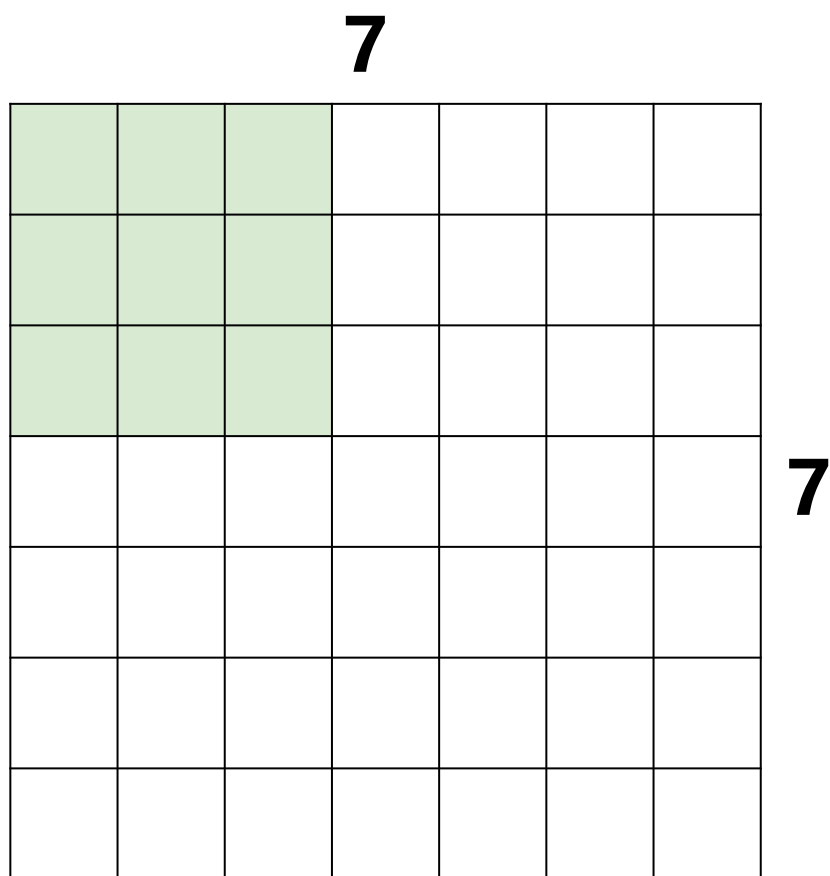


(c) $0 \times 4 + 1 \times 5 + 2 \times 7 + 3 \times 8 = 43$



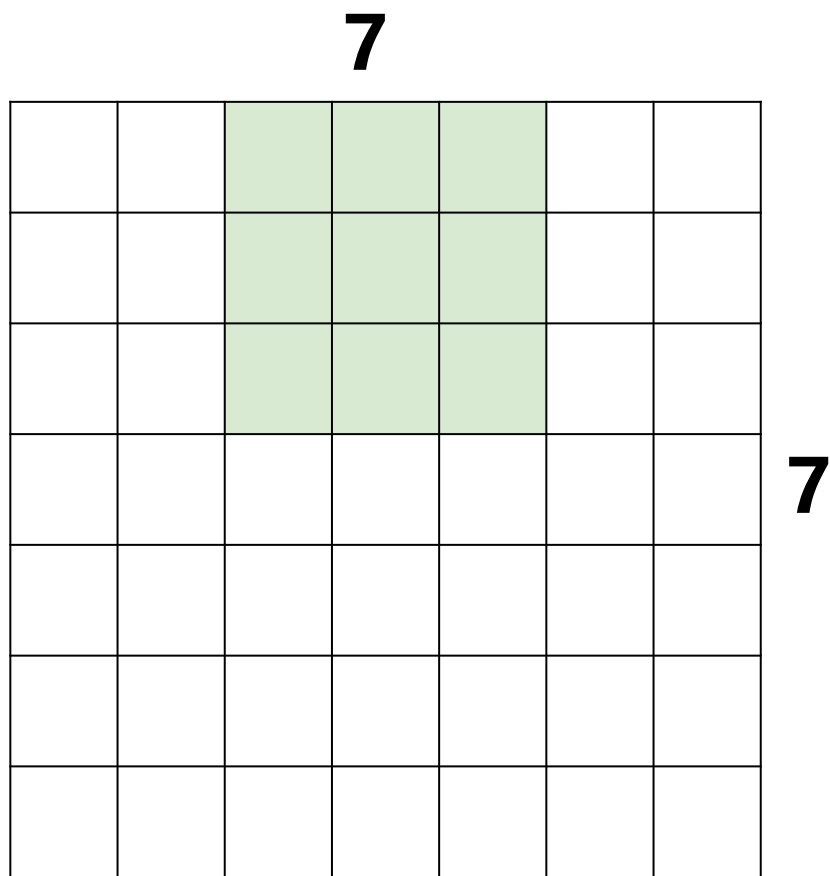
(d) $0 \times 5 + 1 \times 6 + 2 \times 8 + 3 \times 9 = 49$

■ 卷积输出的空间尺寸



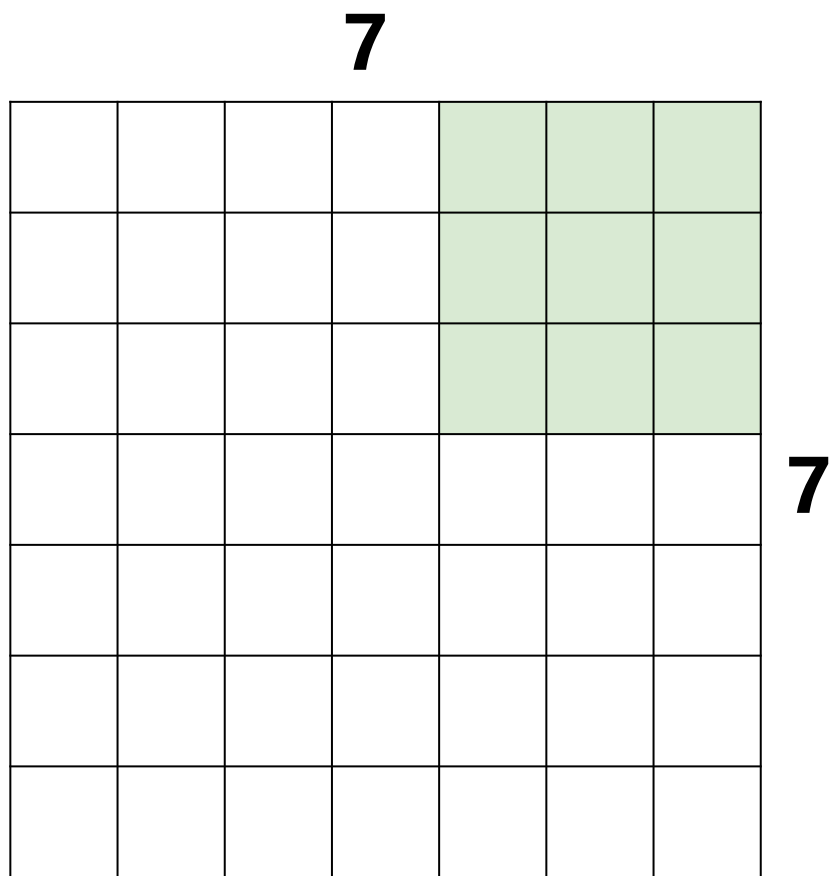
7x7输入
3x3滤波器
2步长

■ 卷积输出的空间尺寸



7x7输入
3x3滤波器
2步长

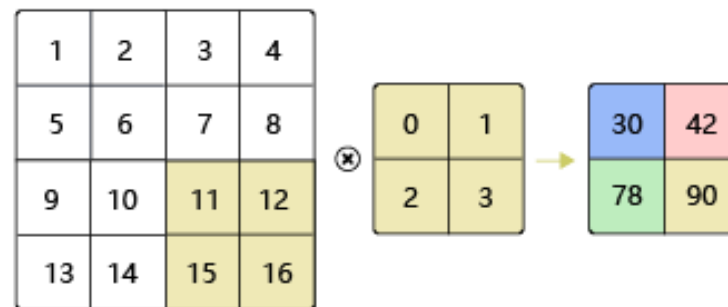
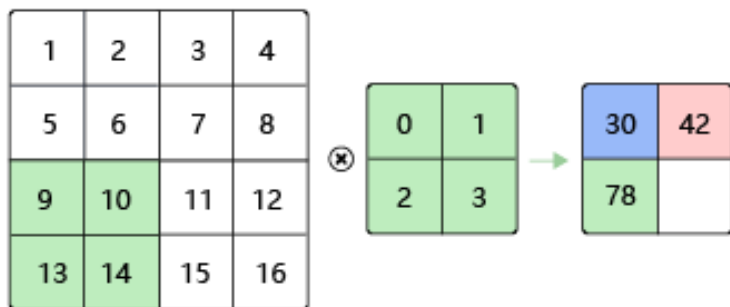
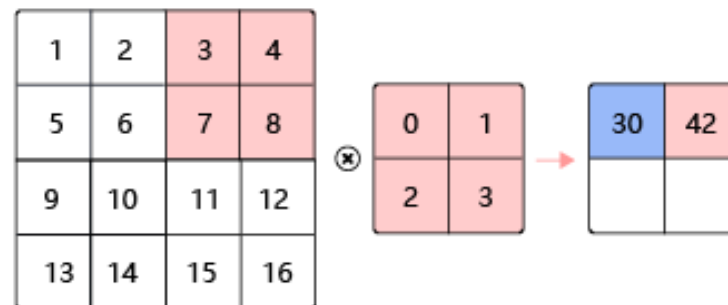
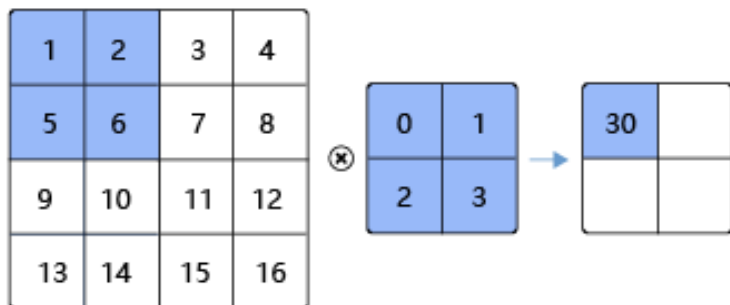
■ 卷积输出的空间尺寸



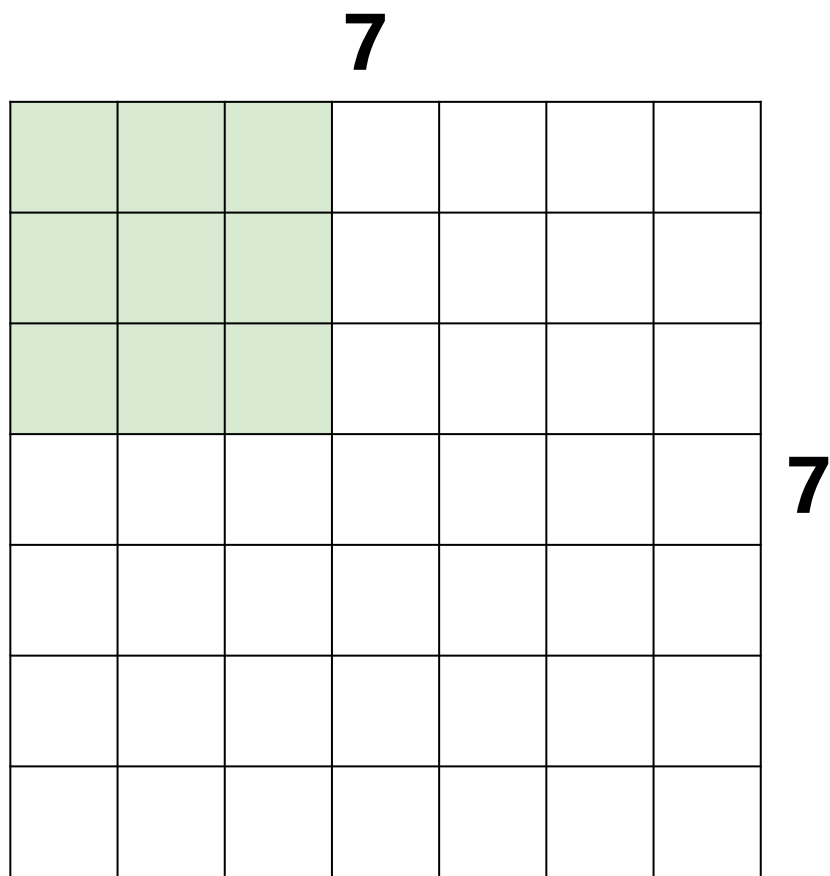
7x7输入
3x3滤波器
2步长

3x3输出

■ 卷积层



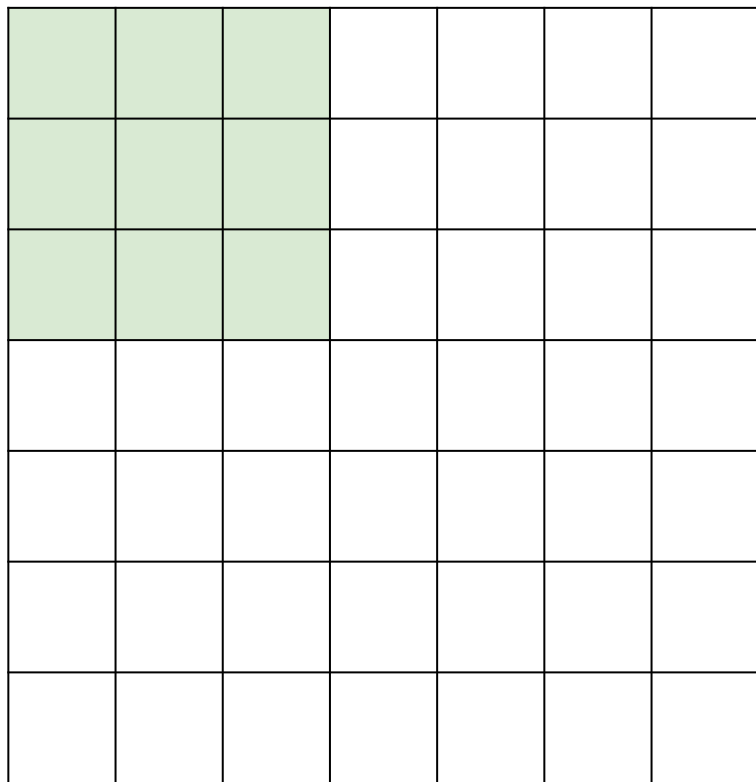
■ 卷积输出的空间尺寸



7x7输入
3x3滤波器
3步长?

不可以!
尺寸不匹配

■ 卷积输出的空间尺寸



NxN 输入

FxF 滤波器

S 步长

输出大小: $(N-F)/S+1$

例如: $N = 7, F = 3$:

步长 1 $\Rightarrow (7 - 3)/1 + 1 = 5$

步长 2 $\Rightarrow (7 - 3)/2 + 1 = 3$

步长 3 $\Rightarrow (7 - 3)/3 + 1 = 2.33$

■ 卷积输出的空间尺寸

0	0	0	0	0	0			
0								
0								
0								
0								

7x7 输入 $N \times N$
3x3 滤波器 $F \times F$
1 步长 S
1 Padding P
输出尺寸?

■ 卷积输出的空间尺寸

0	0	0	0	0	0
0	13	14	15	16	0
0	9	10	11	12	0
0	5	6	7	8	0
0	1	2	3	4	0
0	0	0	0	0	0

(a)padding=1

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	13	14	15	16	0	0
0	0	9	10	11	12	0	0
0	0	5	6	7	8	0	0
0	0	1	2	3	4	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

(b)padding=2

卷积输出的空间尺寸

0	0	0	0	0	0			
0								
0								
0								
0								

7x7 输入 $N \times N$
3x3 滤波器 $F \times F$
1 步长 S
1 Padding P
7x7 输出

输出: $(N+2P-F)/S+1$

如何理解?

■ 卷积输出的空间尺寸

0	0	0	0	0	0			
0								
0								
0								
0								

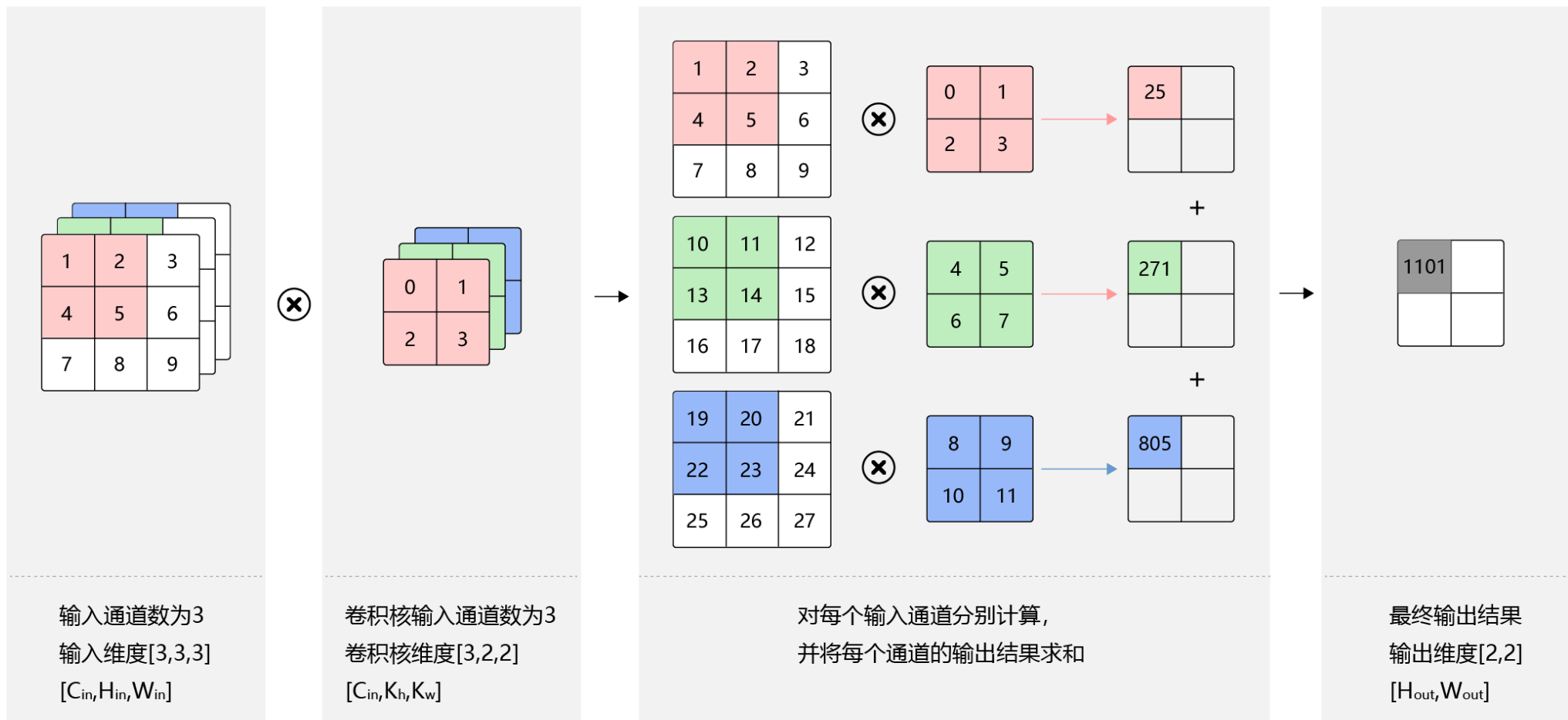
为了保持输出的空间尺寸大小：
CONV层步幅为1
滤波器大小为 $F \times F$
零填充为 $(F-1) / 2$

$F = 3 \Rightarrow$ zero pad with 1

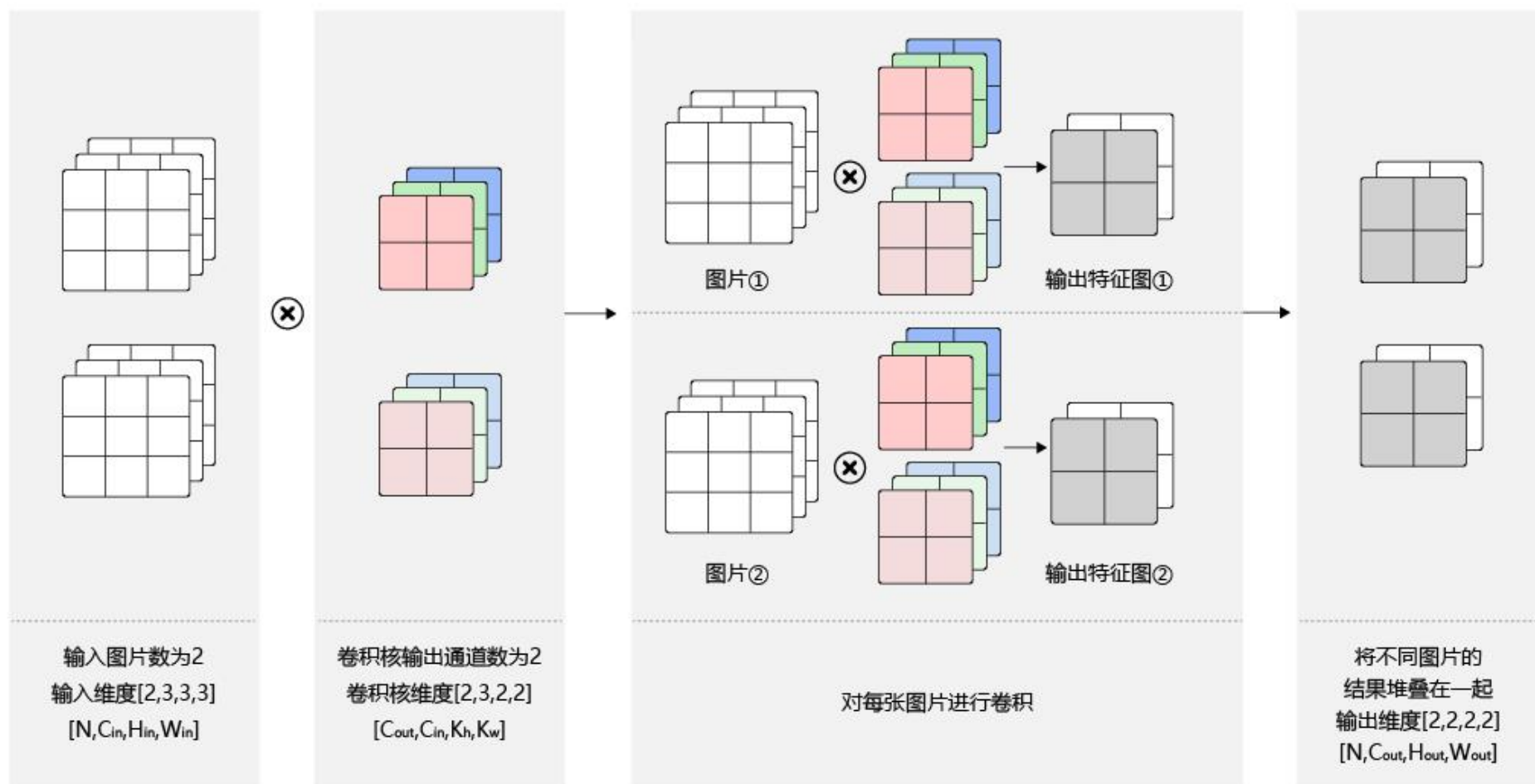
$F = 5 \Rightarrow$ zero pad with 2

$F = 7 \Rightarrow$ zero pad with 3

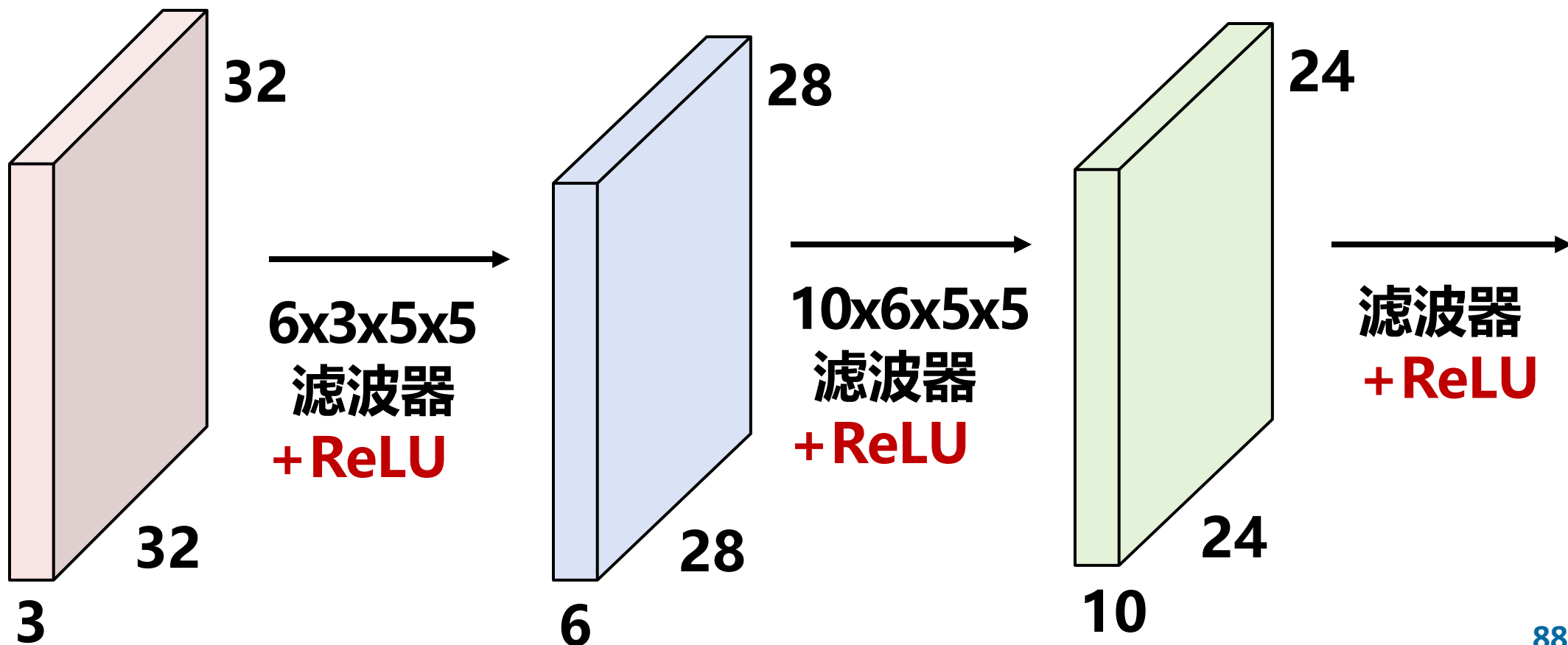
■ 多输入通道卷积



■ 多输出通道卷积



- 激活图尺寸：32x32→28x28→24x24



- 卷积输出的空间尺寸
- 假设：
 - 输入：3x32x32
 - 滤波器：10x5x5
 - 步长：1
 - Padding：2
- 输出多少？

- 卷积输出的空间尺寸

- 假设：

- 输入：3x32x32

- 滤波器：10x5x5

3x

- 步长：1

- Padding：2

- 输出：32x32x10，即 $(32 + 2 * 2 - 5) / 1 + 1 = 32$

- 卷积输出的空间尺寸

- 假设：

- 输入：3x32x32

- 滤波器：10x5x5

- 步长：1

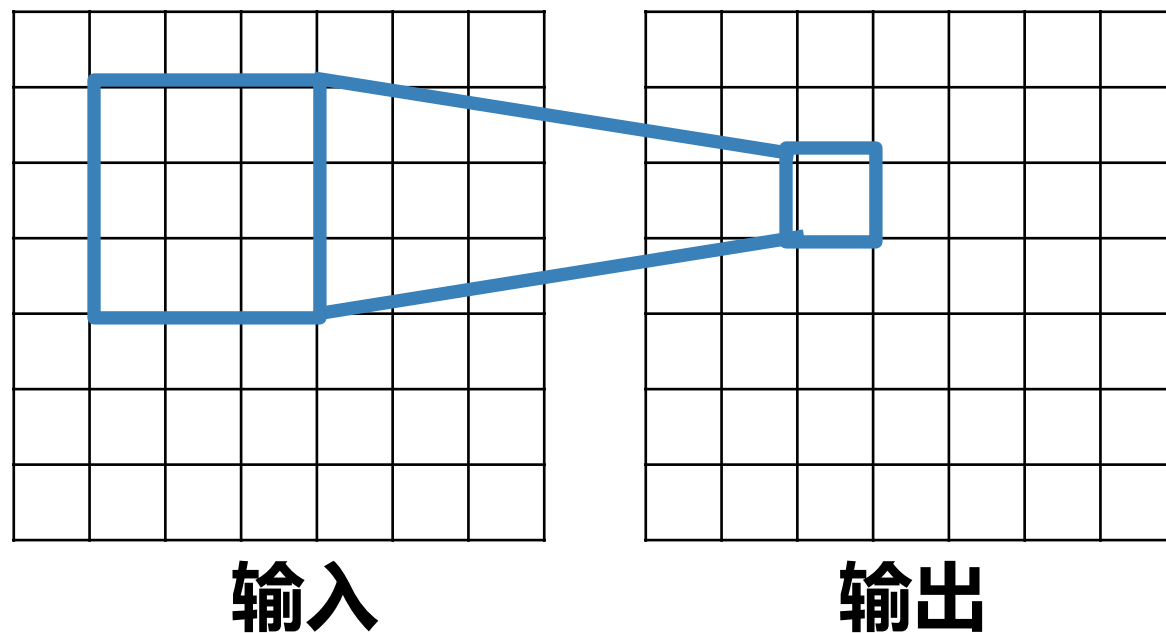
- Padding：2

- 参数量？

- 卷积输出的空间尺寸
- 假设：
 - 输入：3x32x32
 - 滤波器：10x5x5
 - 步长：1
 - Padding：2
 - 参数量：(5x5x3+1)x10



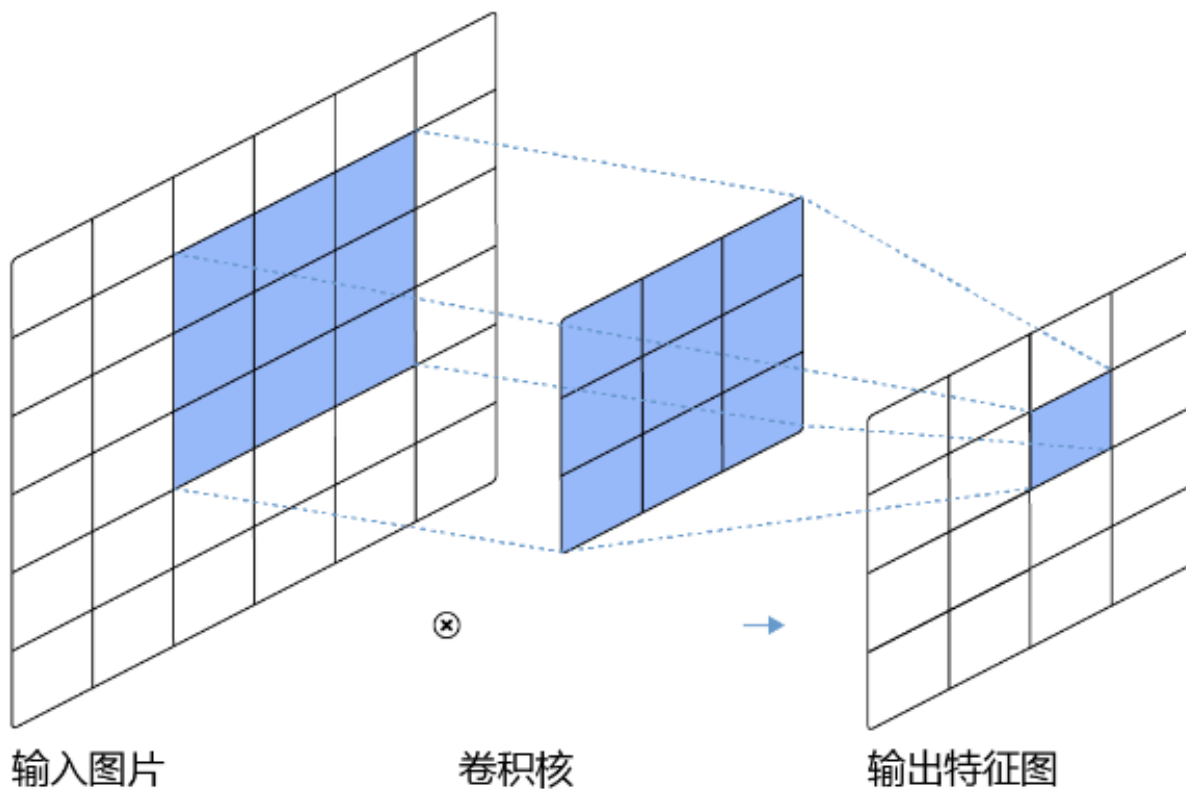
- 卷积层的感受野
- 对于核大小为K的卷积，输出中的每个元素都取决于输入中的 $K \times K$ 感受野



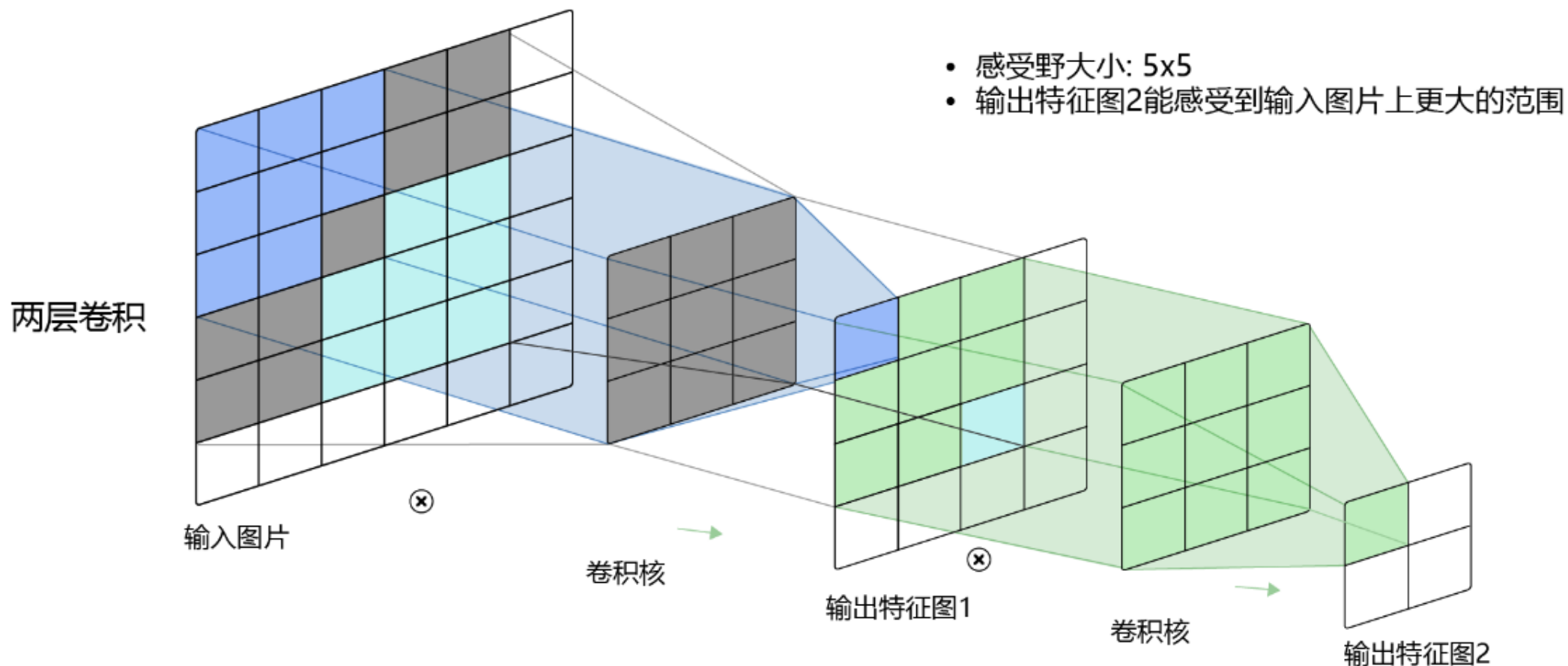
■ 卷积层的感受野

- 感受野大小: 3×3
- 输出特征图上的像素点所能感受到的输入数据的范围

单层卷积



■ 卷积层的感受野

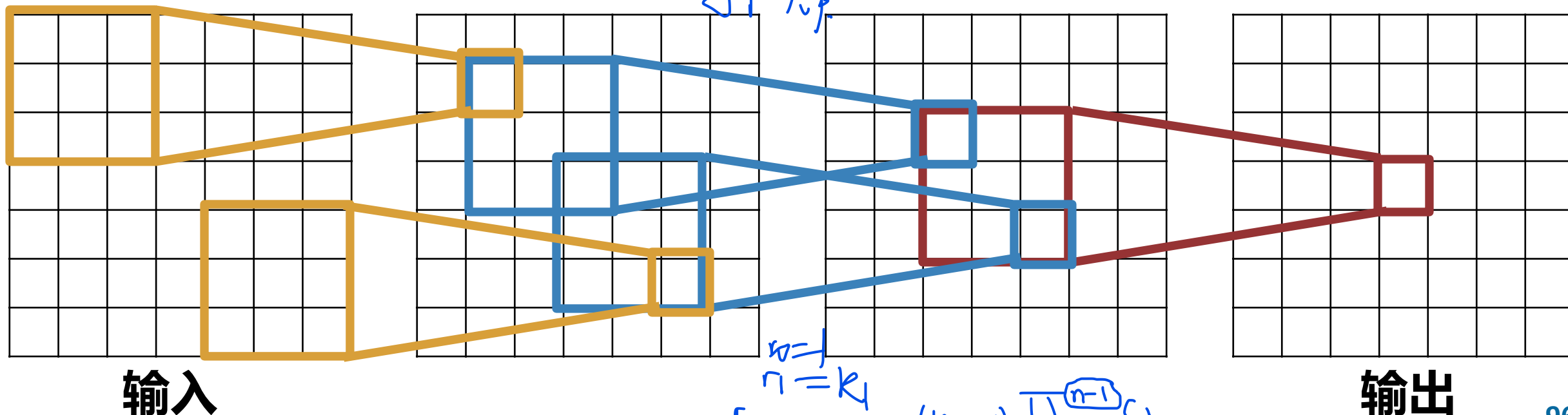


■ 卷积层的感受野

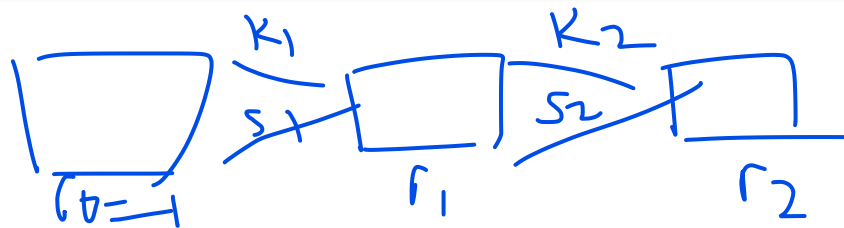
■ 每次连续的卷积都会使感受野大小增加K-1。

对于L层，感受野大小为 $1 + L * (K - 1)$

注意区分：
在前一层上的感受野
在输入图像上的感受野



- 卷积层的感受野



$$\prod_{i=1}^n s_i = 1$$

- 问题：对于大图像，我们需要使用许多卷积层，才能“看到”整个图像

- 解决方案：下采样

- 大步长

- 池化层

■ 卷积层总结

■ 输入: $W_1 \times H_1 \times C$

■ 卷积超参:

■ 滤波器数量 K

■ 滤波器大小 F

■ 步长 S

■ Padding P

■ 输出: $W_2 \times H_2 \times C$

■ $W_2 = (W_1 - F + 2P)/S + 1$

■ $H_2 = (H_1 - F + 2P)/S + 1$

■ 参数量:

■ $F^2CK + K$, 其中 K 是bias数量

■ 卷积层总结

■ 输入: $W_1 \times H_1 \times C$

■ 卷积超参:

■ 滤波器数量 K

■ 滤波器大小 F

■ 步长 S

■ Padding P

■ $K = (2\text{的指数, e.g. } 32, 64, 128, 512)$

■ $F = 3, S = 1, P = 1$

宽 扩展宽度

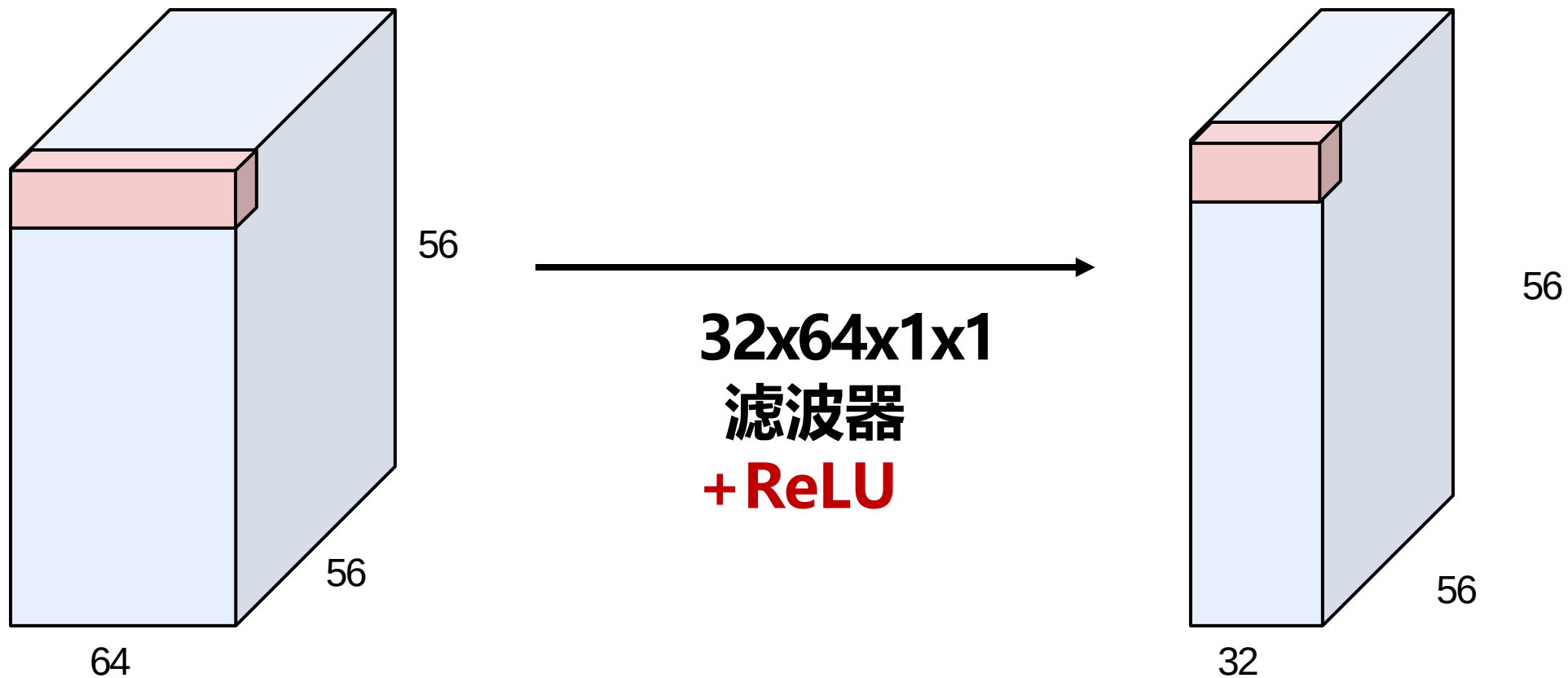
■ $F = 5, S = 1, P = 2$

■ $F = 5, S = 2, P = ?$ (匹配尺寸)

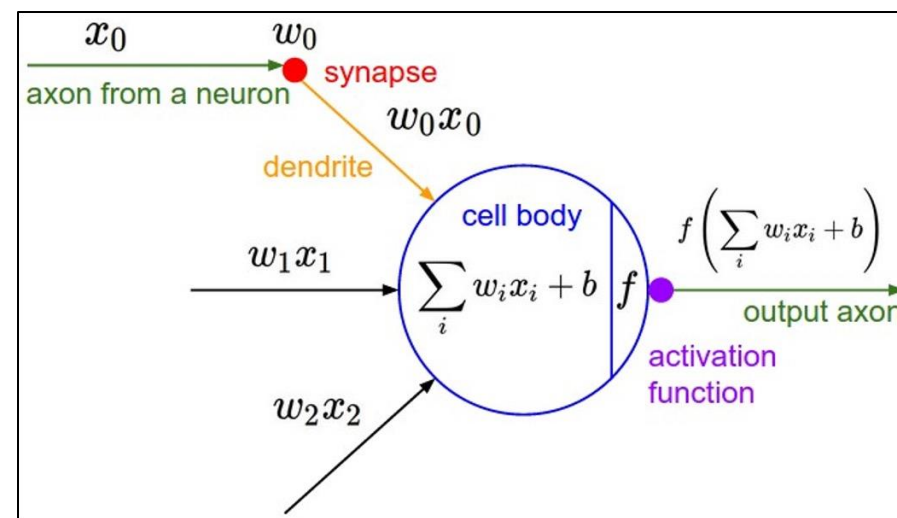
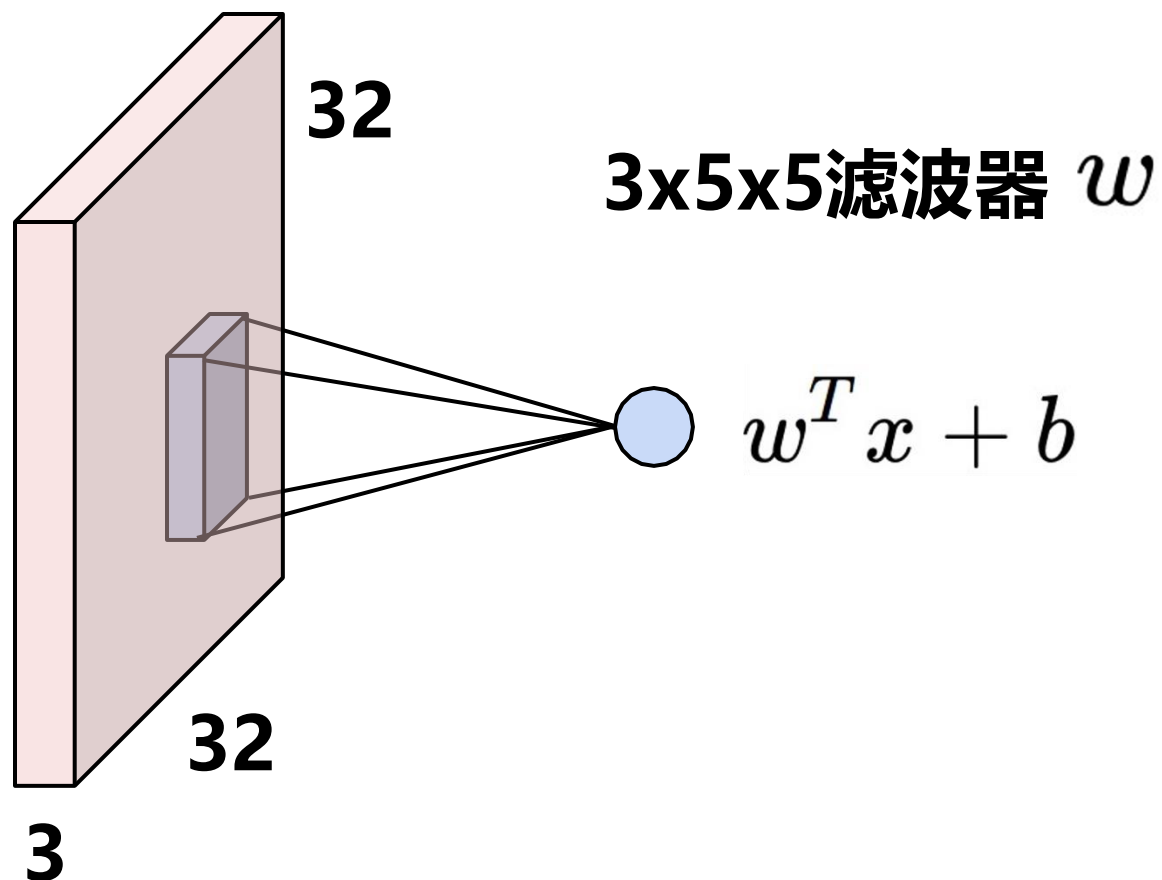
■ $F = 1, S = 1, P = 0$

深 增加通道

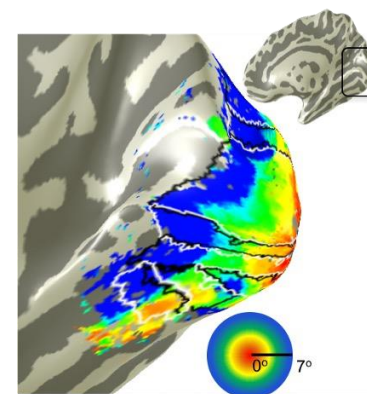
■ 1x1卷积层



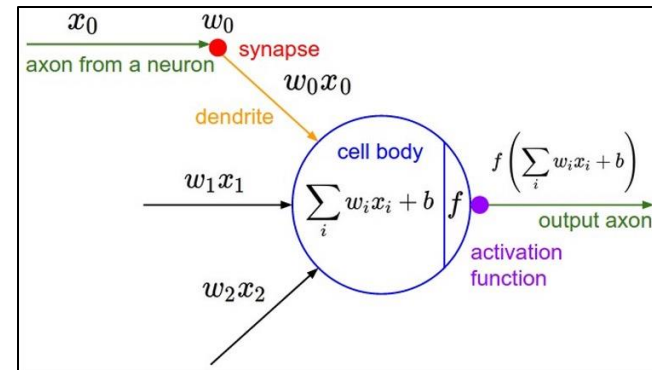
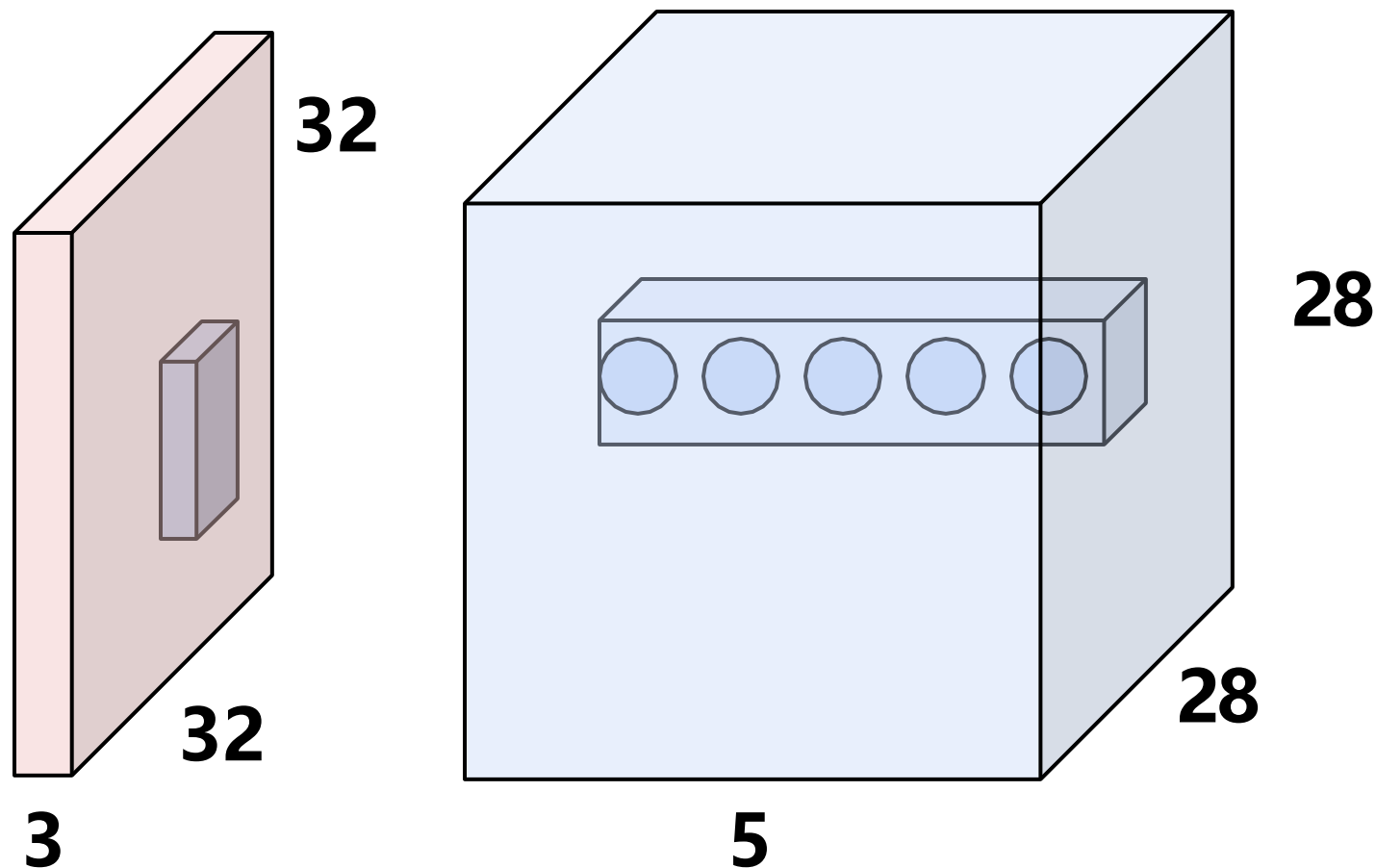
■ 从大脑/神经元角度理解卷积层



具有局部连接
的神经元

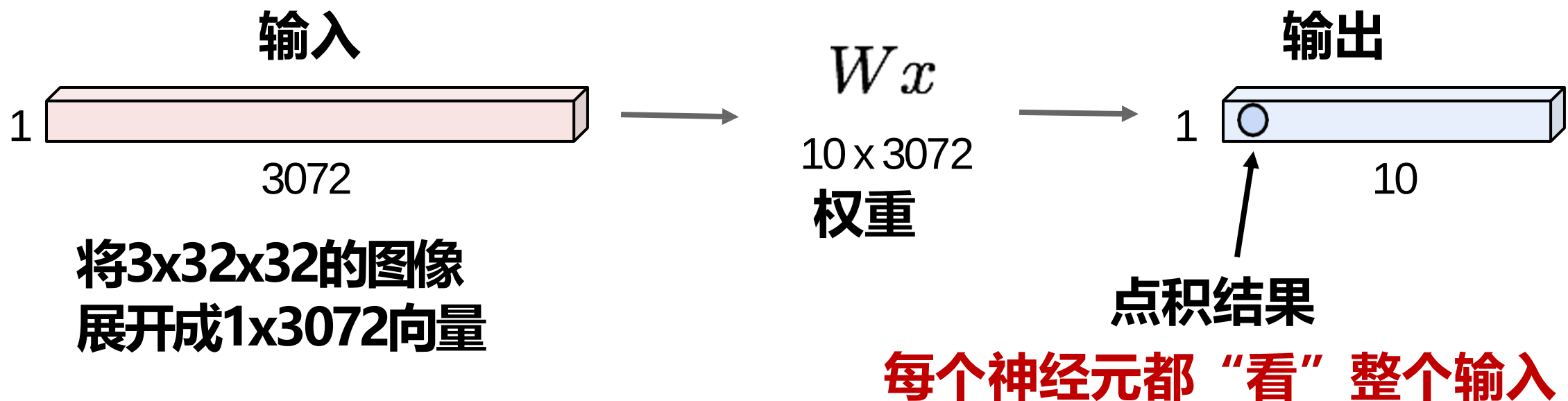


■ 从大脑/神经元角度理解卷积层



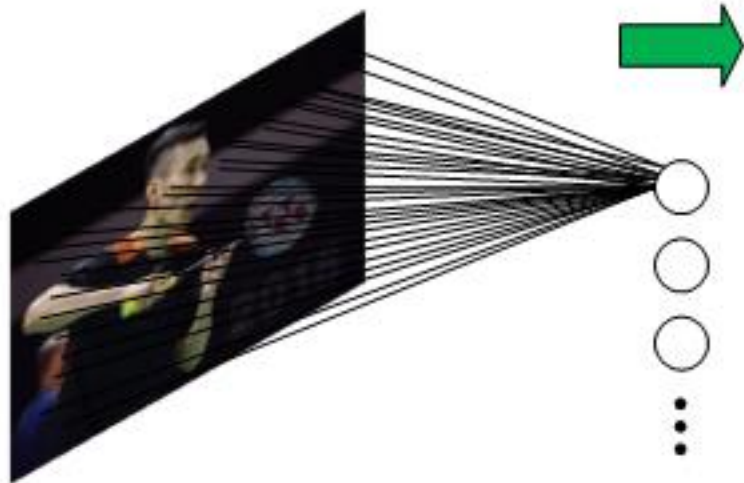
5个不同的神经元都在观察输入体积中的同一区域

■ 从大脑/神经元角度理解全连接层

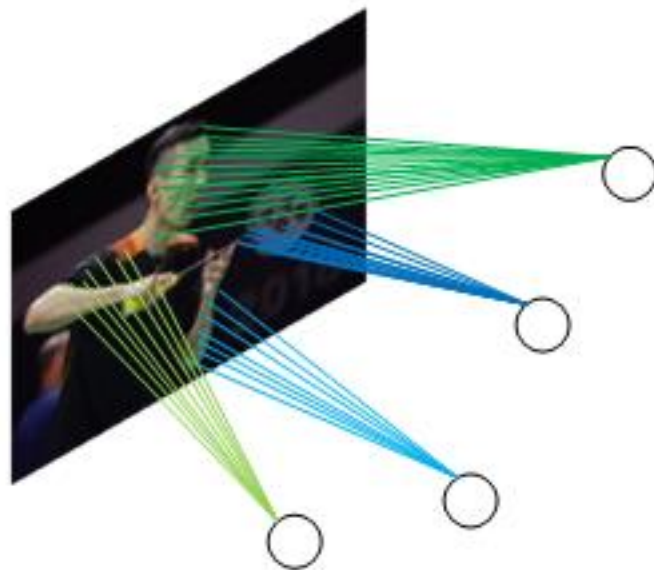


■ 卷积层 v.s. 全连接层

FULLY CONNECTED NEURAL NET

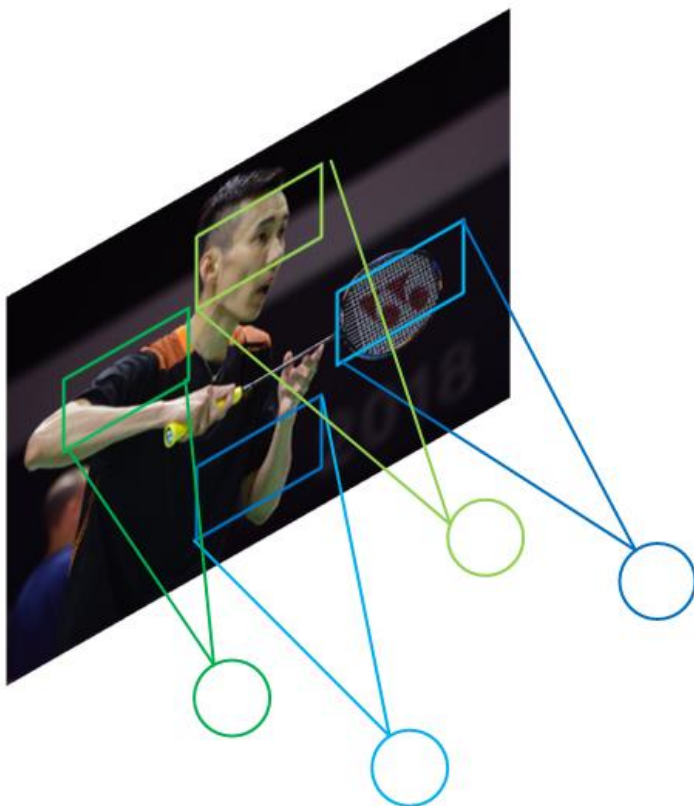


LOCALLY CONNECTED NEURAL NET



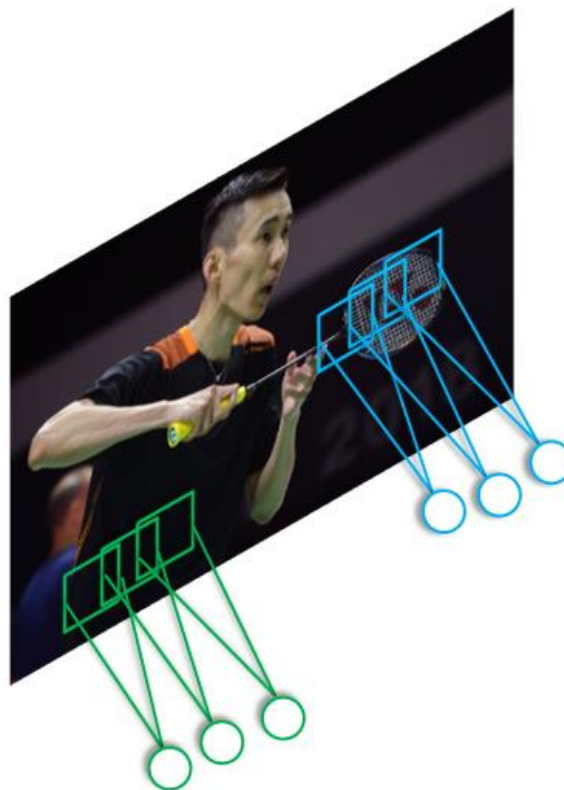
■ 卷积层 v.s. 局部全连接层

LOCALLY CONNECTED NEURAL NET

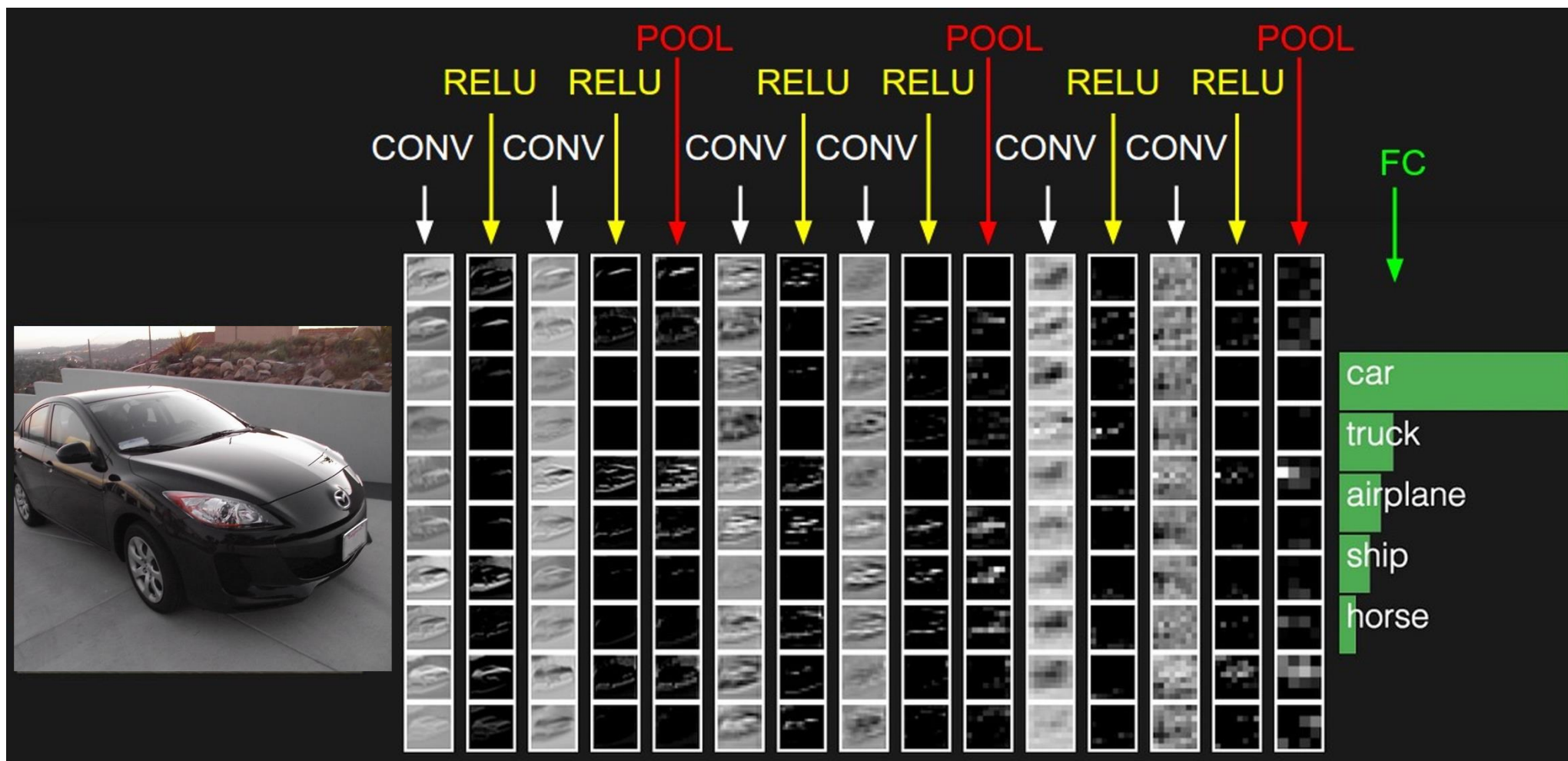


共享权重 参数数量

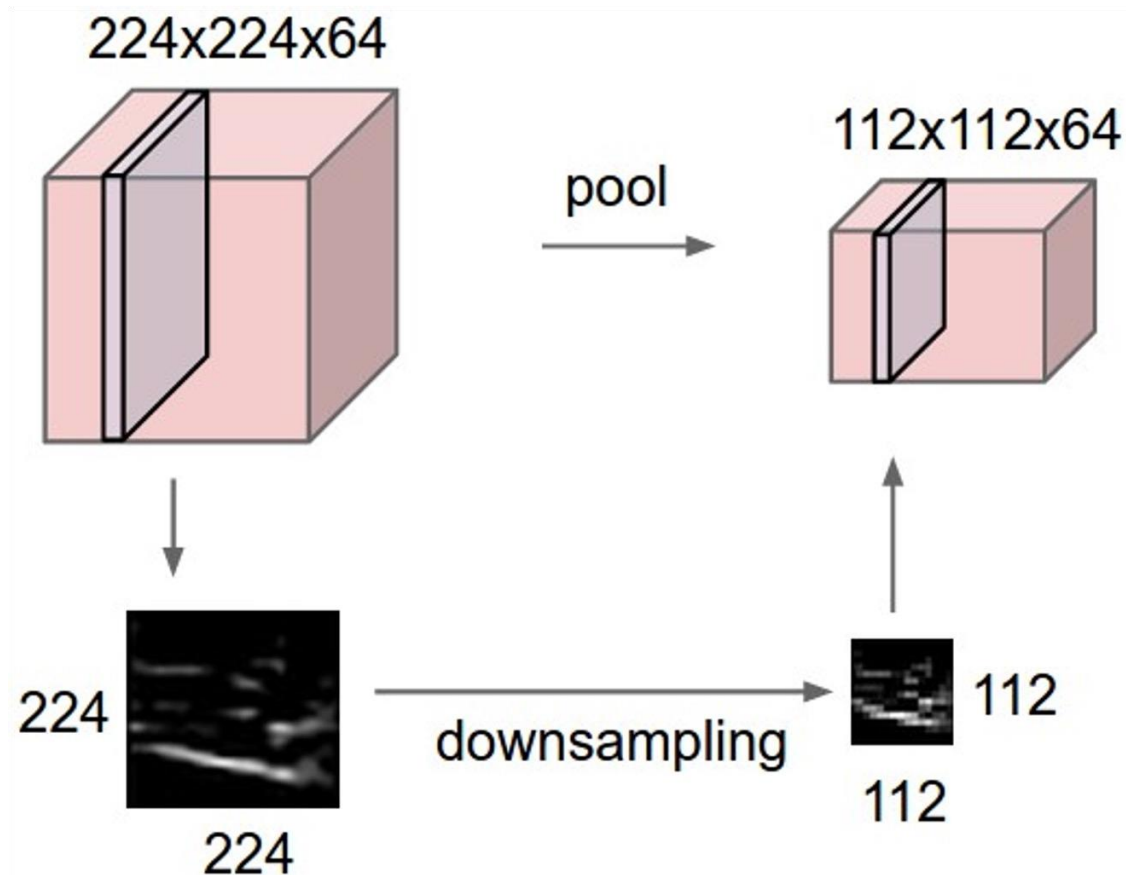
CONVOLUTIONAL NET



■ 池化层



- 池化层
- 使特征更小，更易于处理
- 独立操作每个激活图



■ 最大池化 (Max Pooling)

x ↑

1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

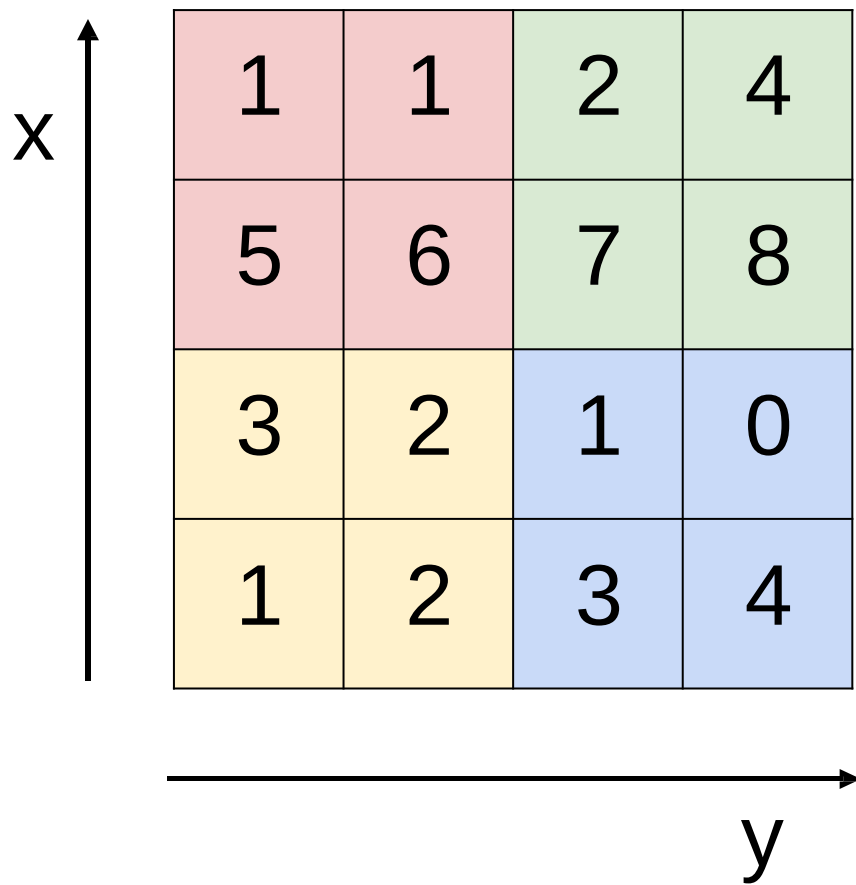
→ y

Max Pooling
2x2 大小和 2 步长

→

6	8
3	4

■ 最大池化 (Max Pooling)



1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

Max Pooling
2x2 大小和 2 步长

6	8
3	4

没有可学习参数
引入空间不变性

■ 池化层总结 不做P

尺寸 ↓

■ 输入: $W_1 \times H_1 \times C$

■ 输出: $W_2 \times H_2 \times C$

■ 卷积超参:

■ $W_2 = (W_1 - F)/S + 1$

■ 滤波器大小 F

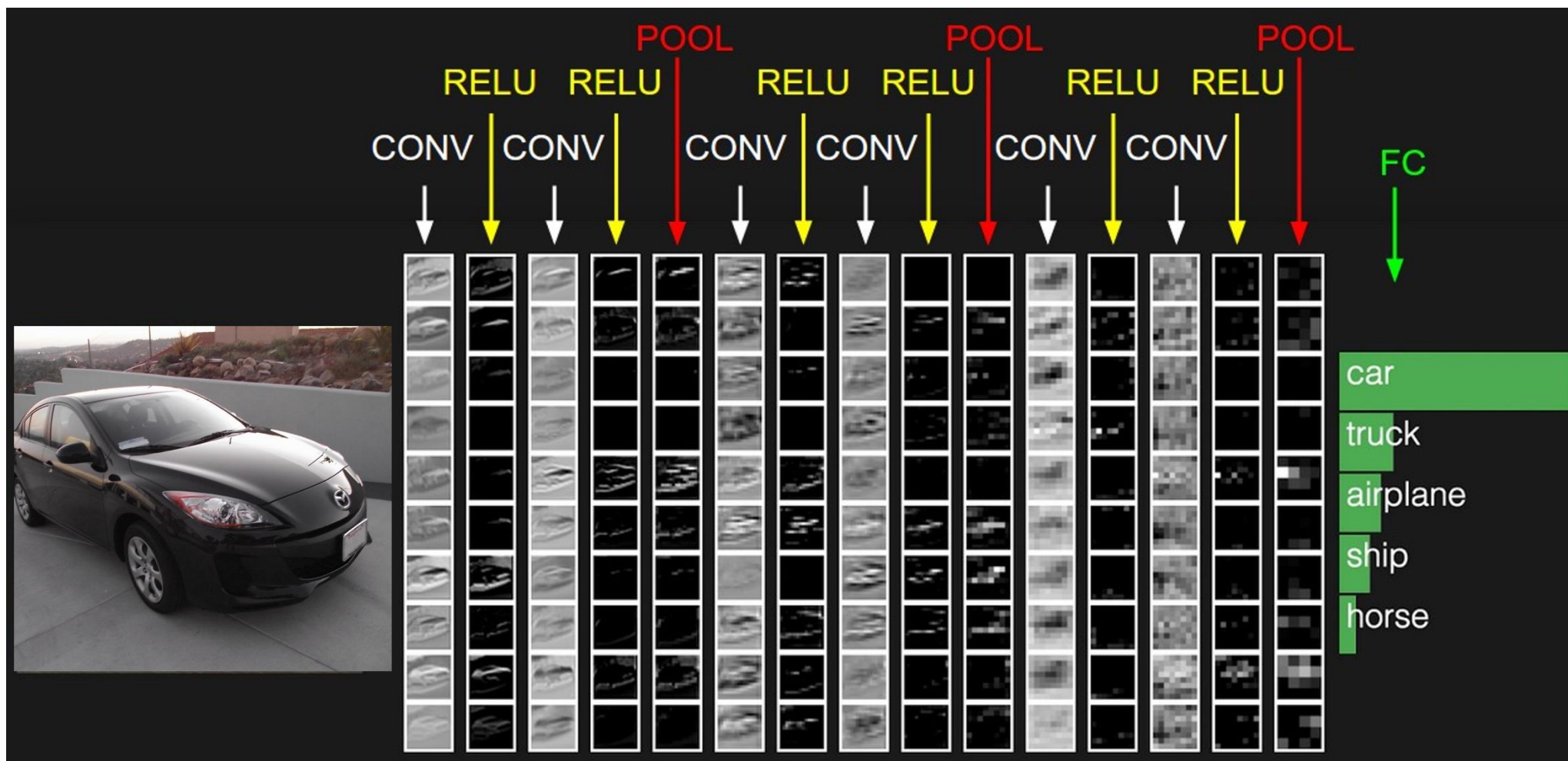
■ $H_2 = (H_1 - F)/S + 1$

■ 步长 S

■ 参数量:

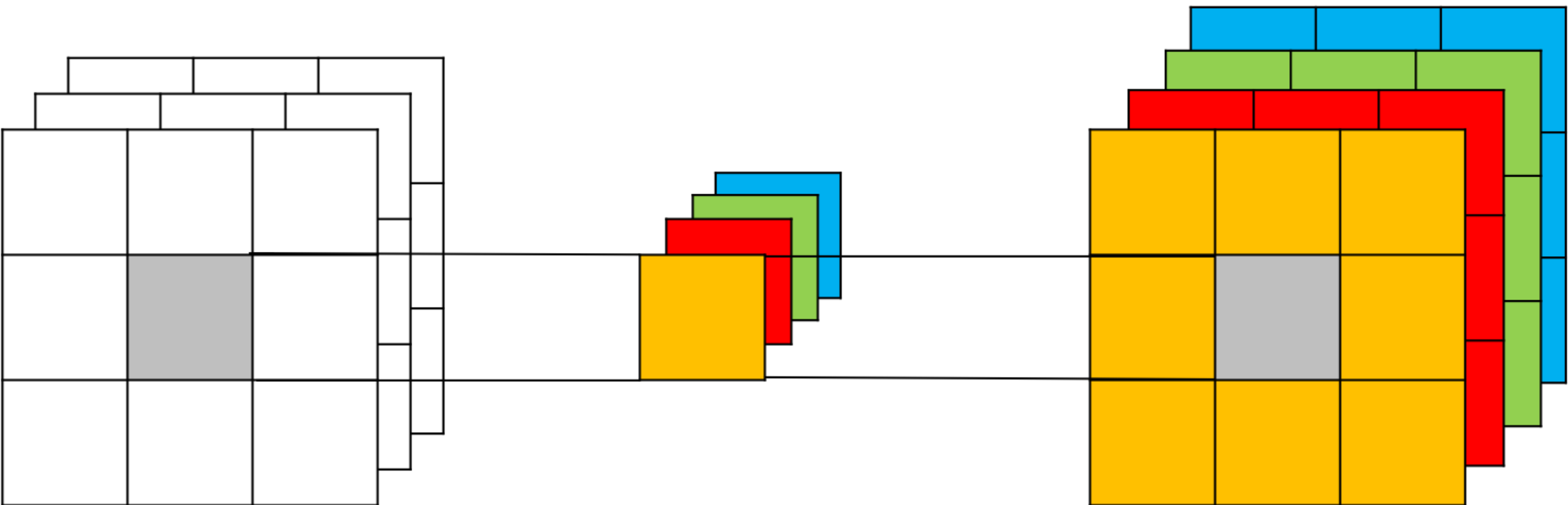
■ 0

■ 全连接层

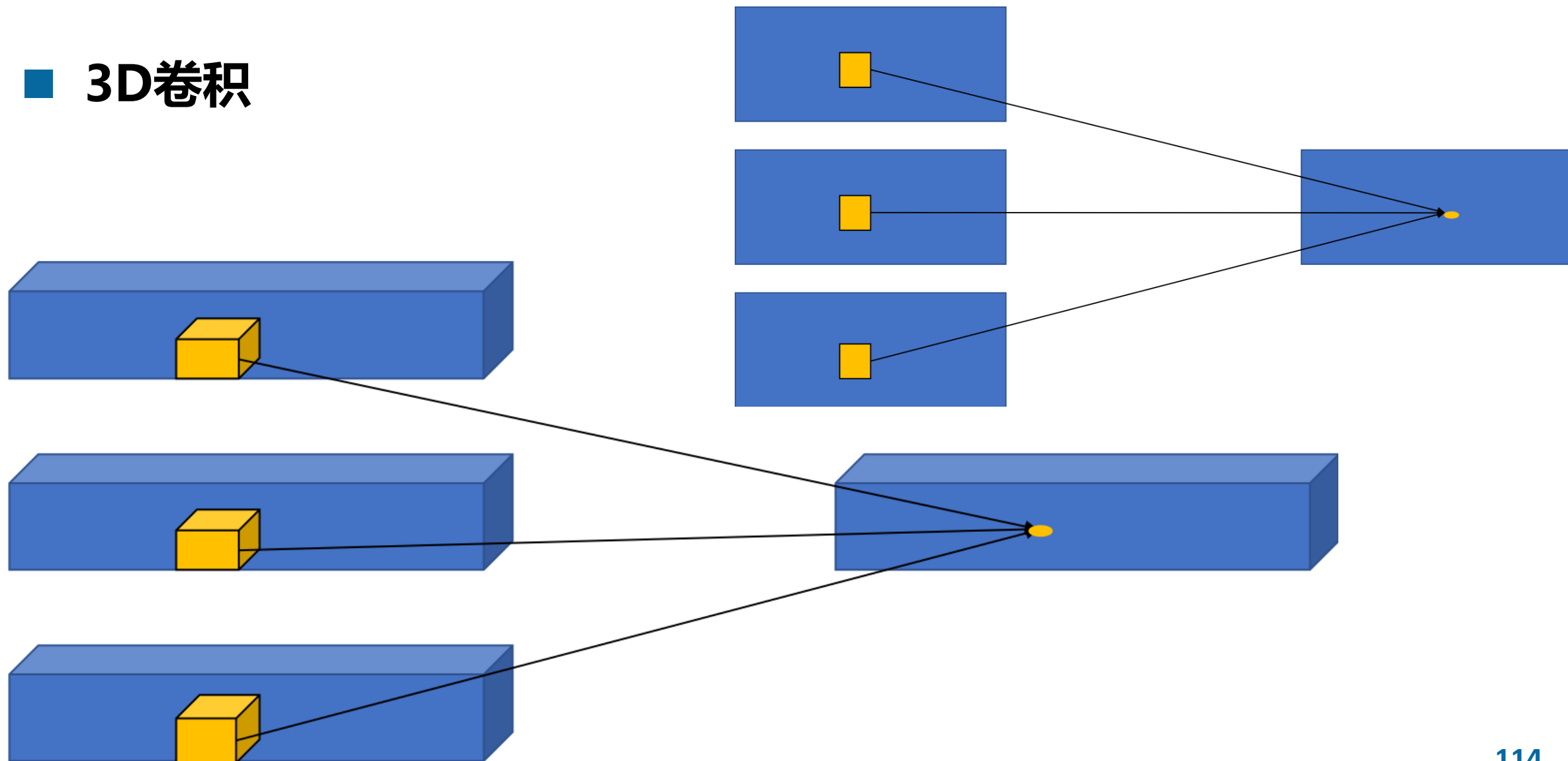


- 卷积层+池化层+全连层+激活函数
- 趋向于更小的滤波器和更深的层 3×3
- 趋向于舍弃池化层和全连接层
→ 共用子最后一层
→ 用可学习的 conv 1/2 代替
- 以往常见的卷积网络架构：
 - $[(\text{Conv-ReLU}) * N - \text{Pooling}] * M - (\text{FC-ReLU}) * K - \text{Softmax}$
 - N一般为~5, M较大, K一般在0~2
- 新型卷积网络 (ResNet, GoogLeNet) 提出了新的架构

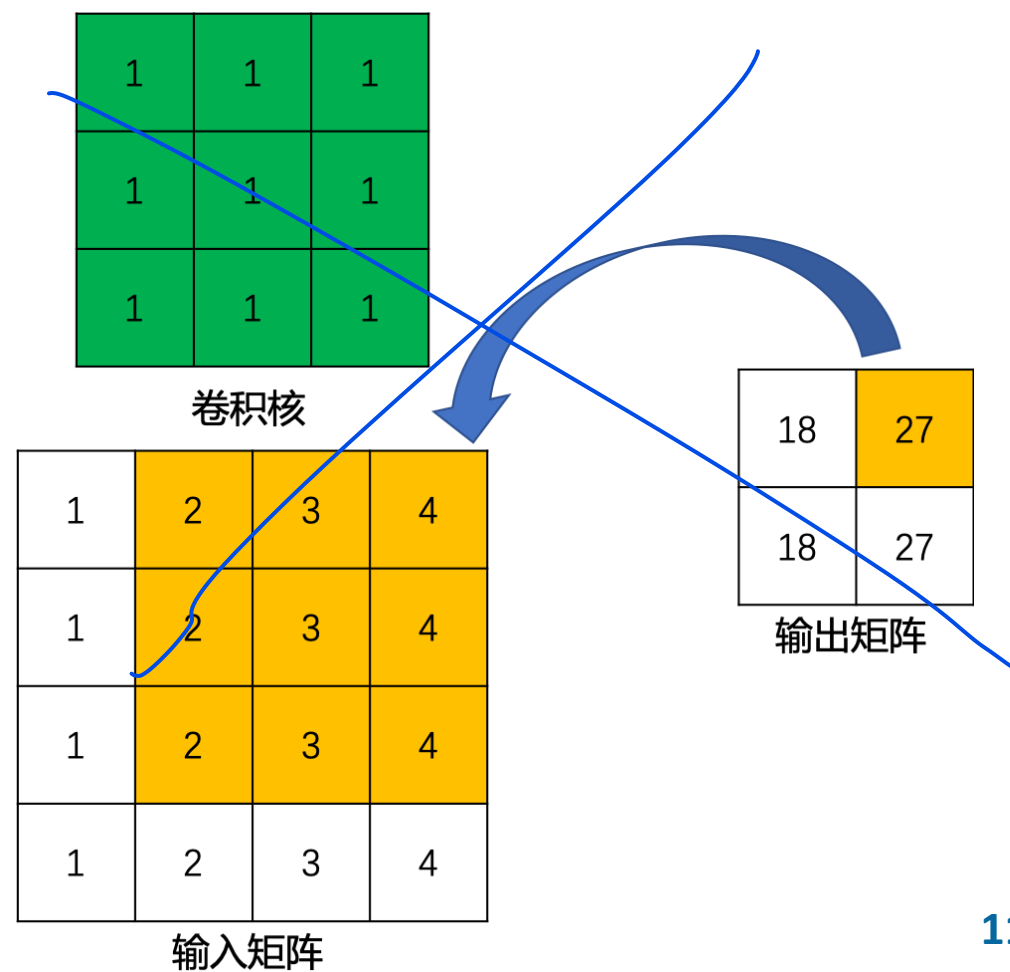
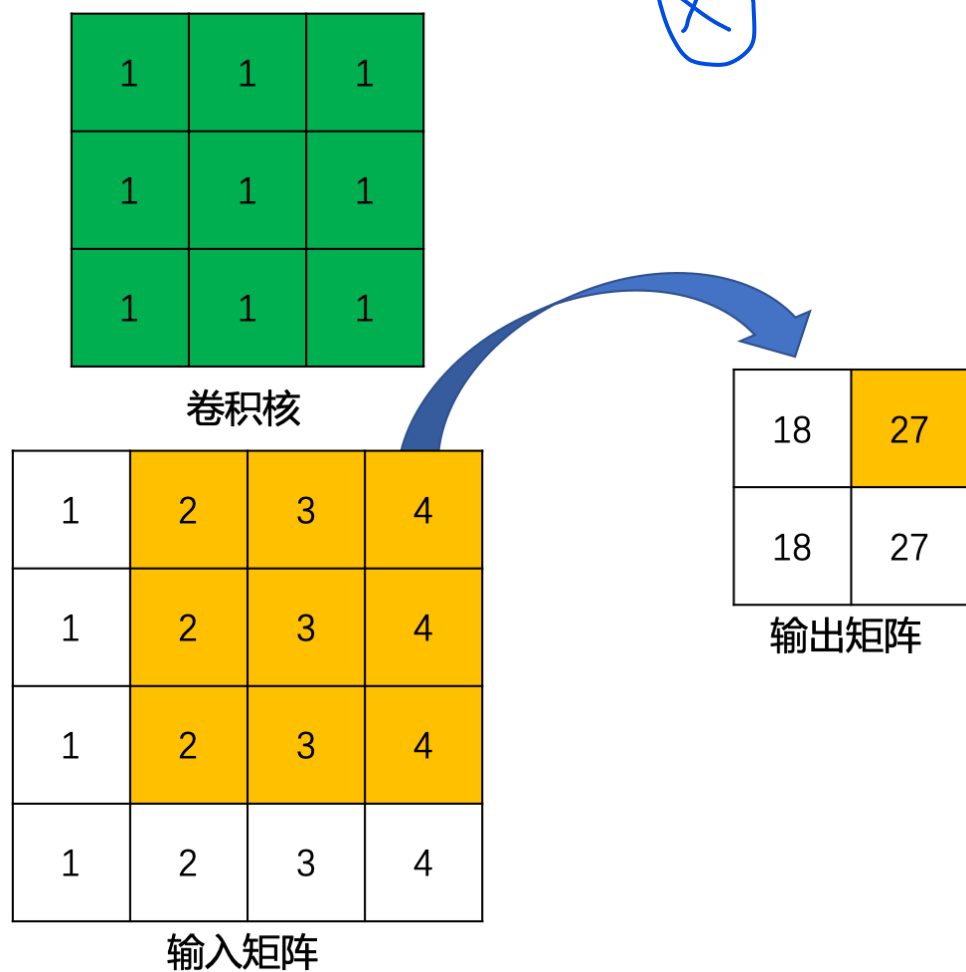
■ 1x1卷积



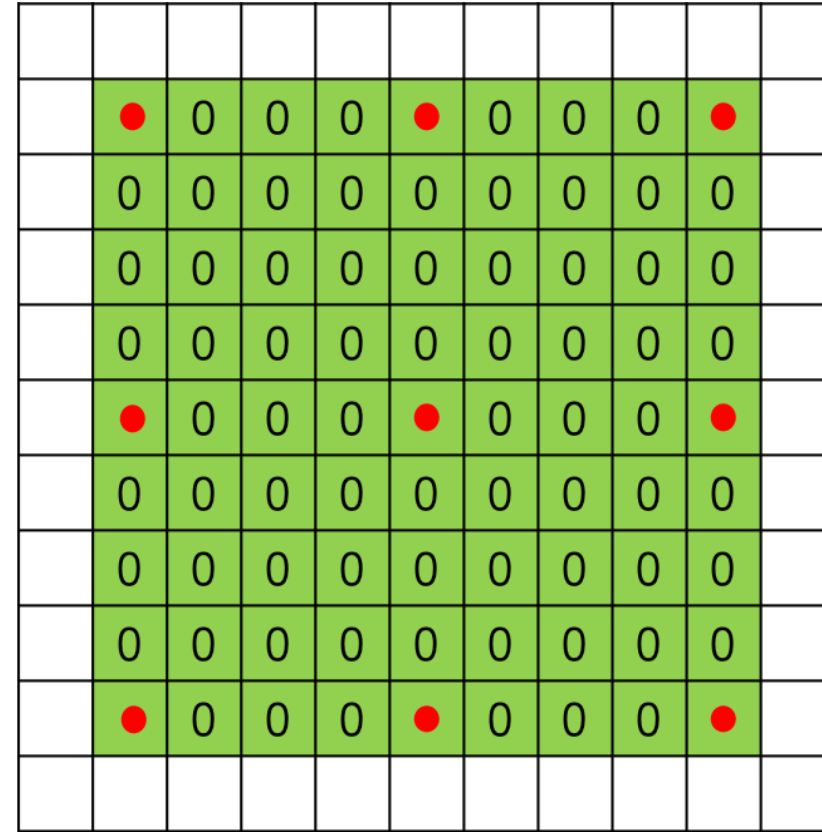
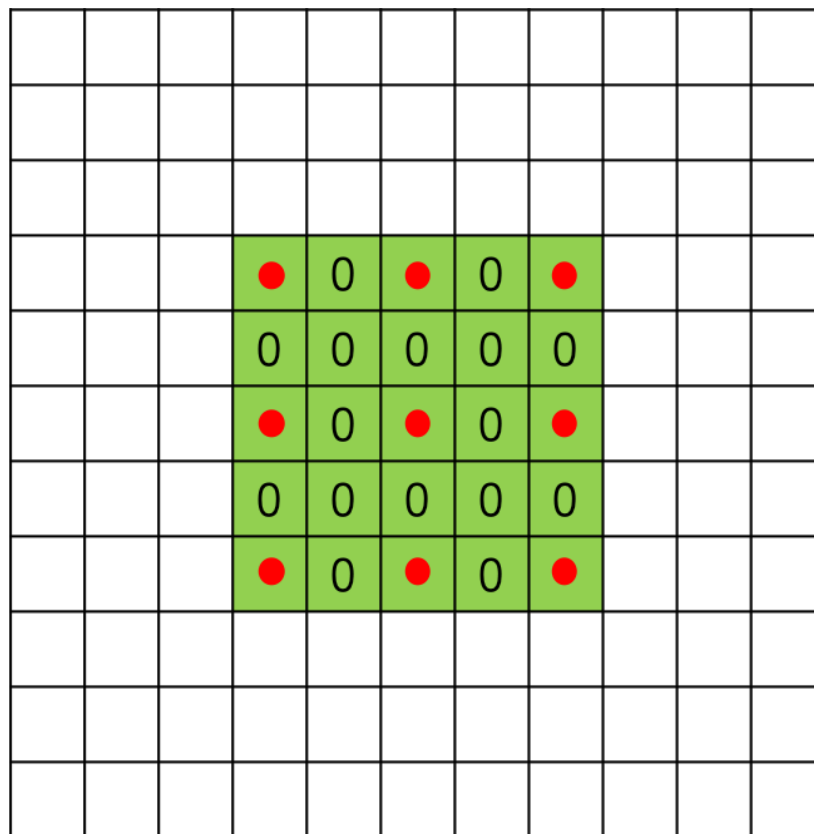
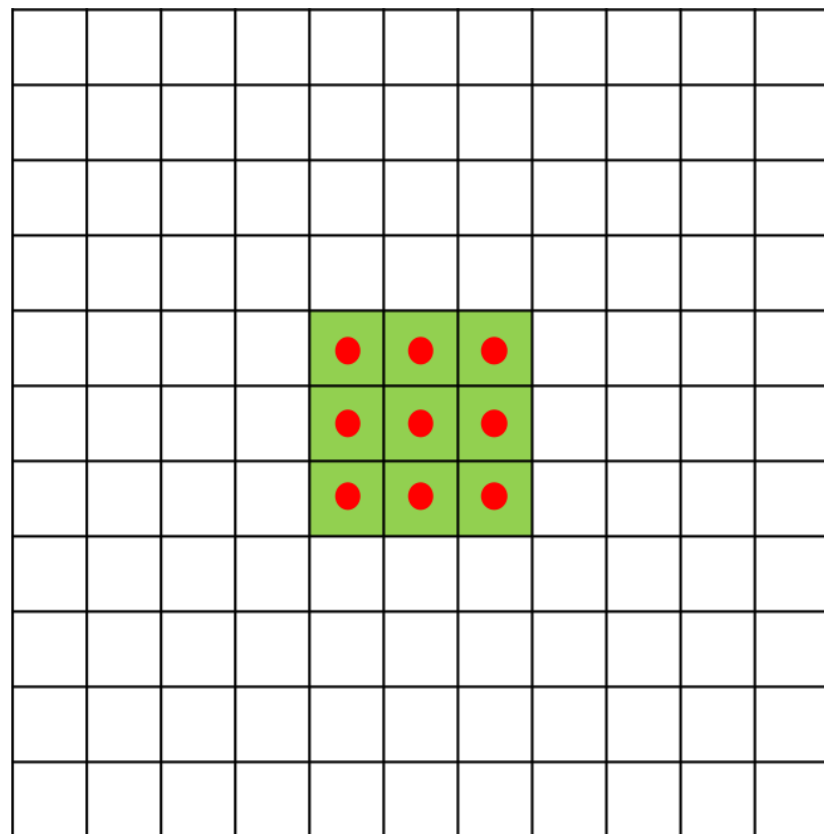
■ 3D卷积



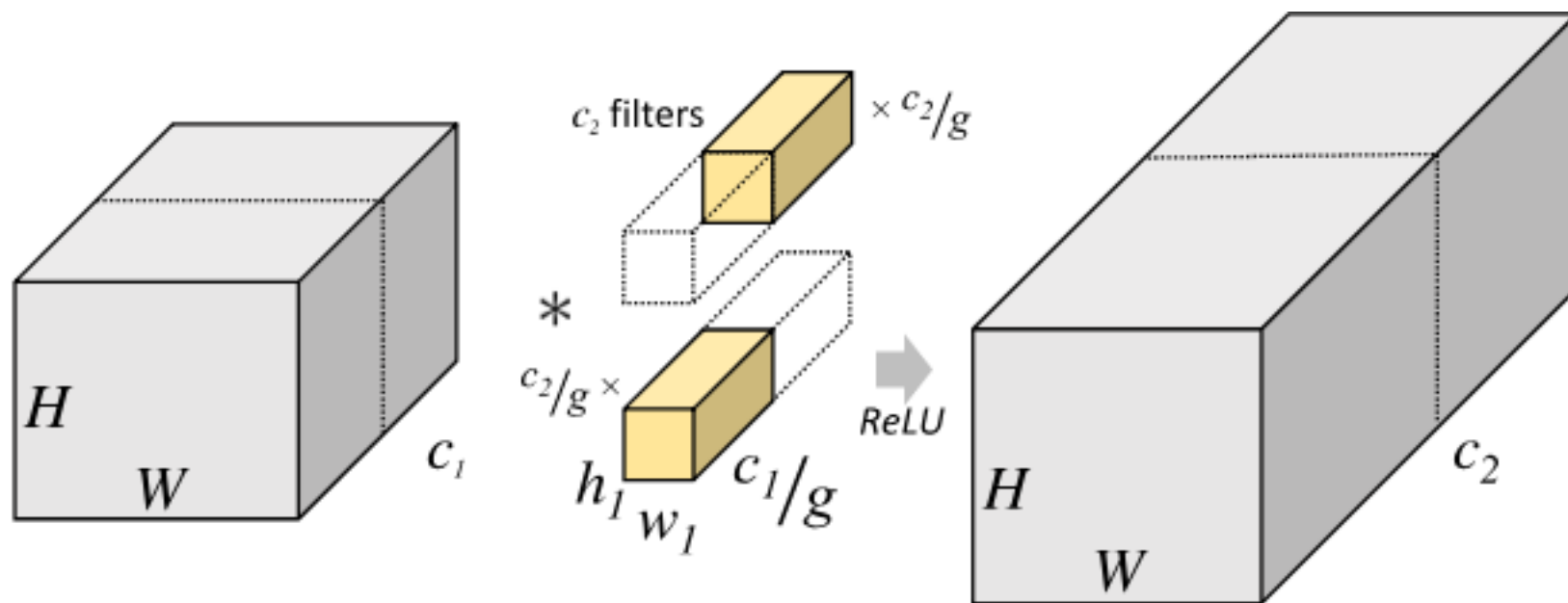
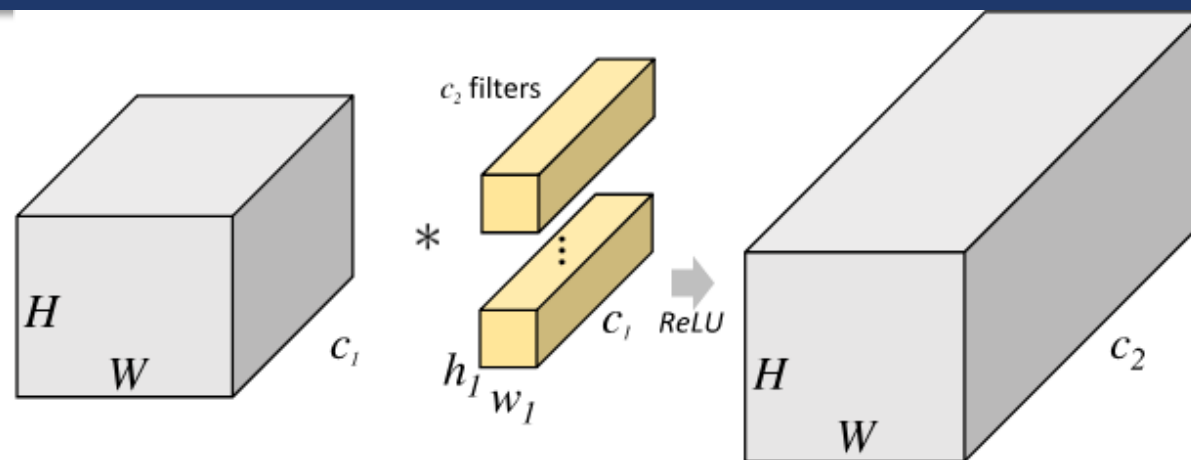
■ 转置卷积



■ 空洞卷积

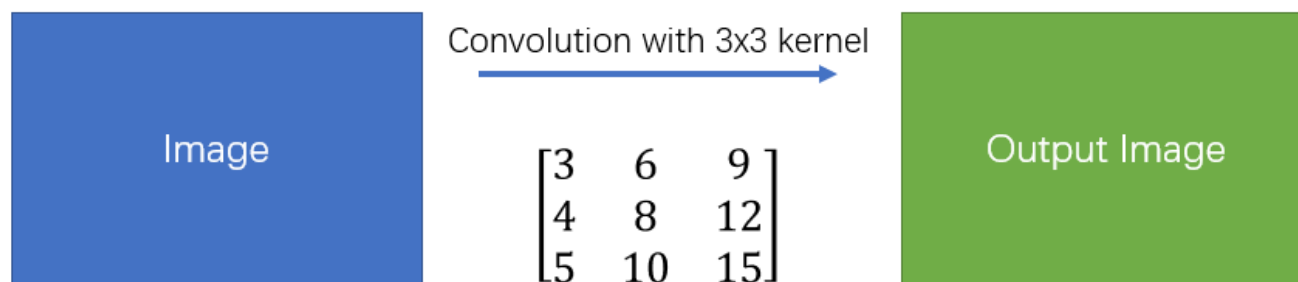


■ 分组卷积

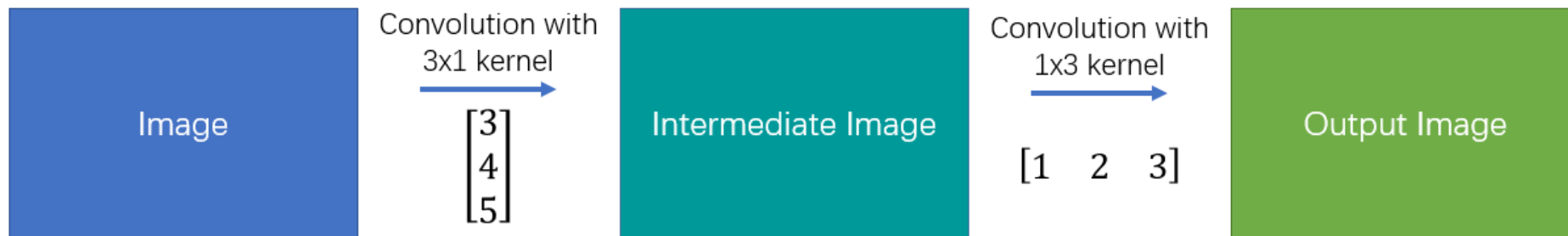


■ 可分离卷积

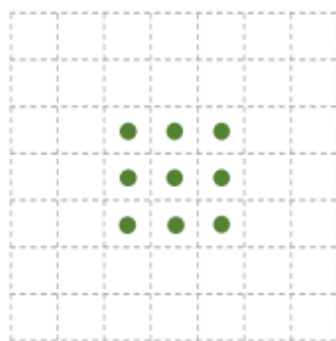
Simple Convolution



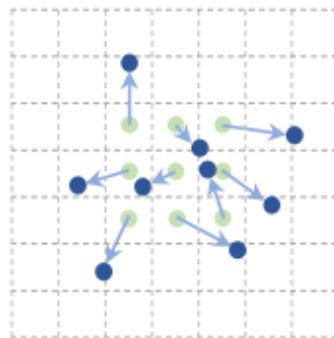
Spatial Separable Convolution



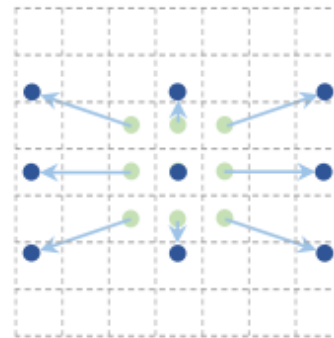
■ 可变形卷积



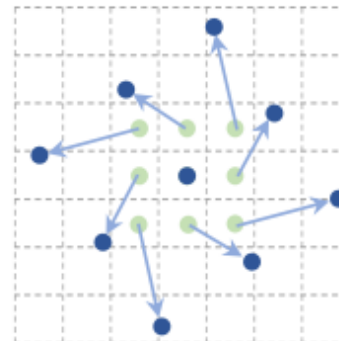
(a)



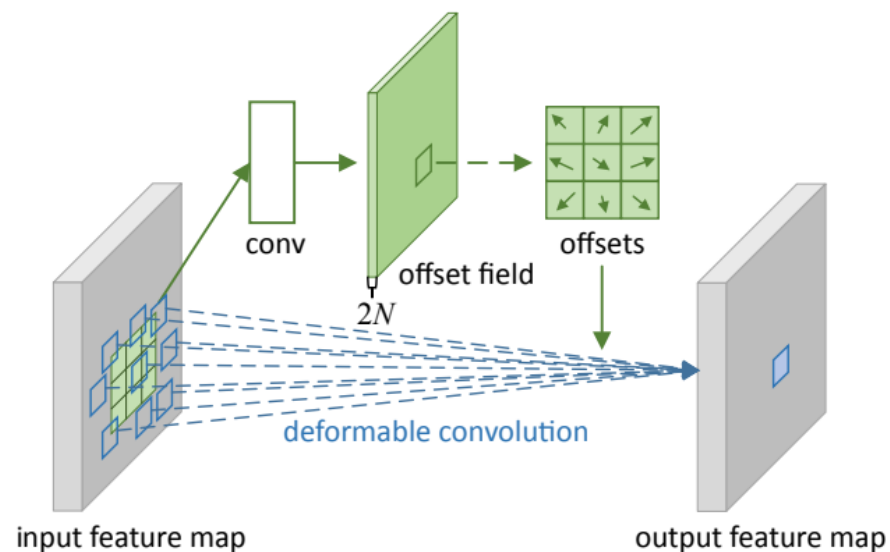
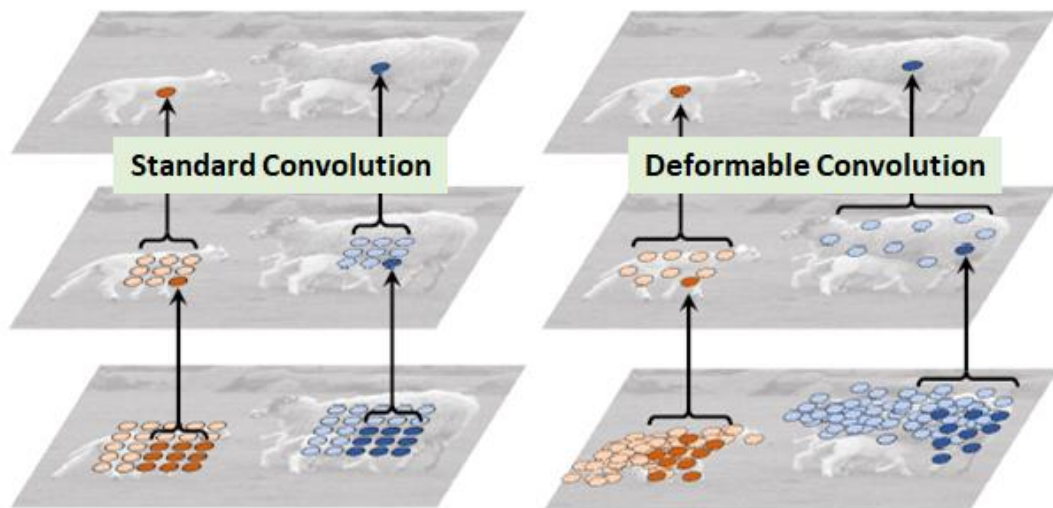
(b)



(c)



(d)



■ 下节课：卷积神经网络架构

